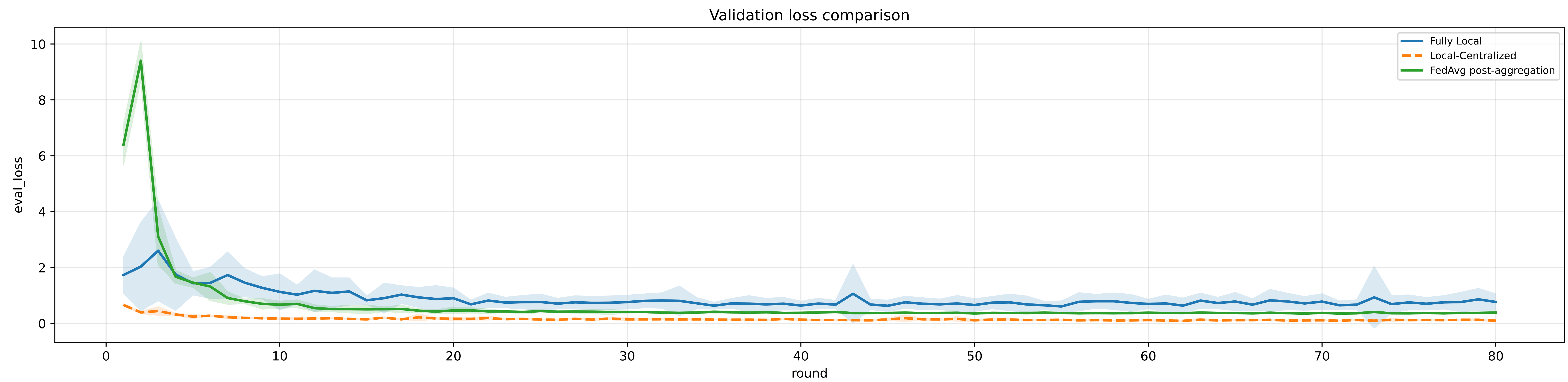
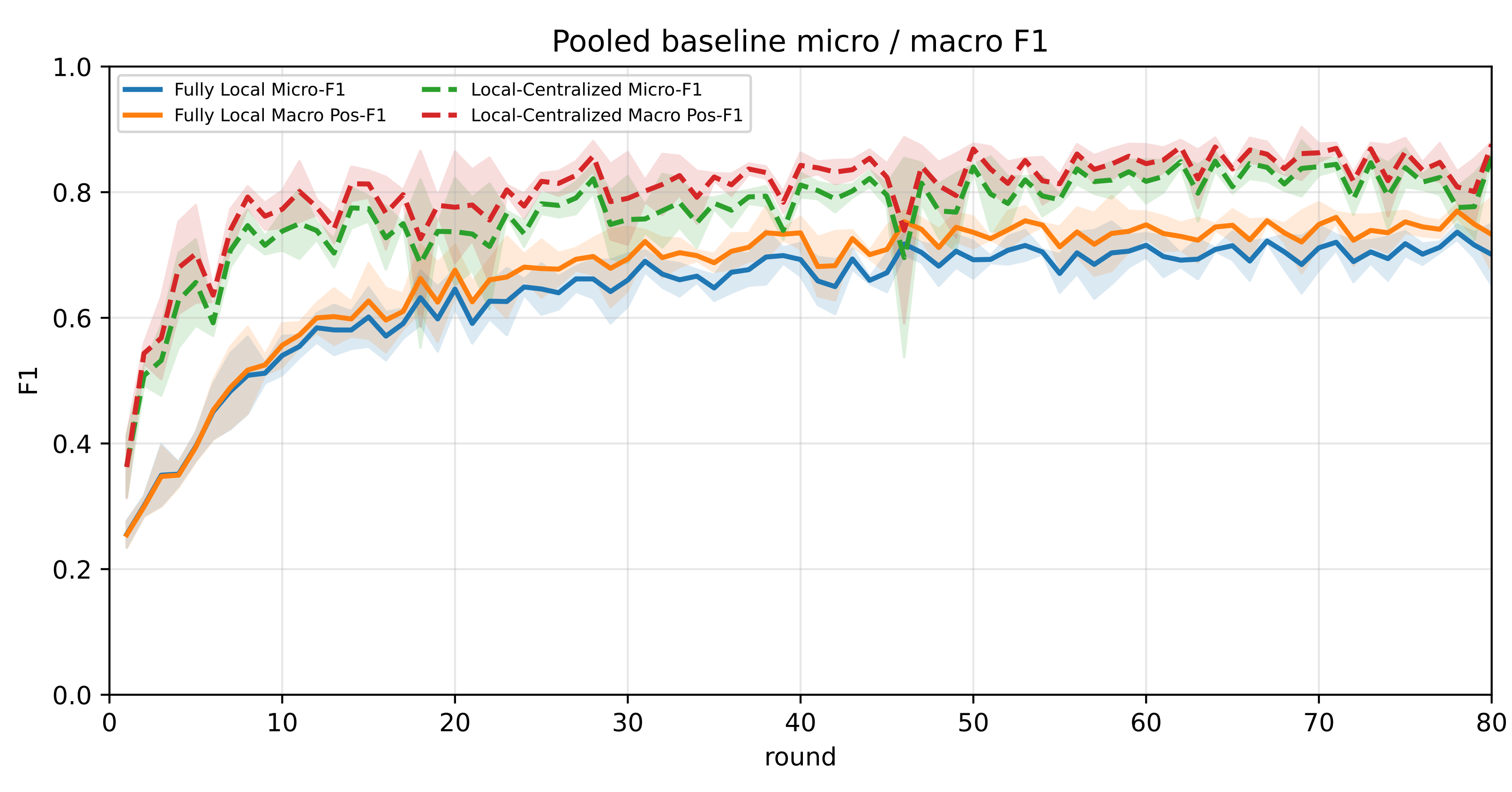
Controlled q-label heterogeneity: iid | FedAvg diagnostics | aggregated / pooled over 5 clients | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64_headmulti

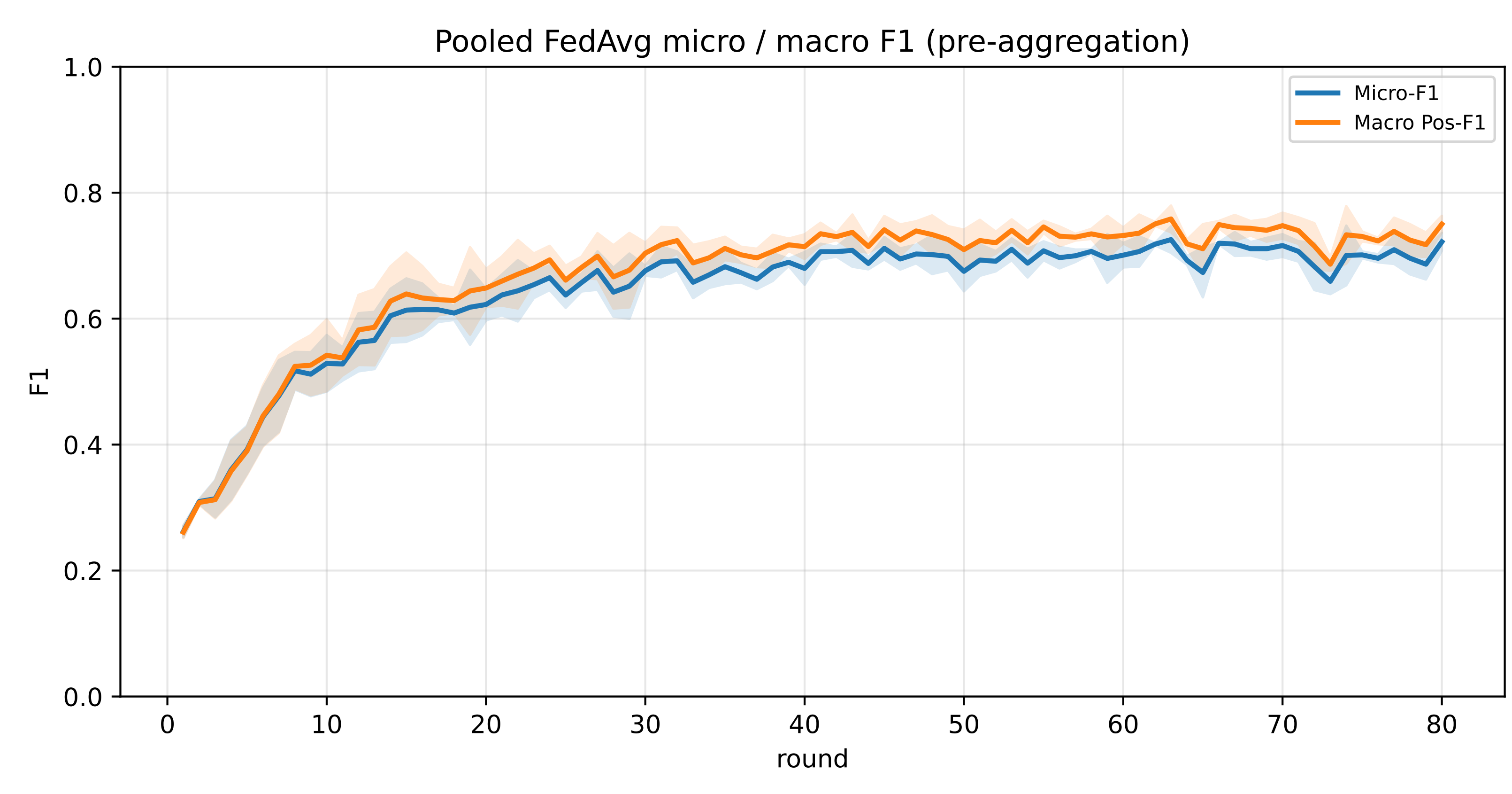
family=q_task_cycle2 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% |
cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle3 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% |
cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle4 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% |
cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle5 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% |
cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
family=q_task_cycle6 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% |
cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
heterogeneity=iid | gamma=0.20 | task_jsd_mean=0.000 | task_jsd_median=0.000 | task_jsd_max=0.000 | family_centroid_jsd_mean=0.000

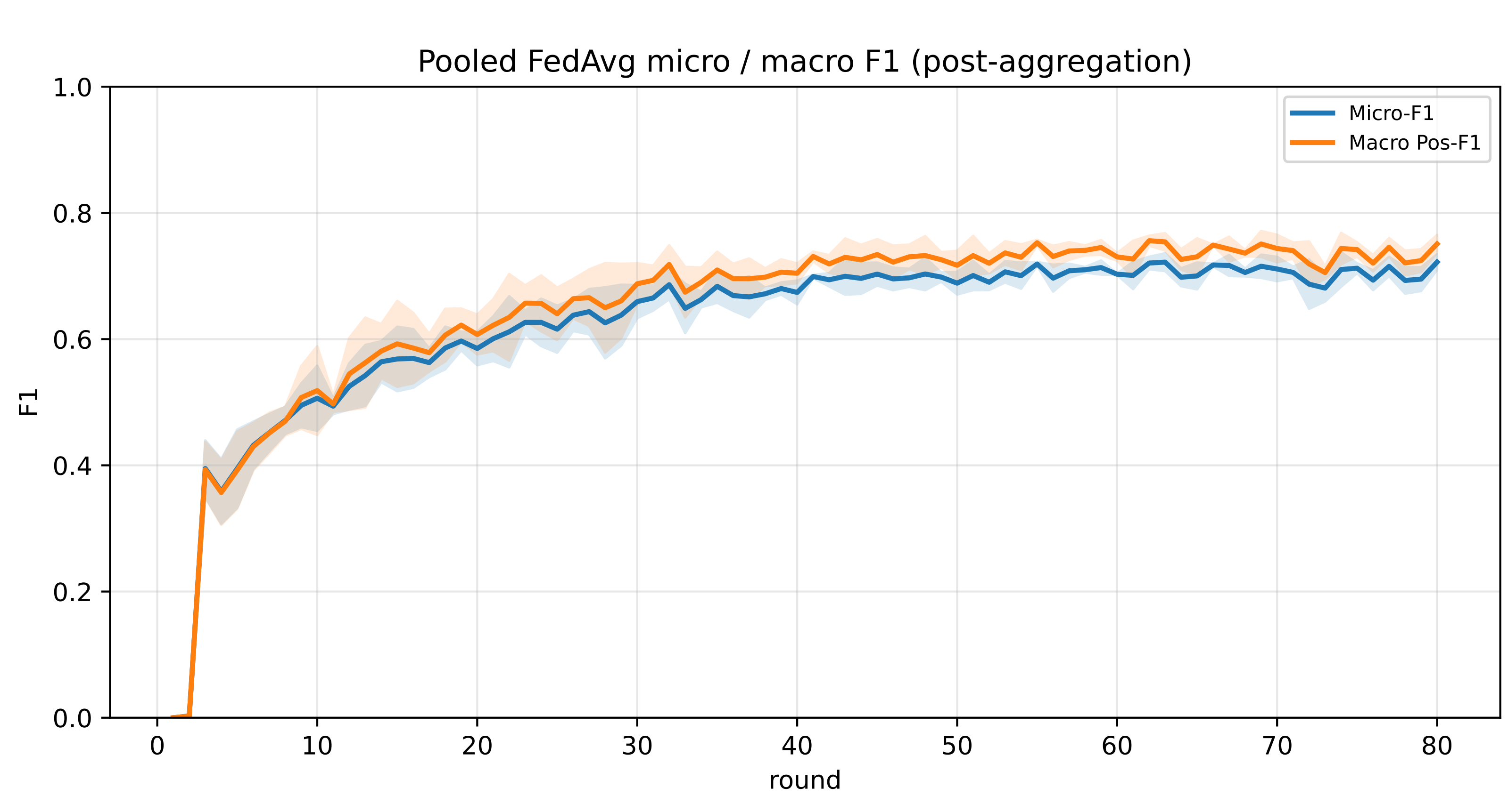
Best logged test performance (Fully Local vs Local-Centralized vs FedAvg)

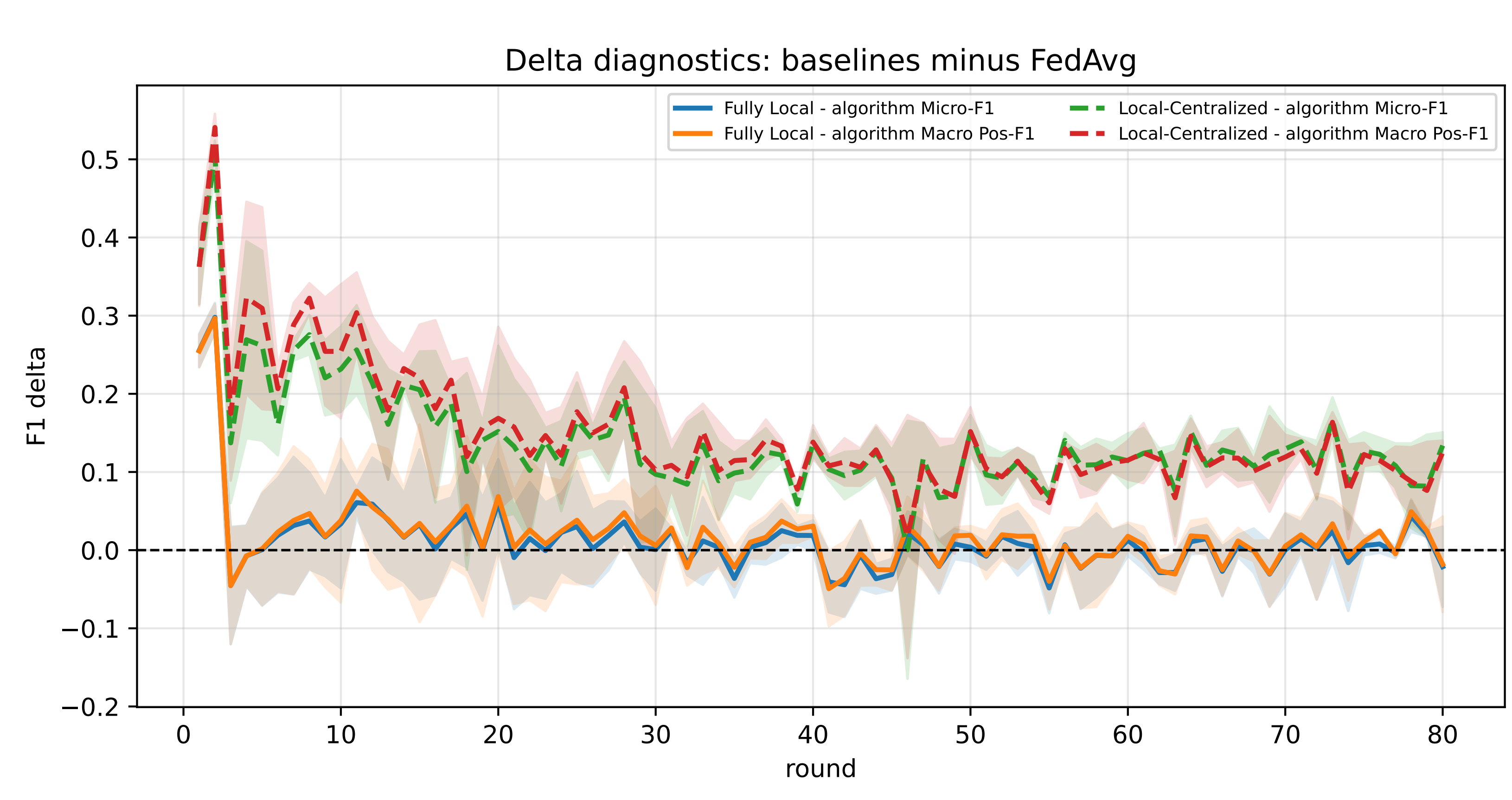
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	67.9% ± 1.6%	72.3% ± 1.5%	90.0% ± 1.5%	85.0% ± 1.7%	67.5% ± 3.5%	66.8% ± 2.2%	50.9% ± 1.2%
Local-Centralized	**85.0% ± 0.7%**	**87.0% ± 0.6%**	**98.0% ± 1.1%**	**94.5% ± 1.4%**	**89.7% ± 1.8%**	**80.3% ± 0.5%**	**72.5% ± 1.3%**
FedAvg	70.2% ± 1.6%	73.5% ± 1.4%	91.2% ± 2.1%	82.8% ± 3.6%	72.4% ± 1.2%	64.1% ± 1.0%	56.9% ± 2.0%









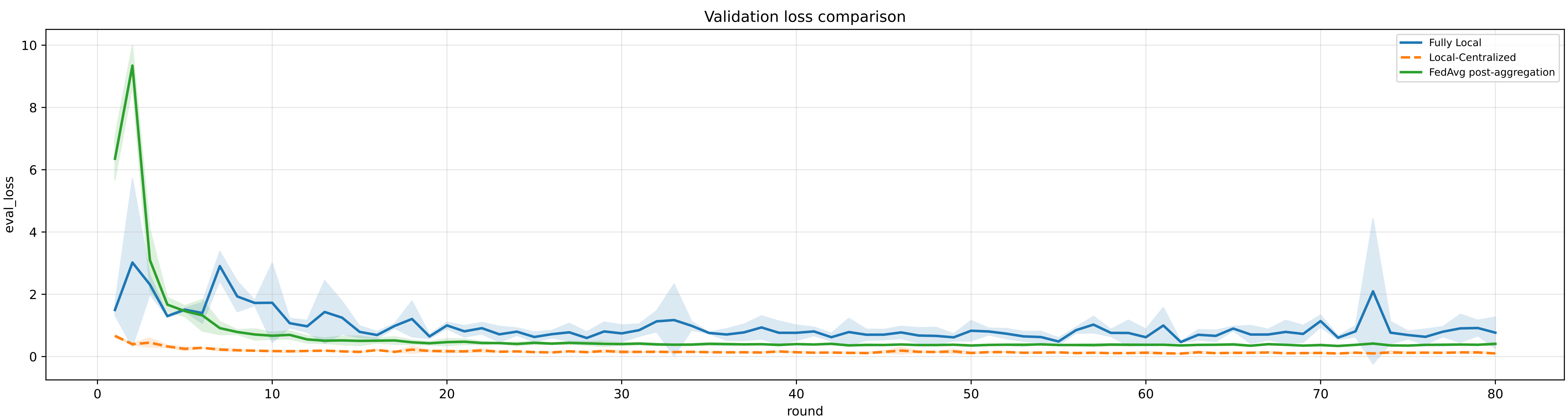


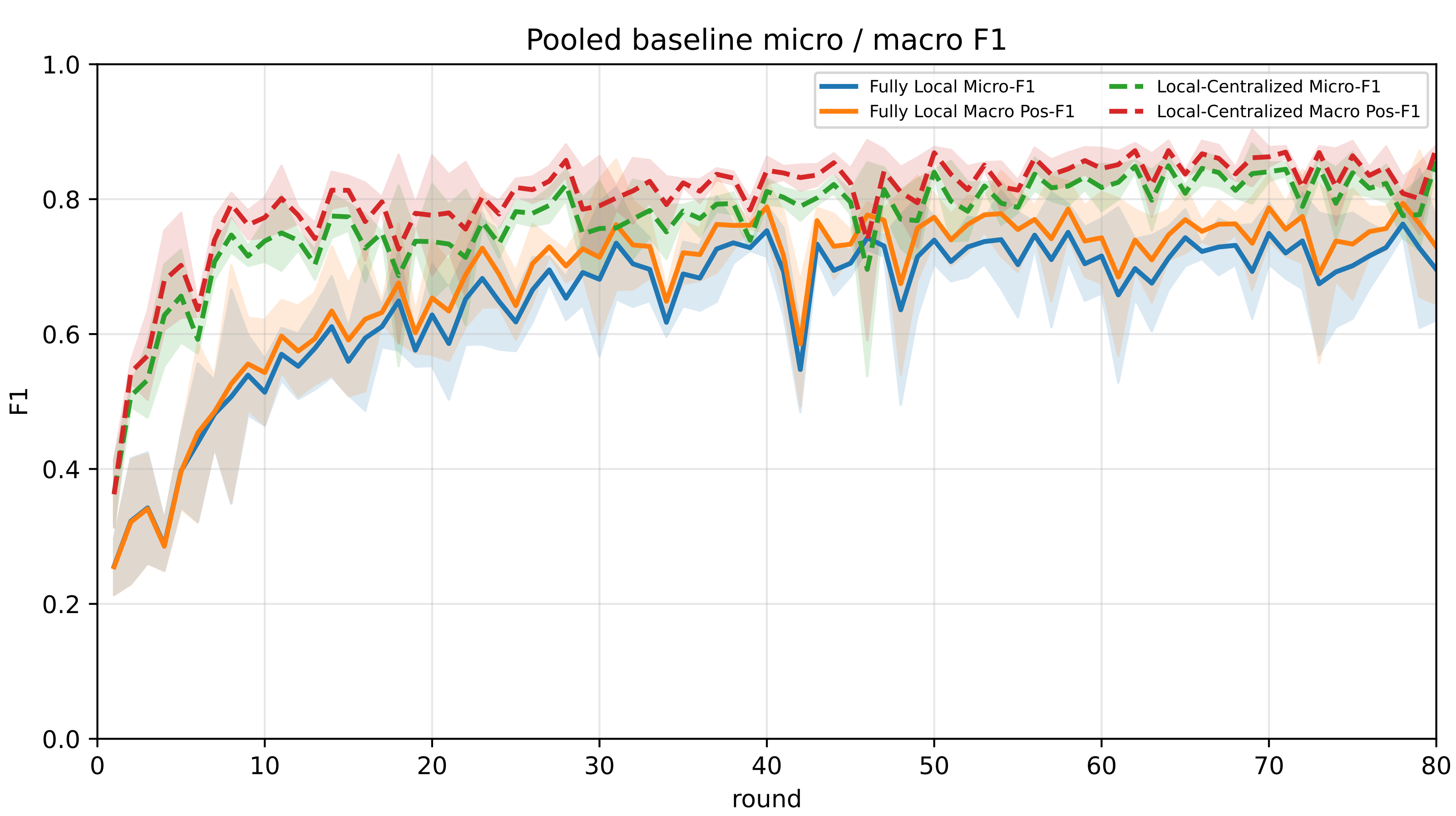
Controlled q-label heterogeneity: iid | FedAvg diagnostics | family=q_task_cycle2 |
n_clients=1 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64_headmulti

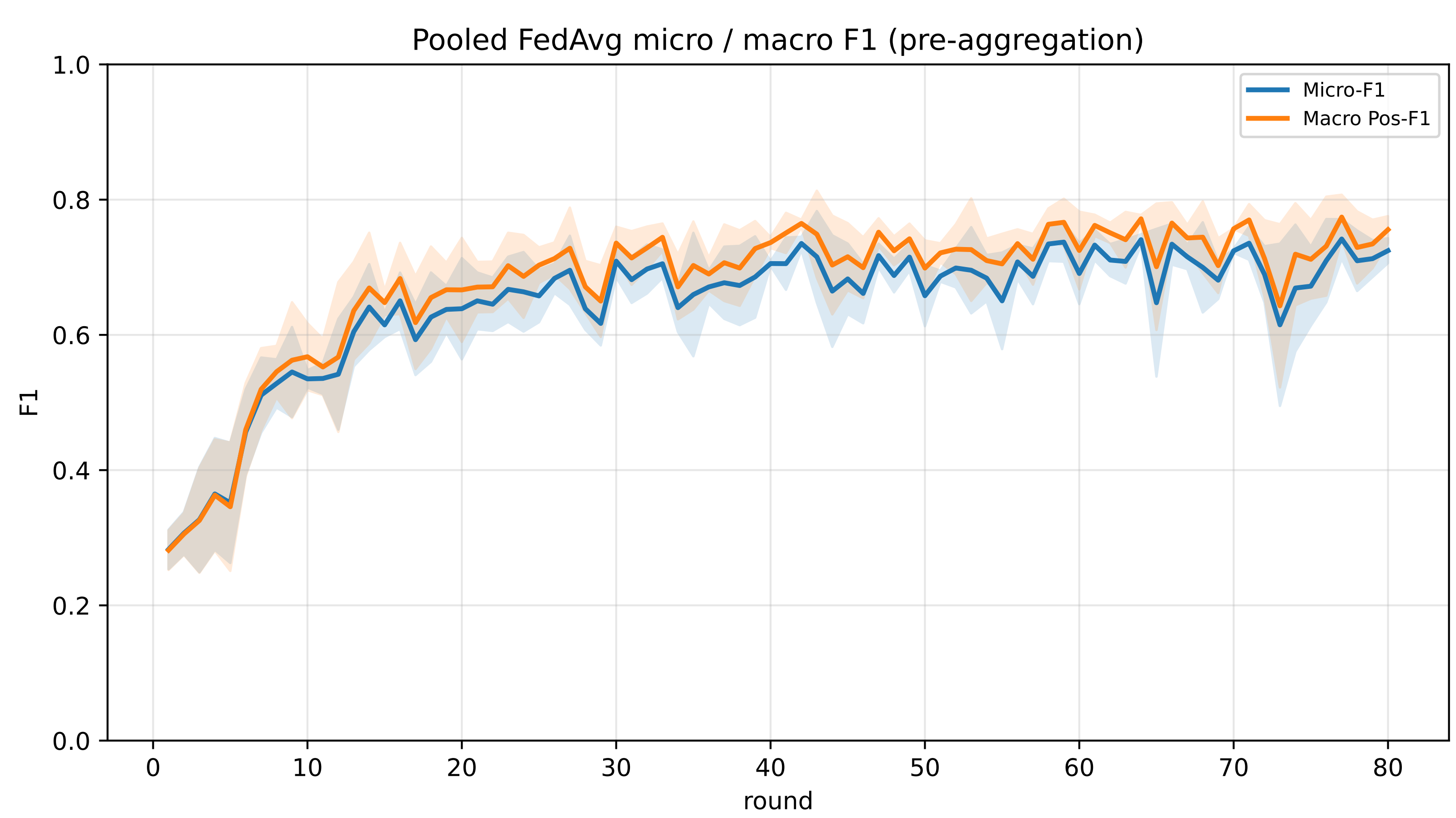
family=q_task_cycle2 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
heterogeneity=iid | gamma=0.20 | task_jsd_mean=0.000 | task_jsd_median=0.000 | task_jsd_max=0.000 | family_centroid_jsd_mean=0.000

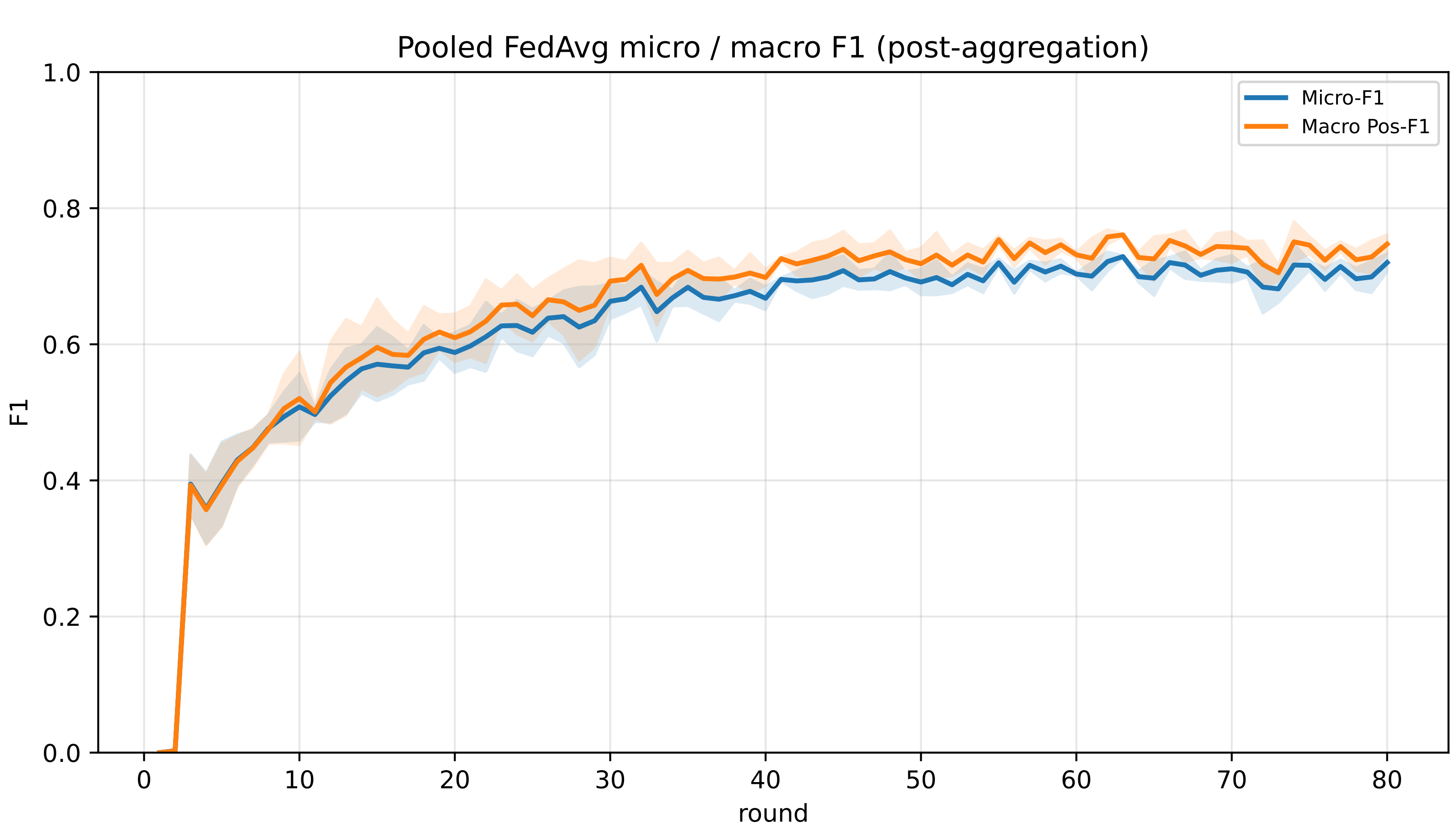
Best logged test performance (Fully Local vs Local-Centralized vs FedAvg)

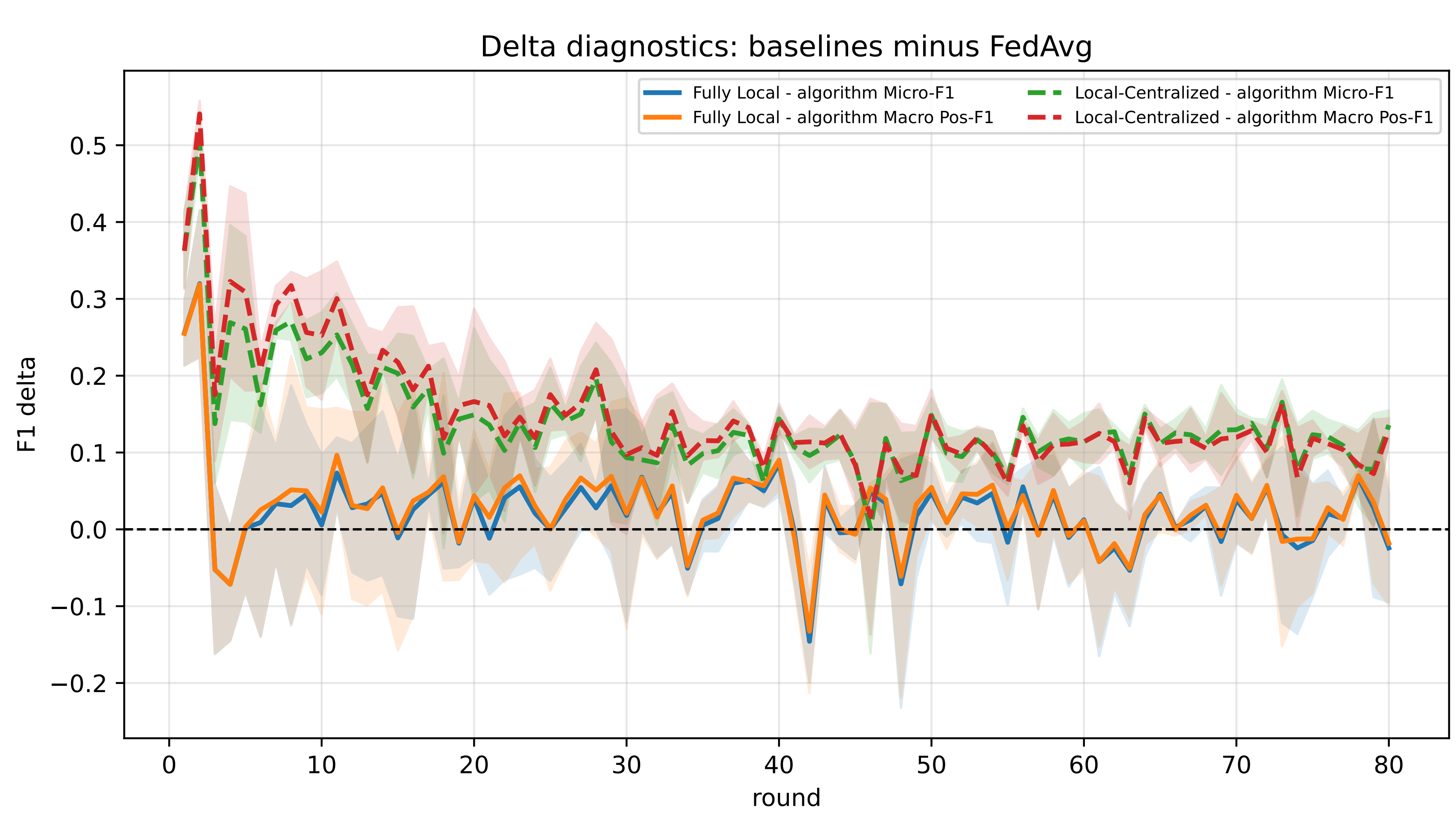
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	70.1% ± 3.2%	74.3% ± 2.5%	93.1% ± 1.1%	86.7% ± 1.6%	70.3% ± 4.6%	67.5% ± 2.2%	53.8% ± 5.0%
Local-Centralized	**85.0% ± 0.7%**	**87.0% ± 0.6%**	**98.0% ± 1.1%**	**94.5% ± 1.4%**	**89.7% ± 1.8%**	**80.3% ± 0.5%**	**72.5% ± 1.3%**
FedAvg	70.3% ± 2.1%	73.5% ± 2.0%	90.3% ± 3.0%	82.8% ± 5.0%	72.6% ± 1.9%	64.5% ± 1.4%	57.1% ± 2.4%









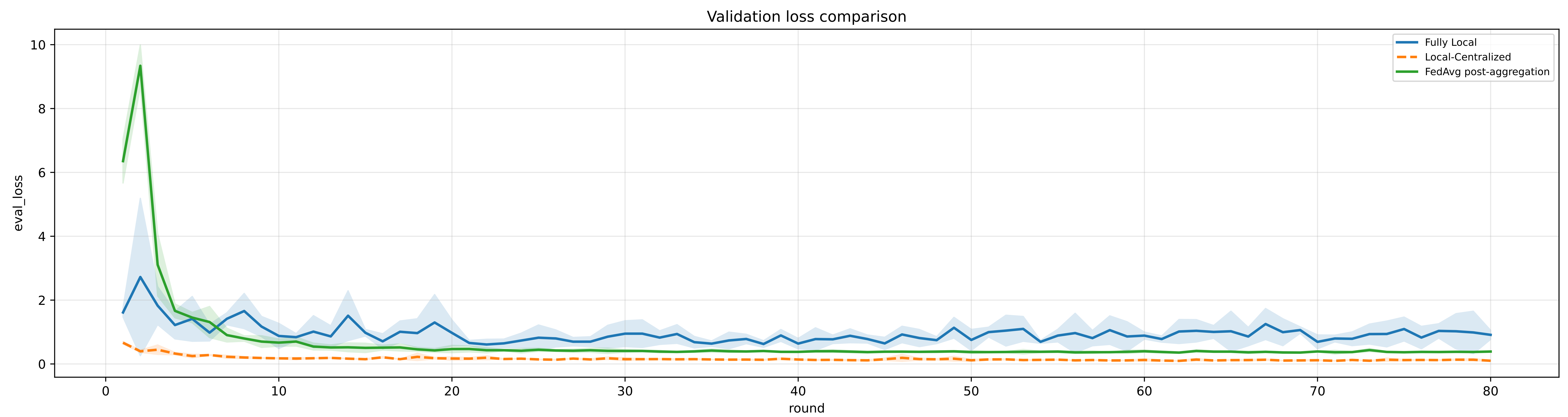


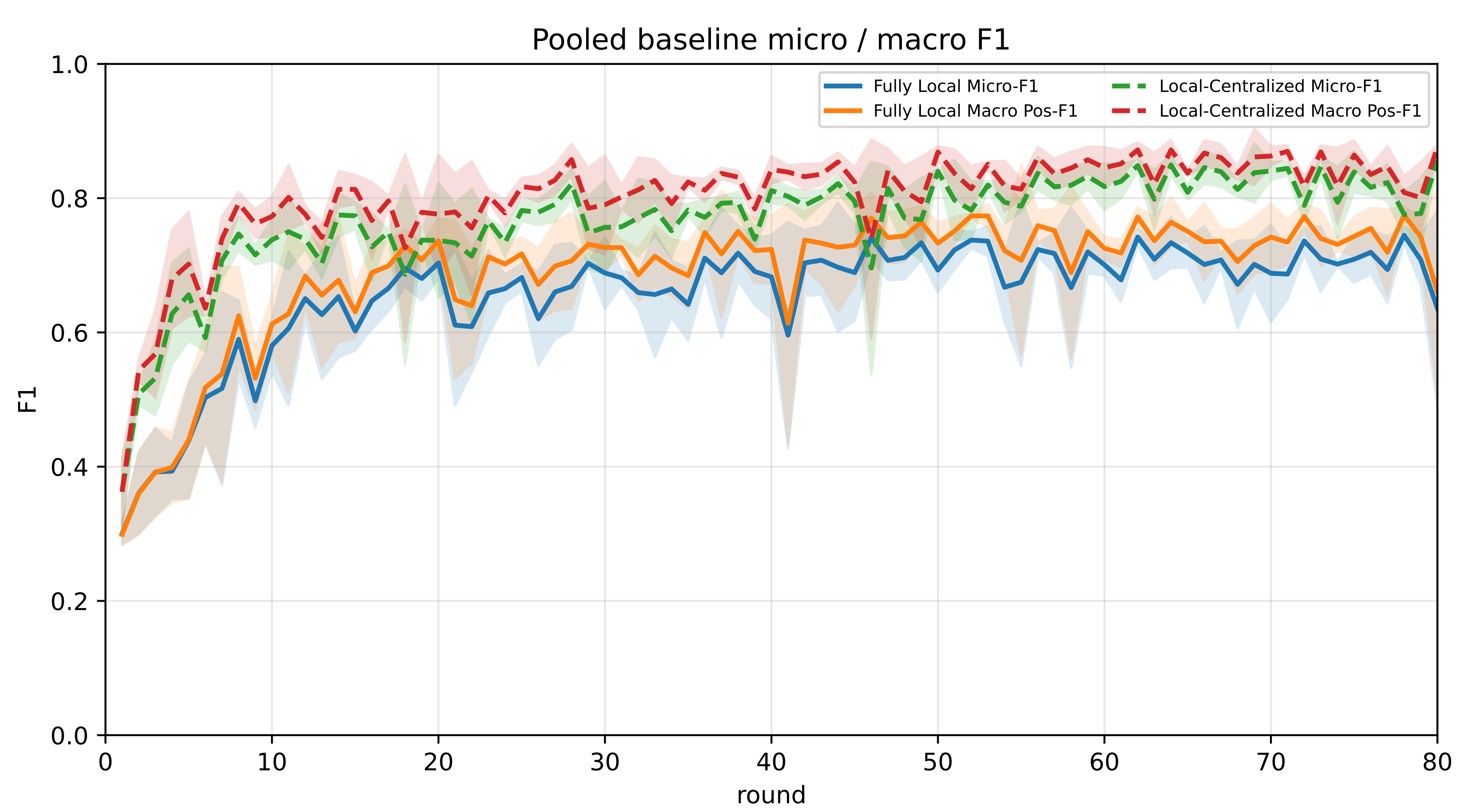
Controlled q-label heterogeneity: iid | FedAvg diagnostics | family=q_task_cycle3 | n_clients=1 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64_headmulti

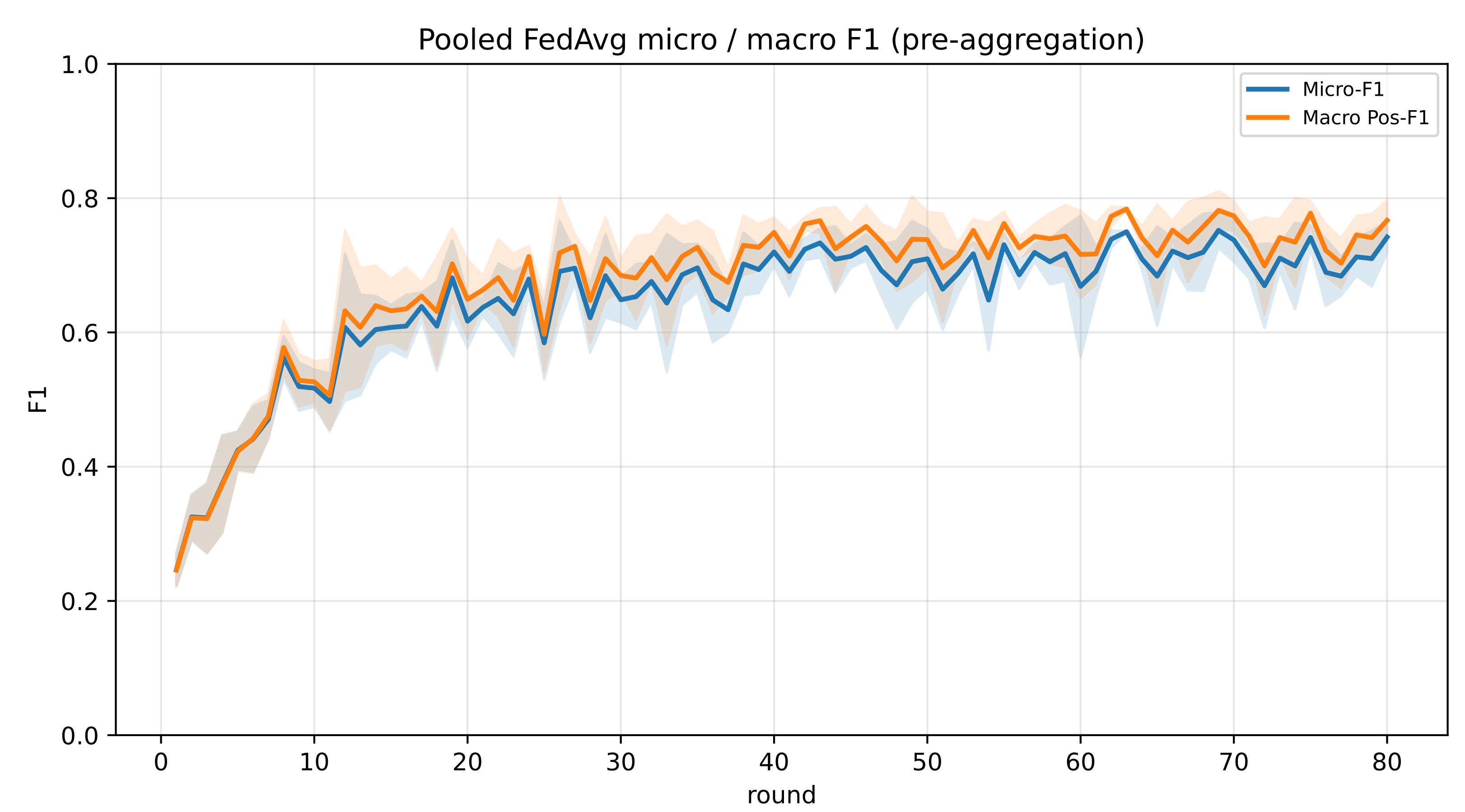
family=q_task_cycle3 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% | cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
heterogeneity=iid | gamma=0.20 | task_jsd_mean=0.000 | task_jsd_median=0.000 | task_jsd_max=0.000 | family_centroid_jsd_mean=0.000

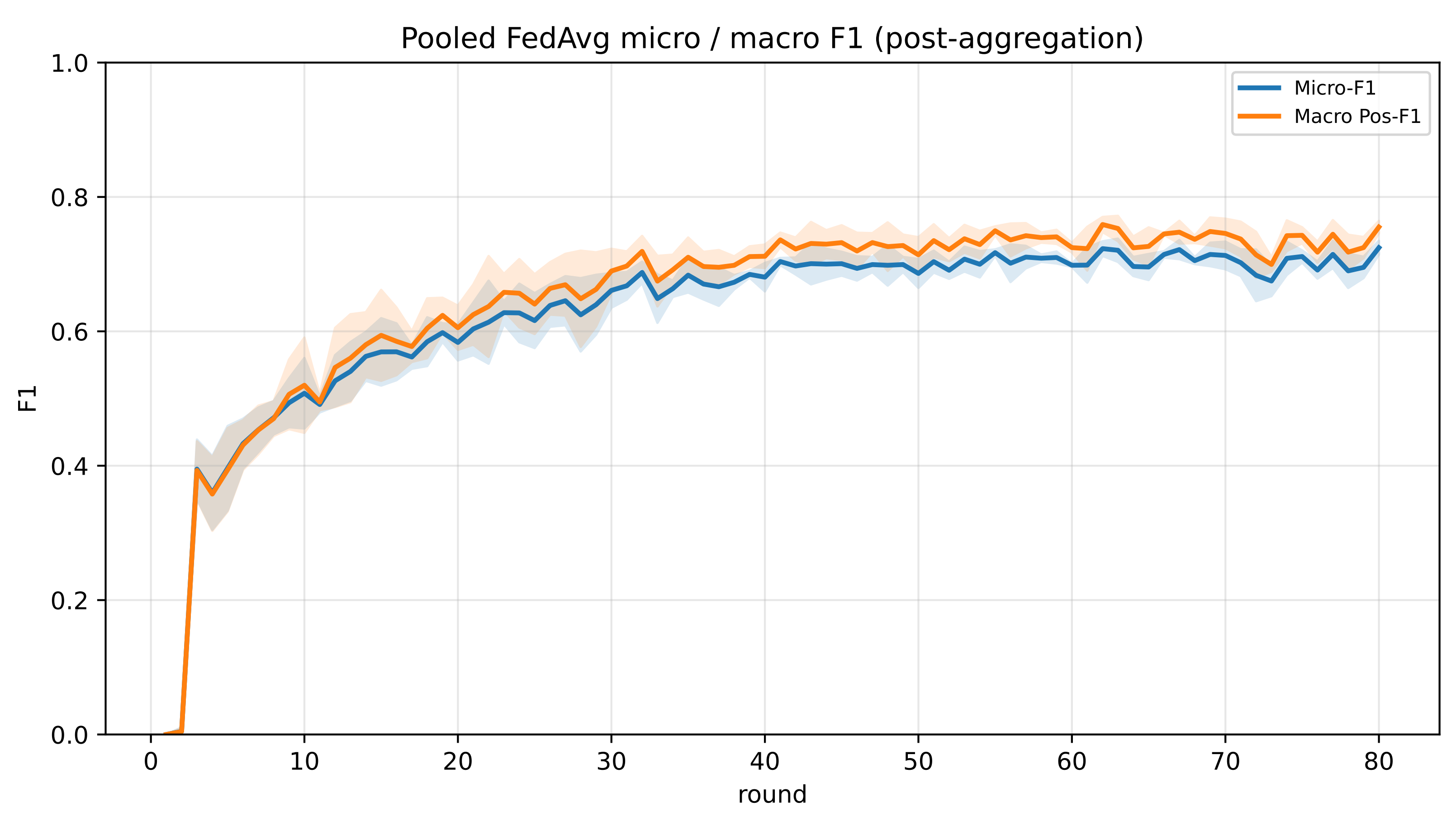
Best logged test performance (Fully Local vs Local-Centralized vs FedAvg)

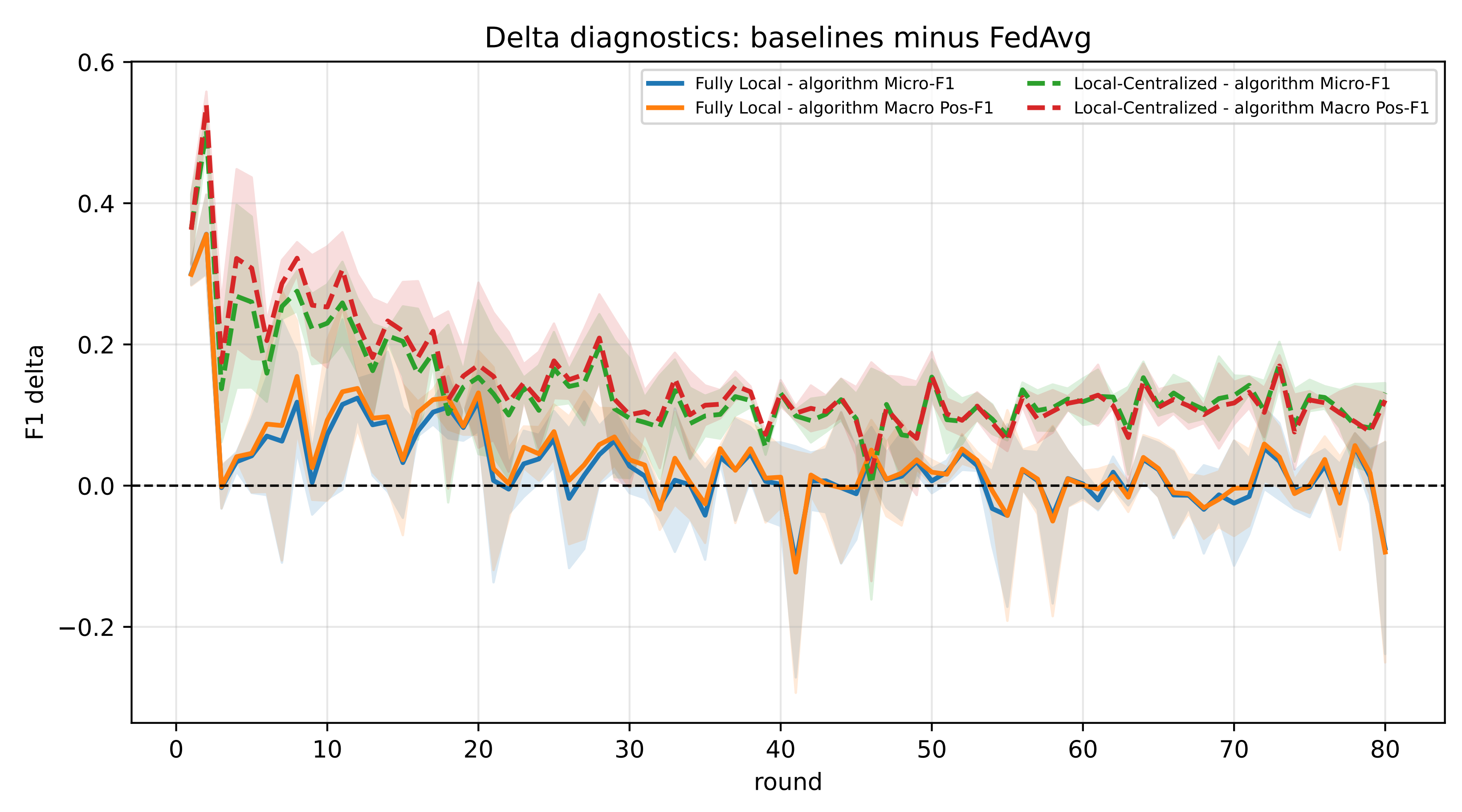
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	69.5% ± 0.8%	73.5% ± 0.6%	90.4% ± 0.9%	85.2% ± 1.6%	67.7% ± 0.6%	71.2% ± 1.1%	52.9% ± 2.1%
Local-Centralized	**85.0% ± 0.7%**	**87.0% ± 0.6%**	**98.0% ± 1.1%**	**94.5% ± 1.4%**	**89.7% ± 1.8%**	**80.3% ± 0.5%**	**72.5% ± 1.3%**
FedAvg	70.5% ± 1.8%	73.9% ± 1.5%	92.8% ± 1.8%	82.1% ± 3.6%	73.3% ± 1.6%	64.4% ± 1.5%	56.9% ± 2.3%









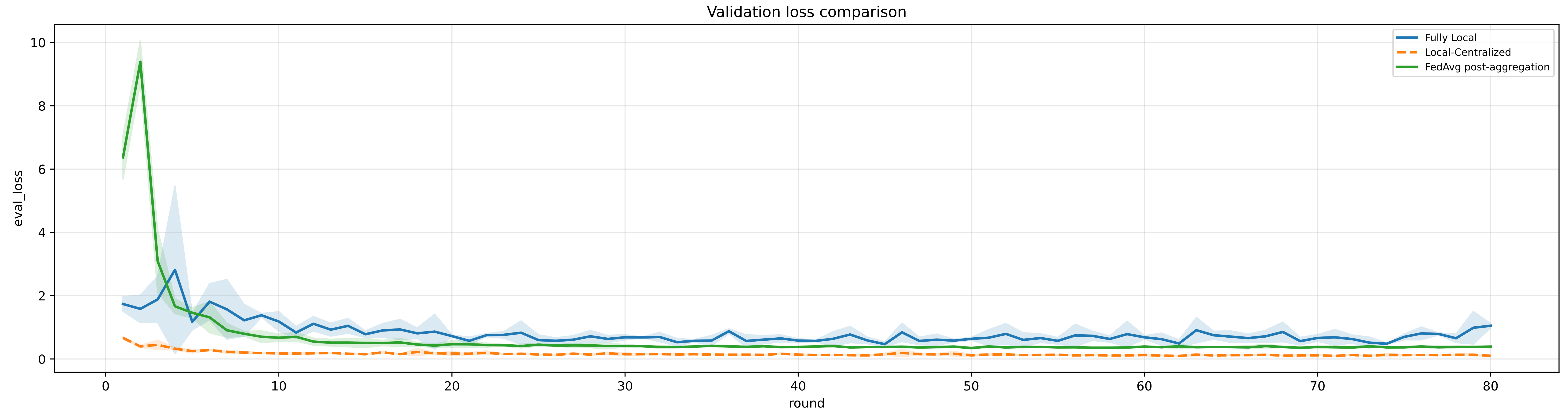


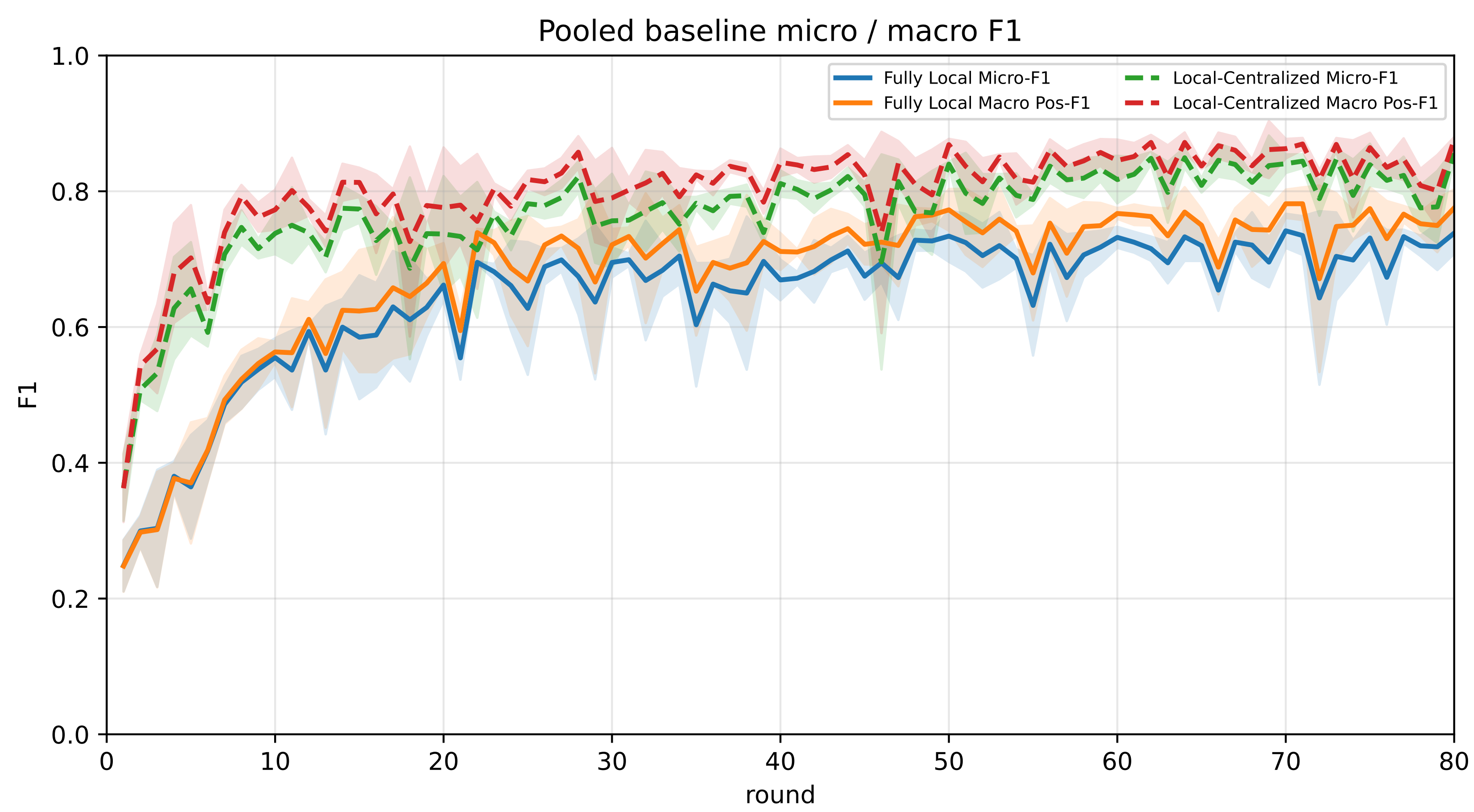
Controlled q-label heterogeneity: iid | FedAvg diagnostics | family=q_task_cycle4 |
n_clients=1 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64_headmulti

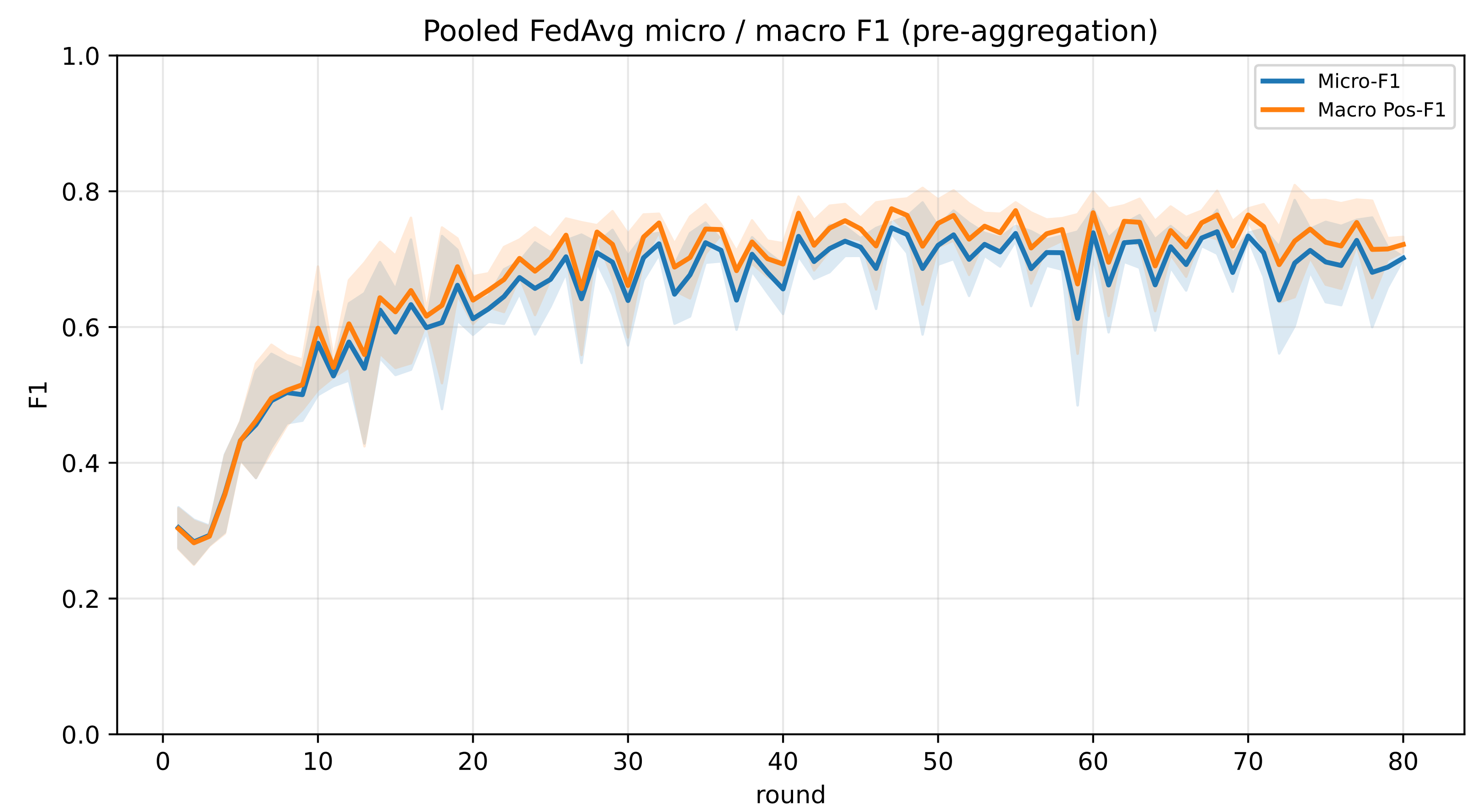
family=q_task_cycle4 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% |
cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
heterogeneity=iid | gamma=0.20 | task_jsd_mean=0.000 | task_jsd_median=0.000 | task_jsd_max=0.000 | family_centroid_jsd_mean=0.000

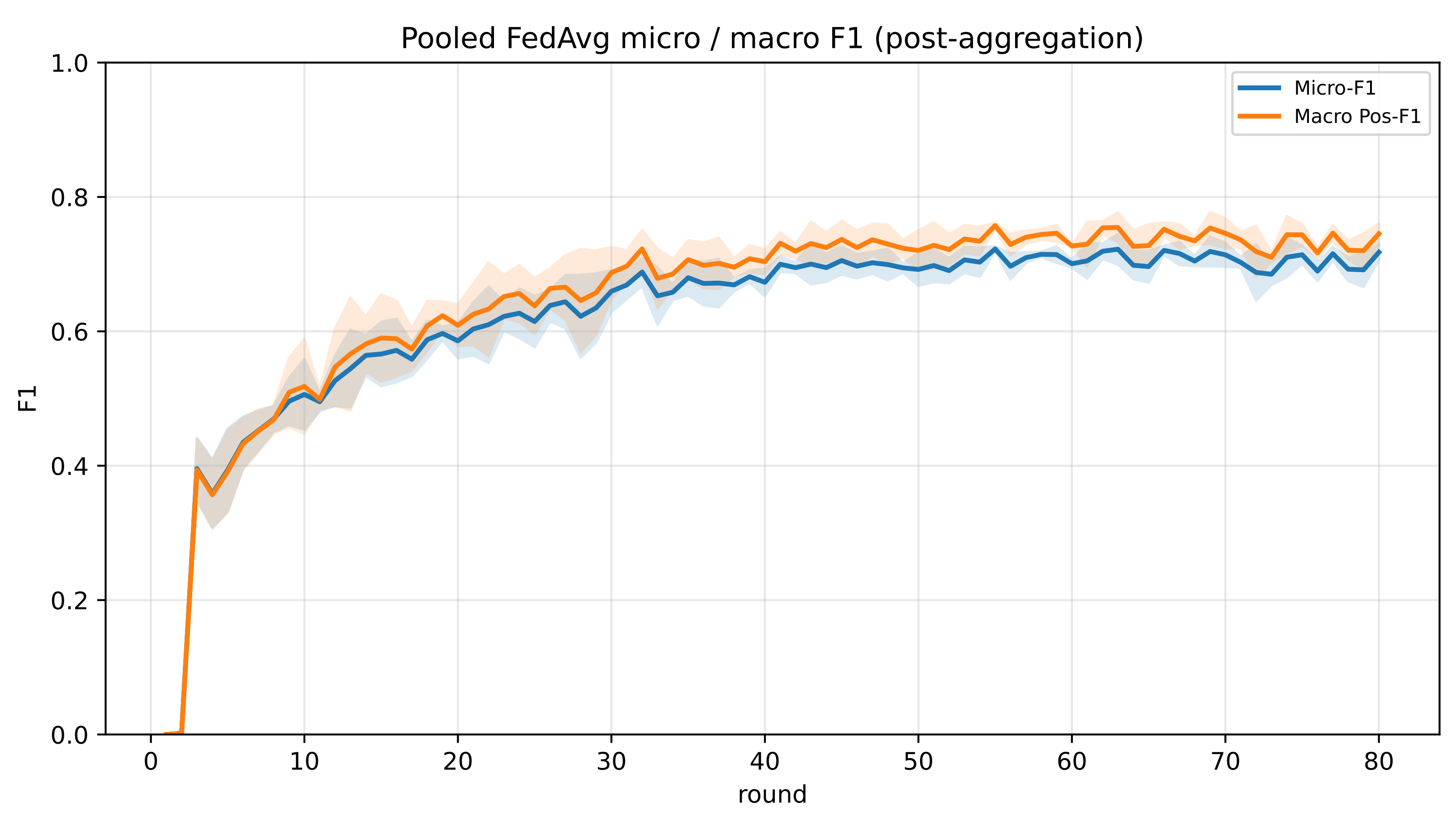
Best logged test performance (Fully Local vs Local-Centralized vs FedAvg)

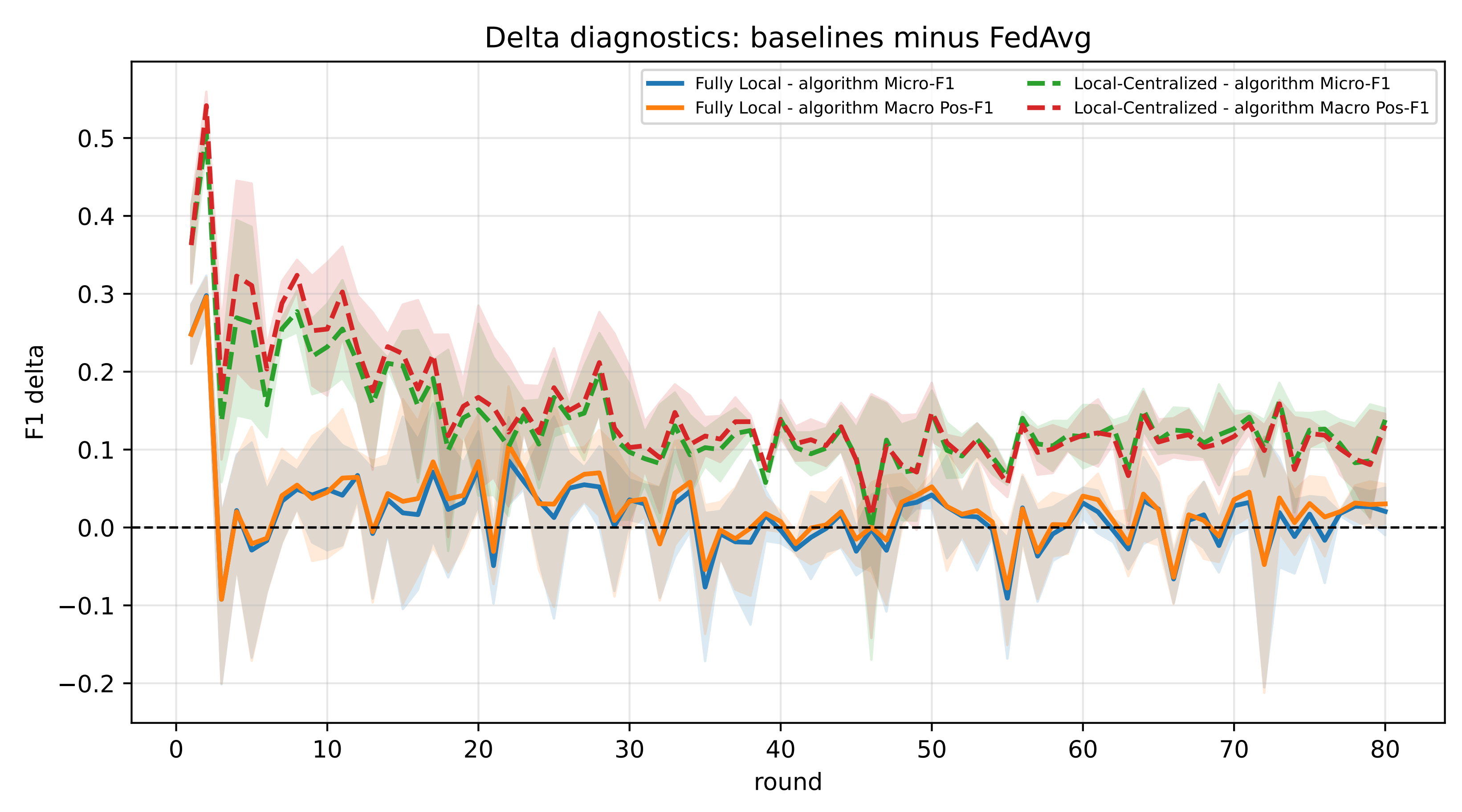
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	64.8% ± 3.3%	70.1% ± 2.7%	92.2% ± 3.4%	82.7% ± 3.5%	65.7% ± 8.3%	60.9% ± 7.0%	48.9% ± 2.7%
Local-Centralized	**85.0% ± 0.7%**	**87.0% ± 0.6%**	**98.0% ± 1.1%**	**94.5% ± 1.4%**	**89.7% ± 1.8%**	**80.3% ± 0.5%**	**72.5% ± 1.3%**
FedAvg	69.8% ± 1.8%	73.1% ± 1.7%	91.5% ± 1.7%	82.7% ± 3.3%	71.2% ± 1.4%	63.3% ± 1.3%	57.0% ± 2.1%









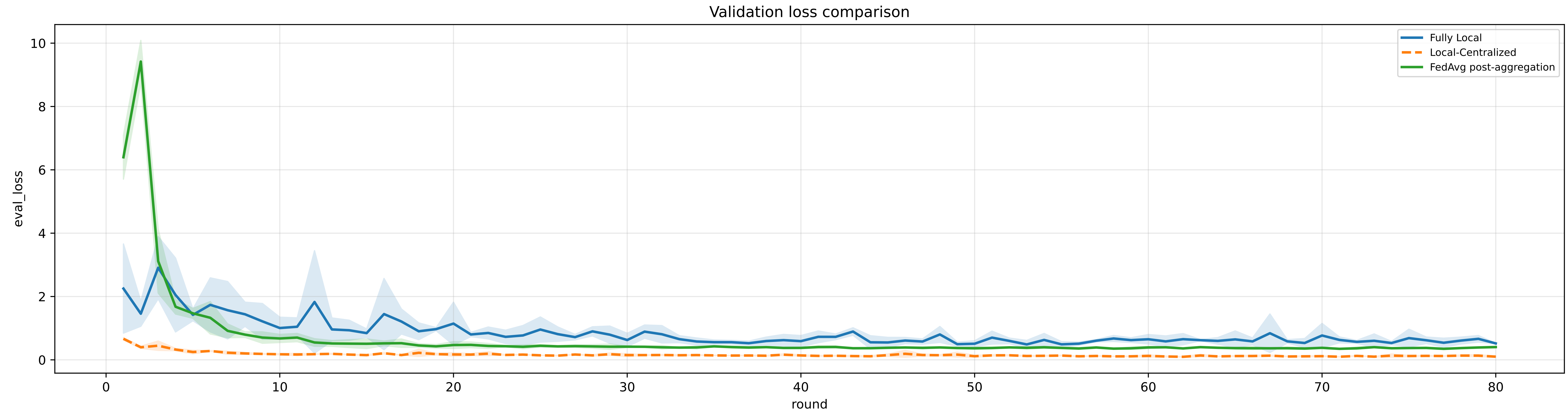


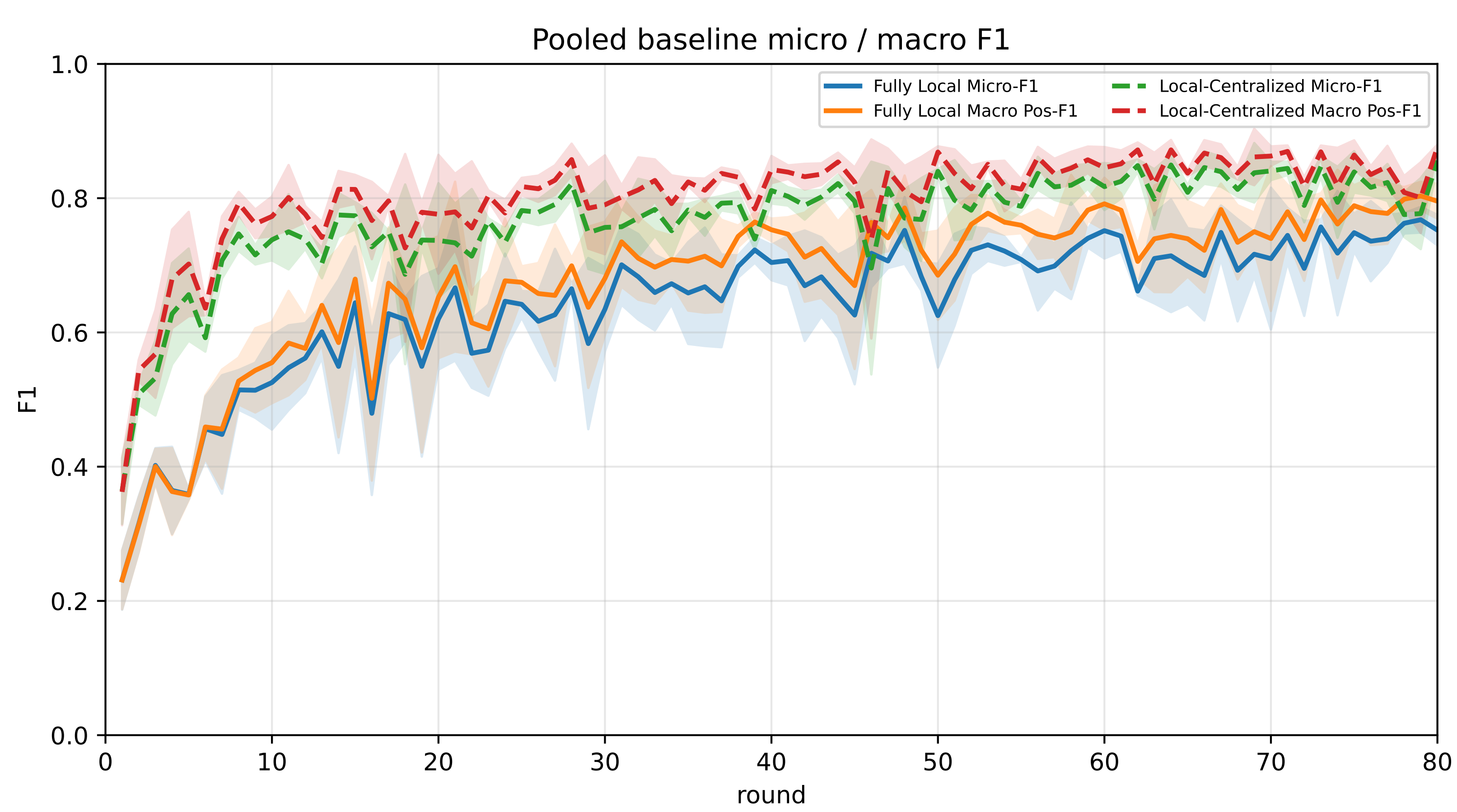
Controlled q-label heterogeneity: iid | FedAvg diagnostics | family=q_task_cycle5 |
n_clients=1 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64_headmulti

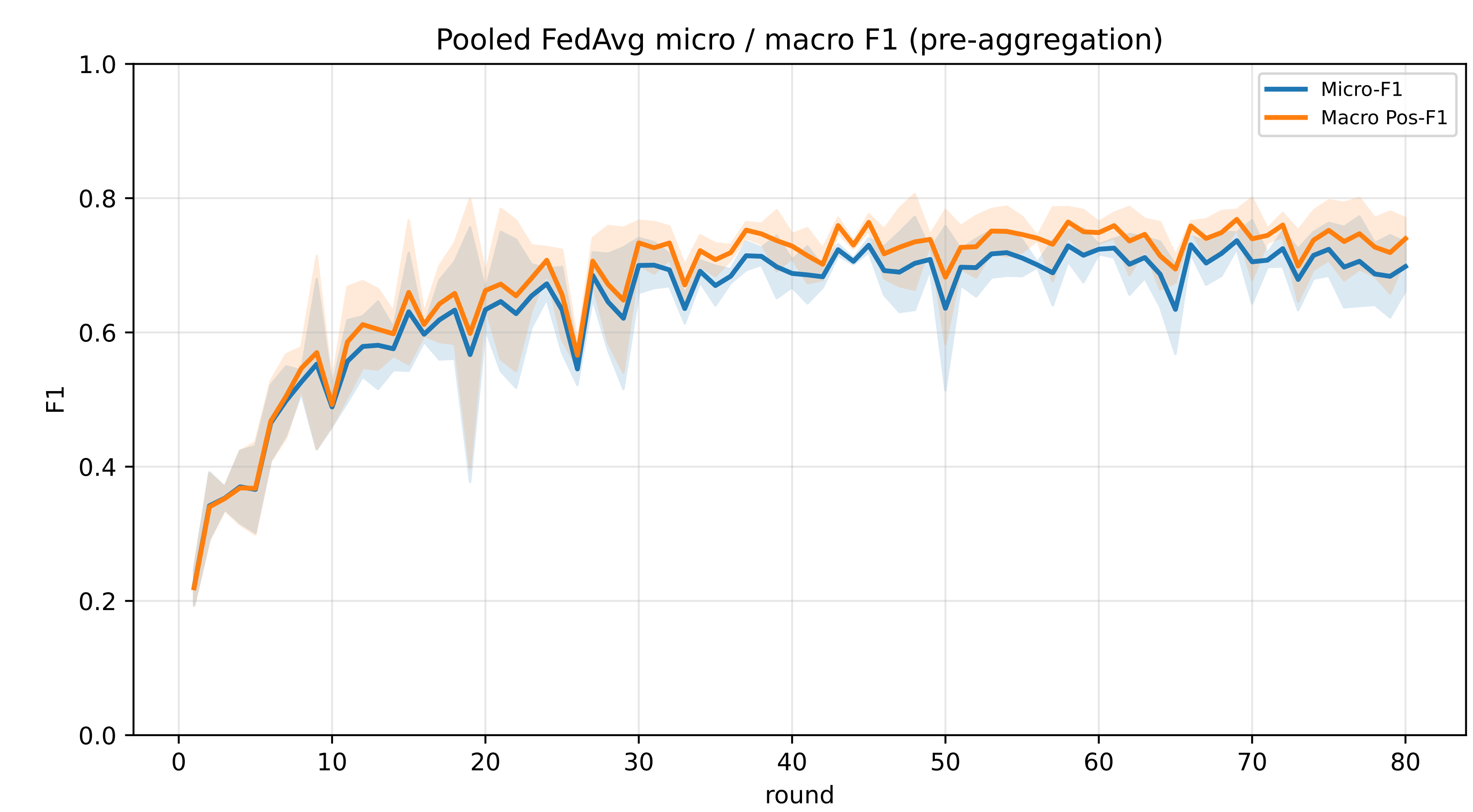
family=q_task_cycle5 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% |
cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
heterogeneity=iid | gamma=0.20 | task_jsd_mean=0.000 | task_jsd_median=0.000 | task_jsd_max=0.000 | family_centroid_jsd_mean=0.000

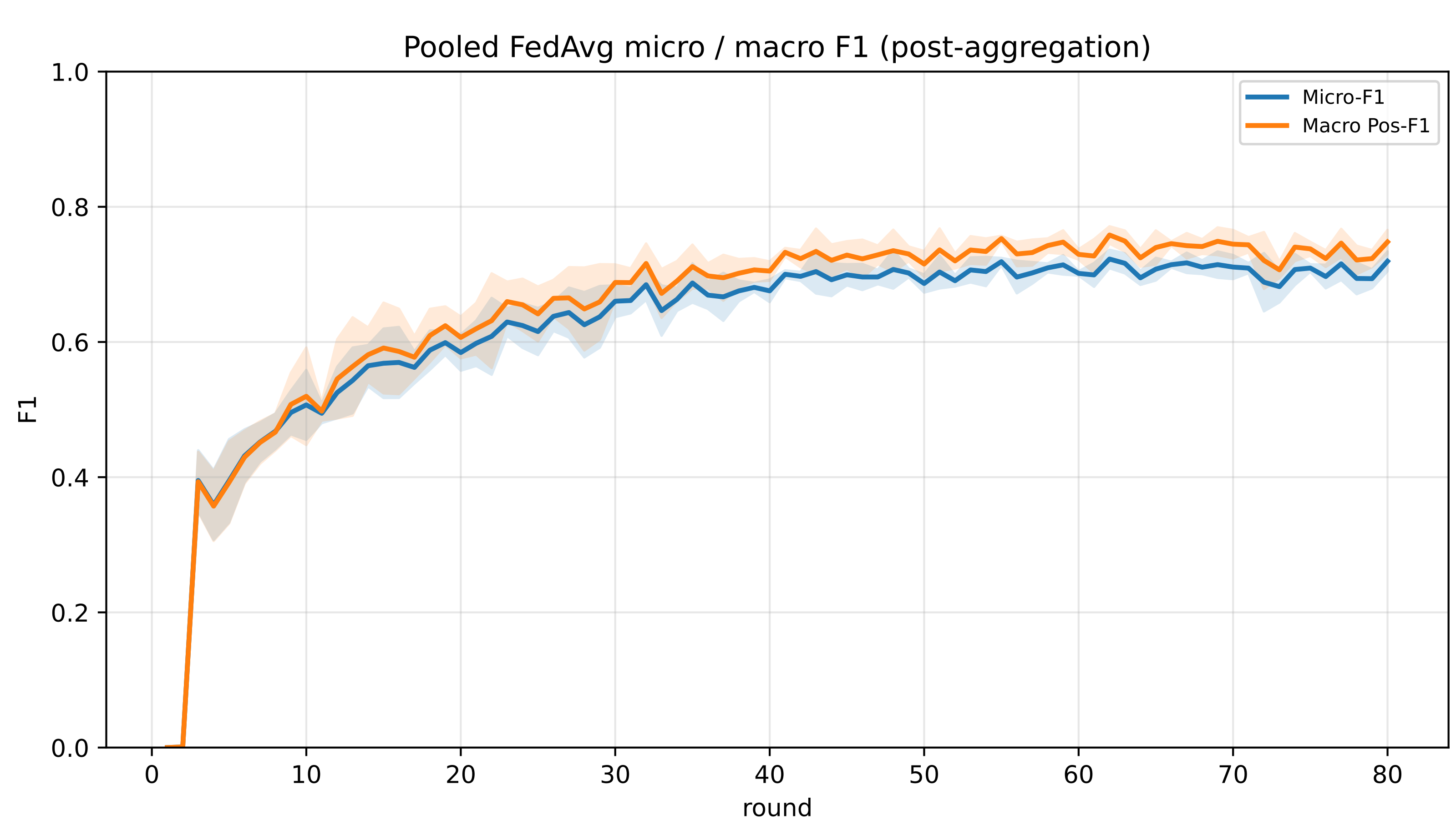
Best logged test performance (Fully Local vs Local-Centralized vs FedAvg)

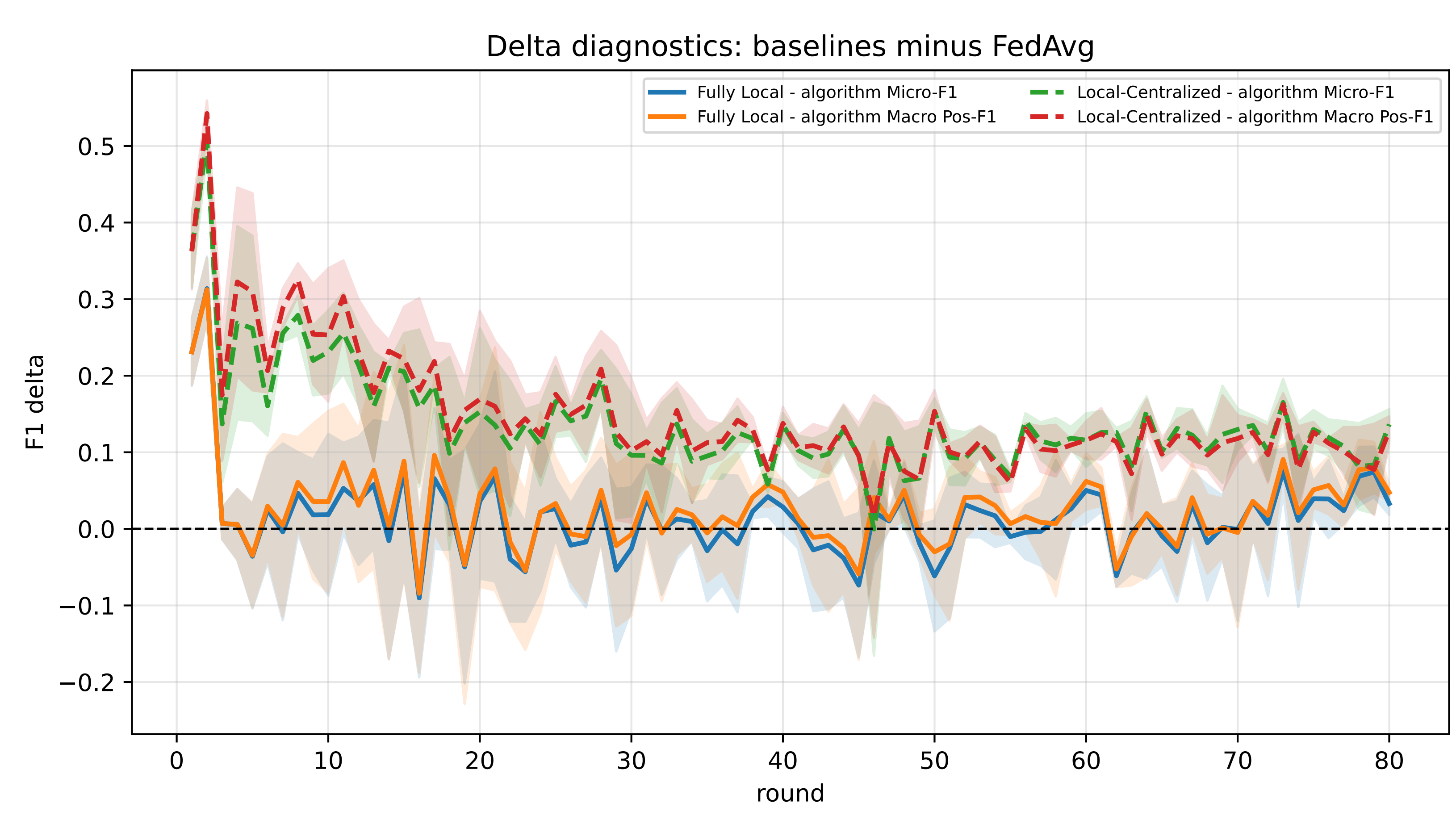
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	70.5% ± 1.9%	74.2% ± 2.2%	87.8% ± 6.8%	86.4% ± 2.4%	74.0% ± 2.0%	69.7% ± 1.8%	53.3% ± 1.7%
Local-Centralized	**85.0% ± 0.7%**	**87.0% ± 0.6%**	**98.0% ± 1.1%**	**94.5% ± 1.4%**	**89.7% ± 1.8%**	**80.3% ± 0.5%**	**72.5% ± 1.3%**
FedAvg	70.3% ± 1.1%	73.4% ± 0.8%	89.6% ± 2.1%	83.4% ± 2.9%	72.9% ± 1.2%	64.1% ± 0.9%	57.2% ± 1.6%









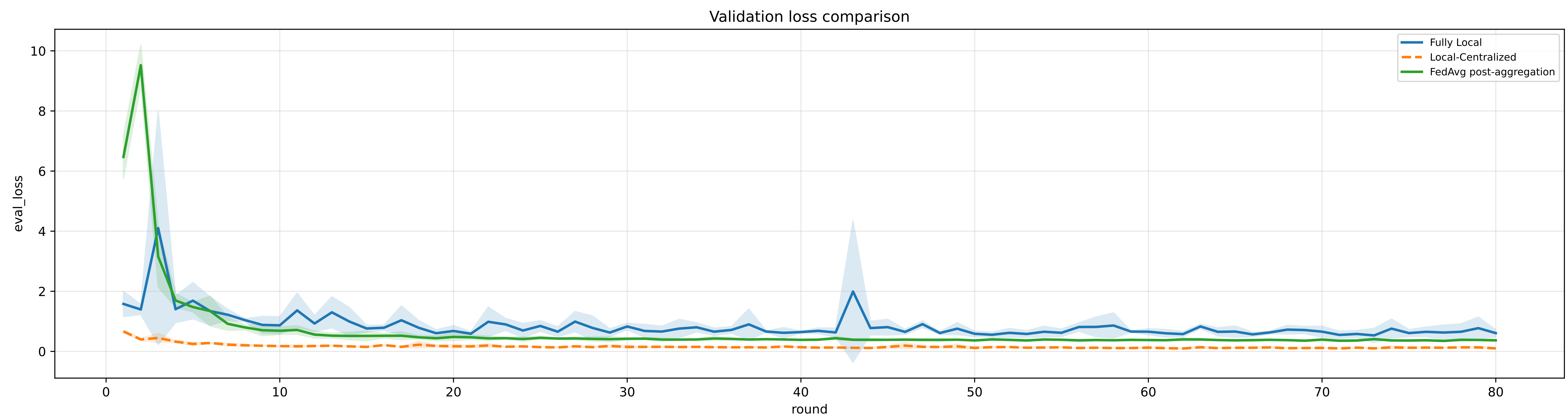


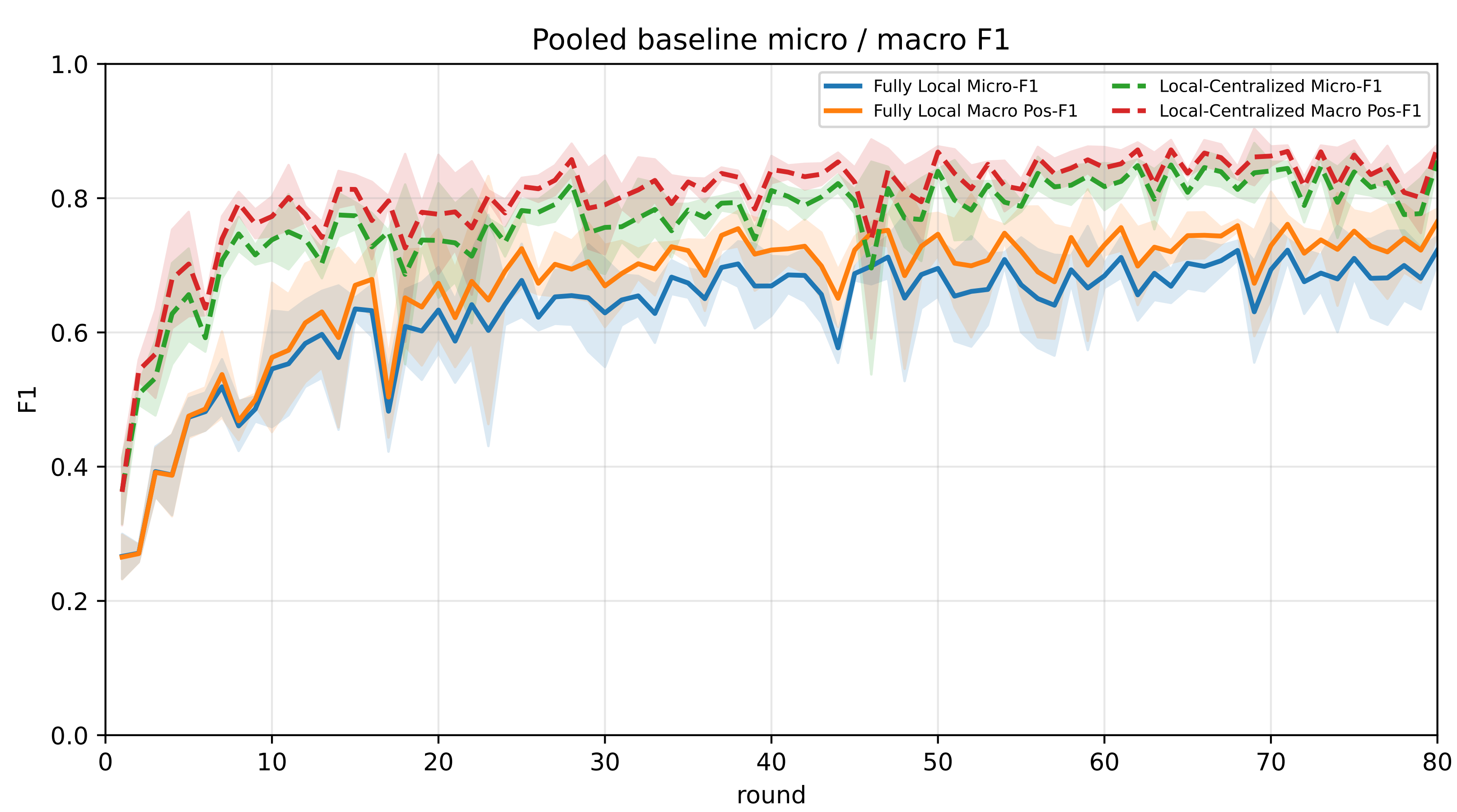
Controlled q-label heterogeneity: iid | FedAvg diagnostics | family=q_task_cycle6 |
n_clients=1 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64_headmulti

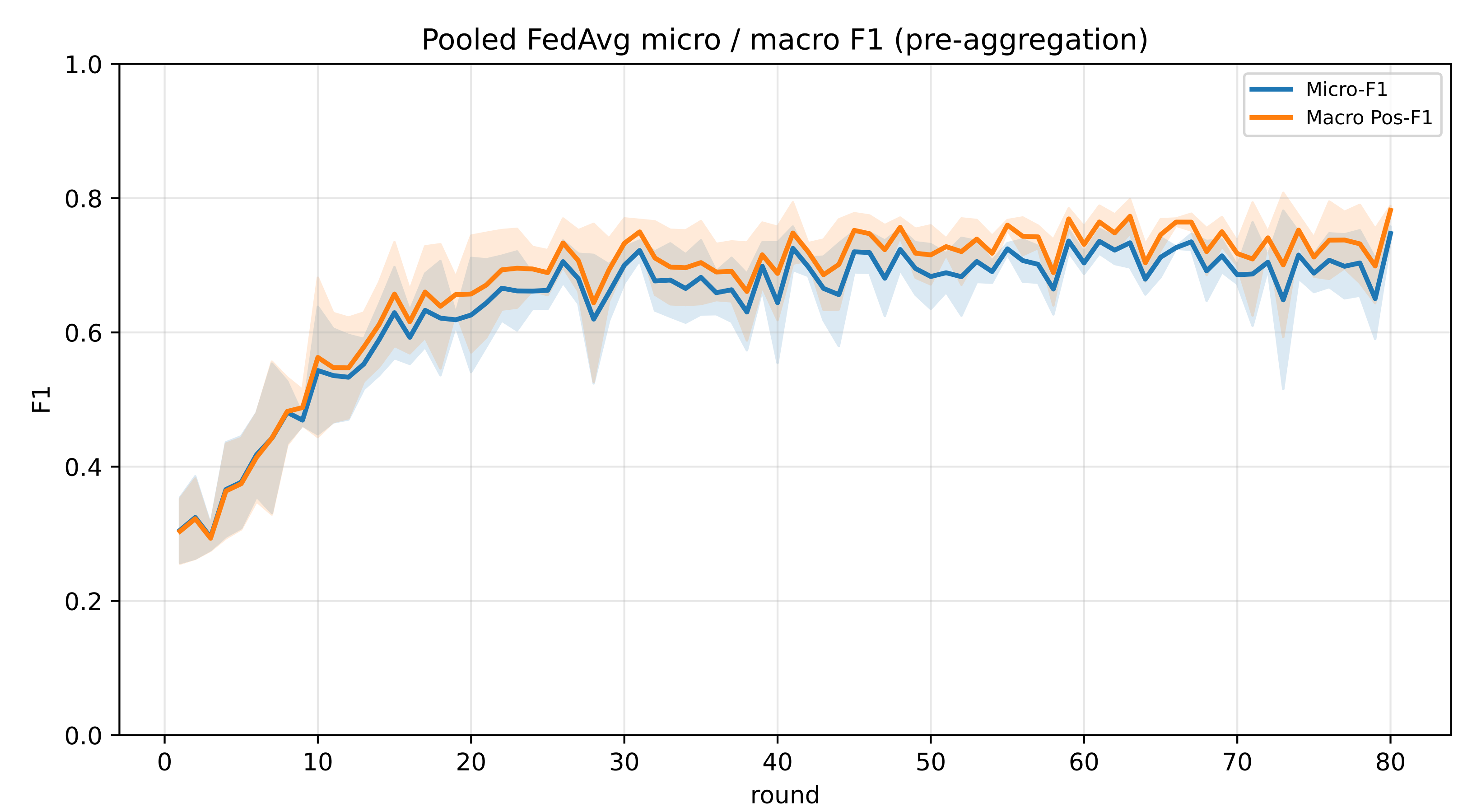
family=q_task_cycle6 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% |
cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
heterogeneity=iid | gamma=0.20 | task_jsd_mean=0.000 | task_jsd_median=0.000 | task_jsd_max=0.000 | family_centroid_jsd_mean=0.000

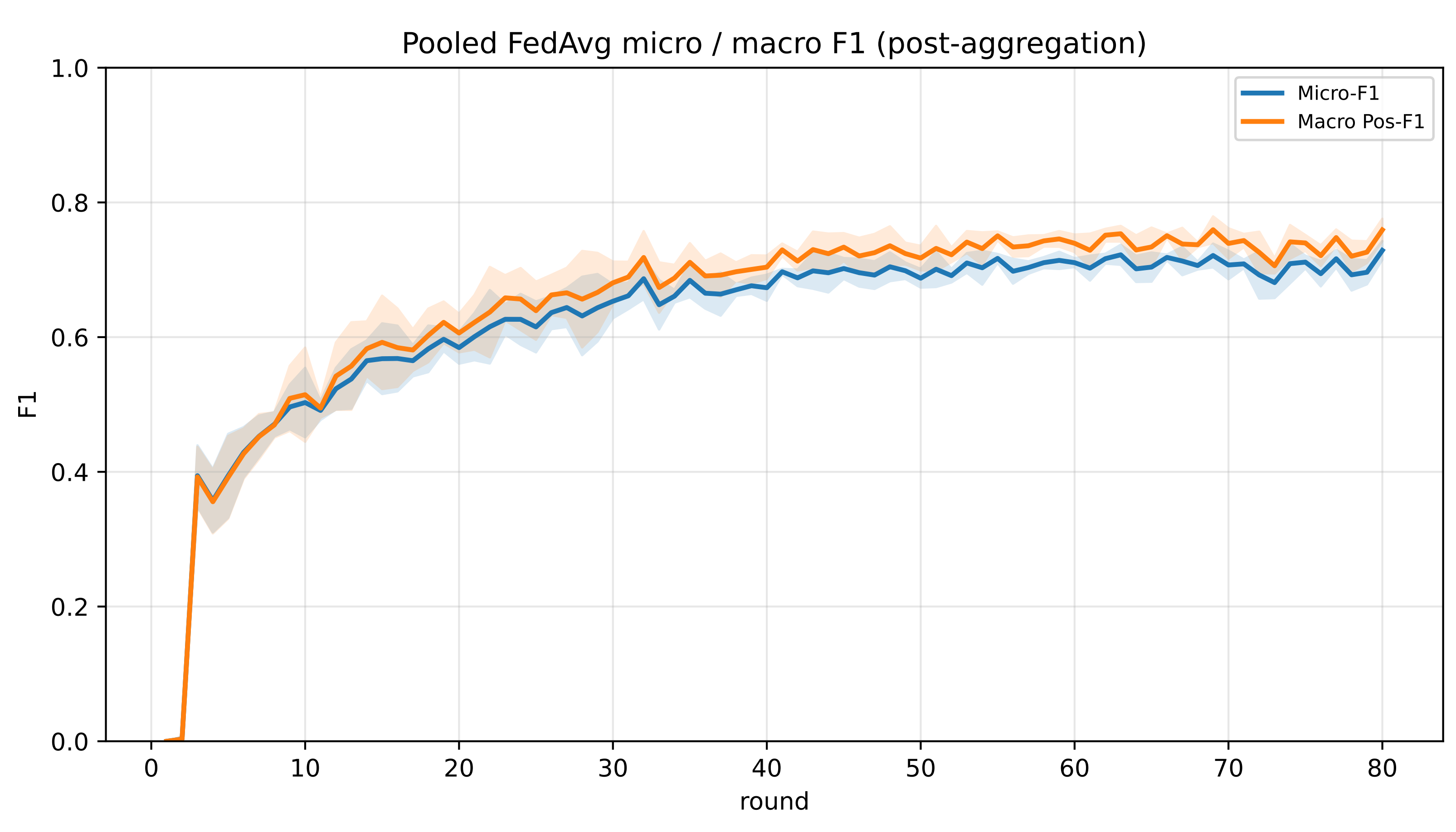
Best logged test performance (Fully Local vs Local-Centralized vs FedAvg)

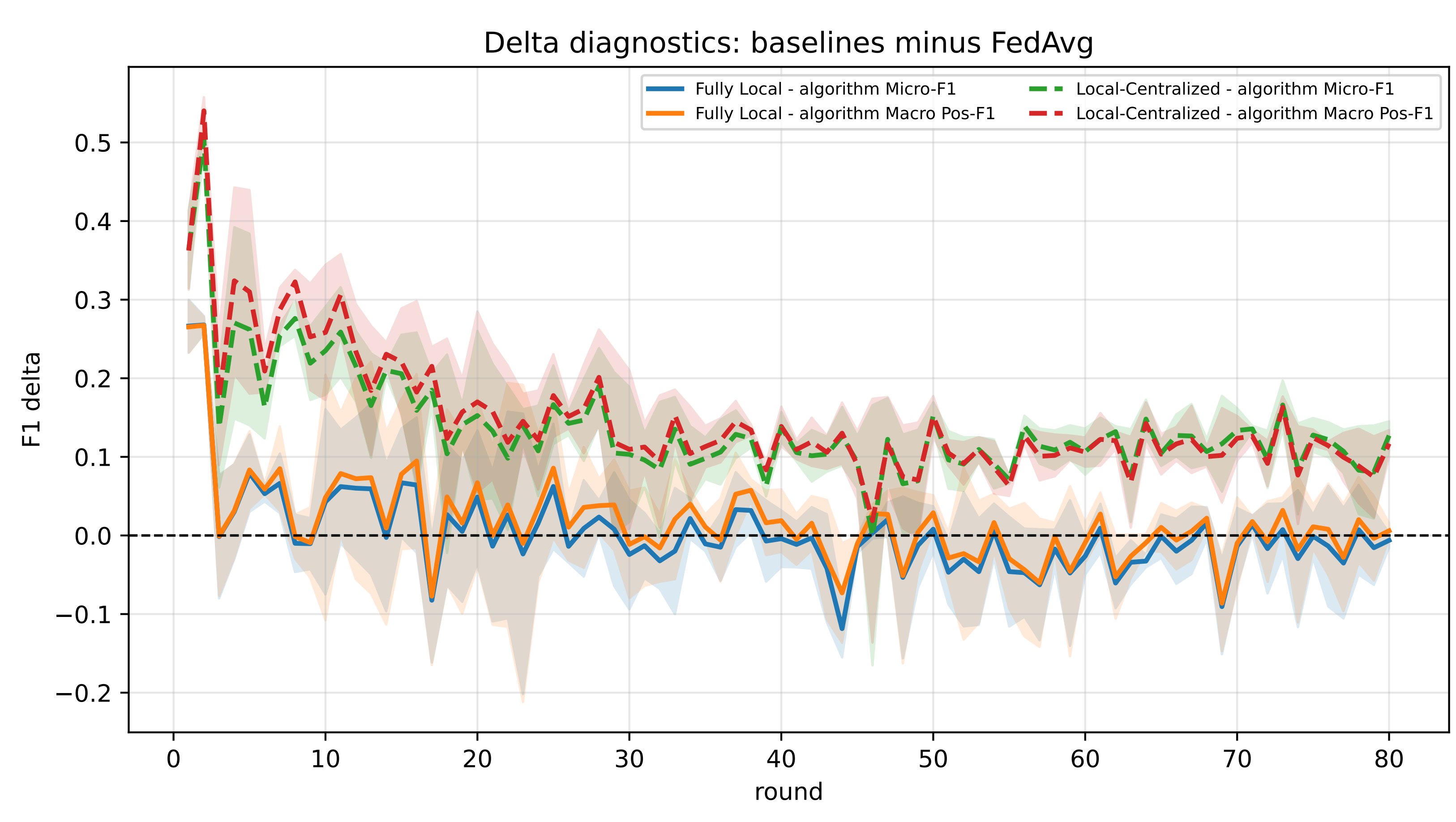
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	64.5% ± 3.5%	69.6% ± 3.3%	87.3% ± 2.7%	84.4% ± 3.4%	62.5% ± 7.2%	67.0% ± 5.0%	46.8% ± 1.7%
Local-Centralized	**85.0% ± 0.7%**	**87.0% ± 0.6%**	**98.0% ± 1.1%**	**94.5% ± 1.4%**	**89.7% ± 1.8%**	**80.3% ± 0.5%**	**72.5% ± 1.3%**
FedAvg	70.1% ± 1.0%	73.5% ± 1.1%	91.7% ± 2.3%	83.0% ± 3.3%	72.2% ± 0.9%	64.2% ± 0.5%	56.6% ± 1.6%









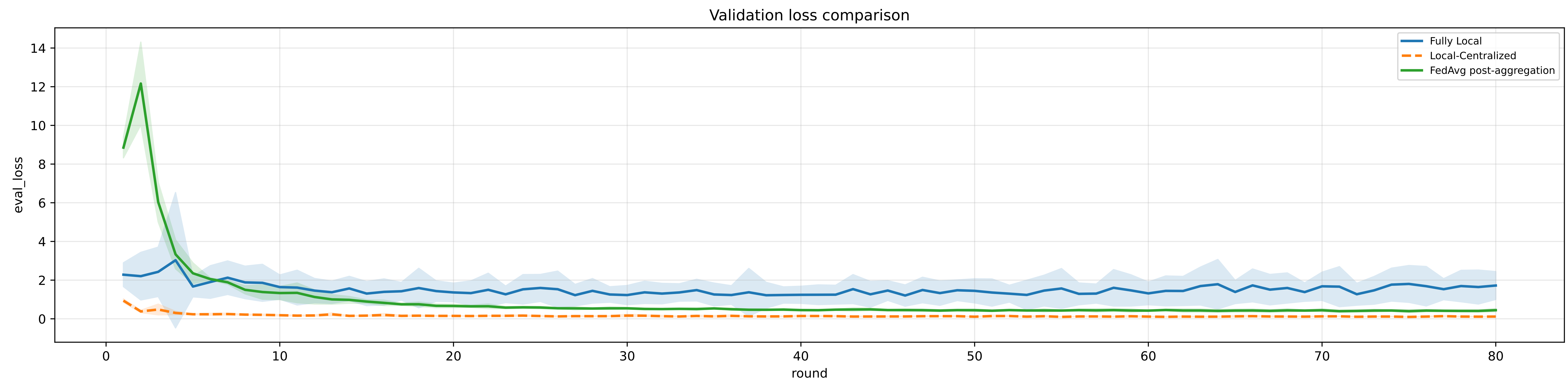


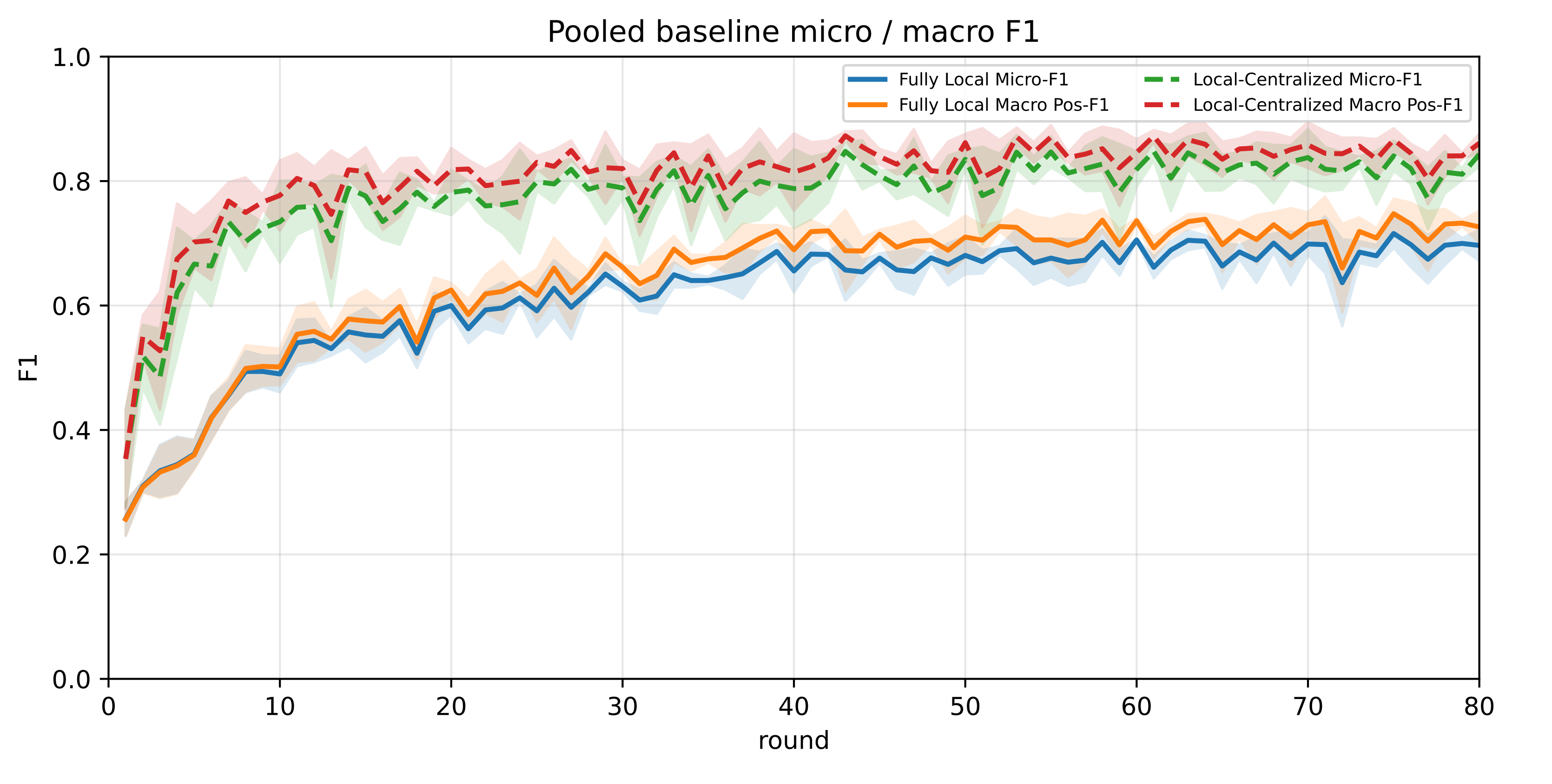
Controlled q-label heterogeneity: Mid | FedAvg diagnostics | aggregated / pooled over 5 clients | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64_headmulti

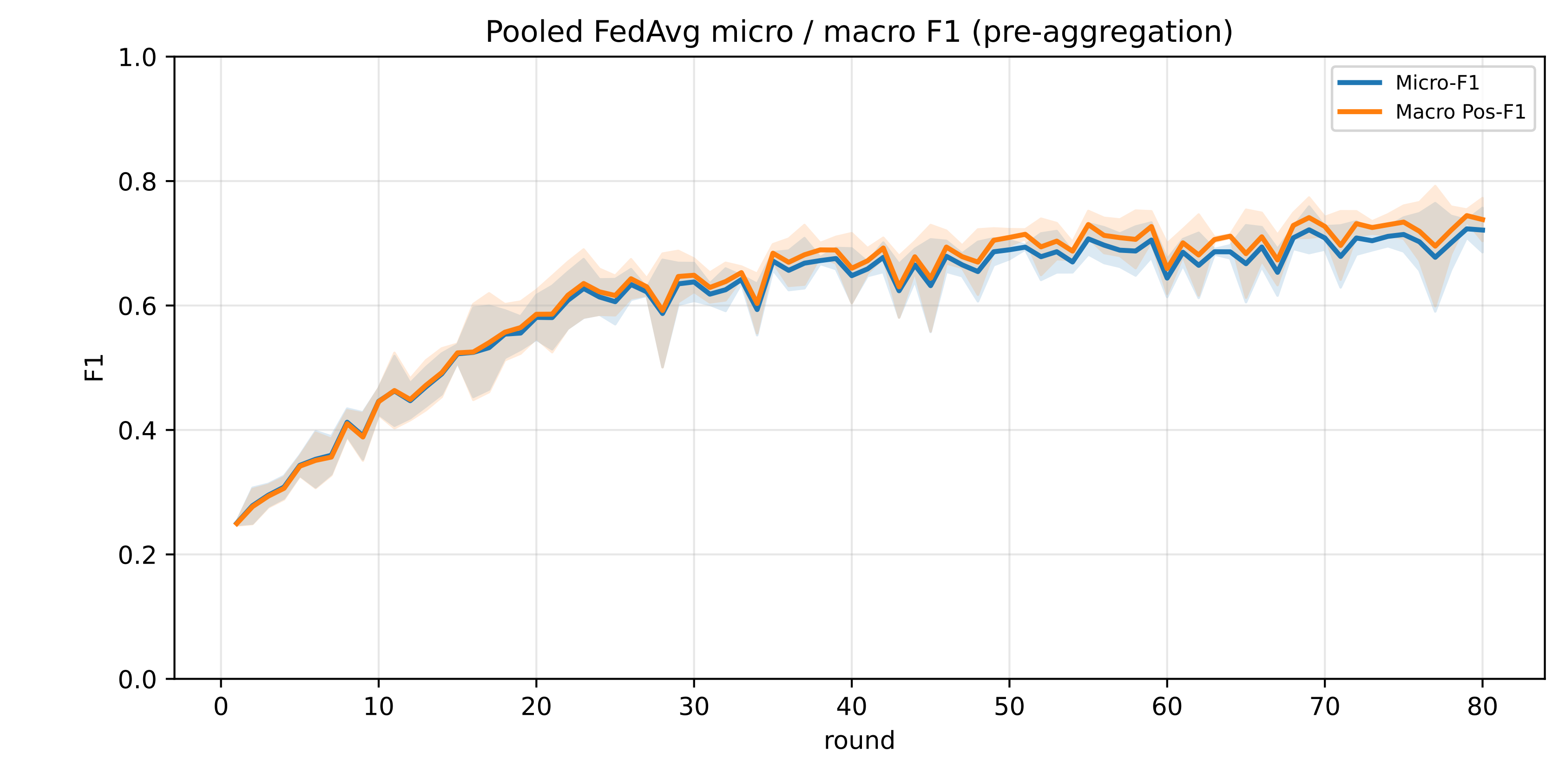
family=q_task_cycle2 | n_clients=1 | cycle2=visible 2.9%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle3 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 3.1%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle4 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 3.3%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle5 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 3.2%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
family=q_task_cycle6 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 3.1%±0.0% / true 6.3%
heterogeneity=mid | gamma=0.50 | task_jsd_mean=0.120 | task_jsd_median=0.121 | task_jsd_max=0.123 | family_centroid_jsd_mean=0.120

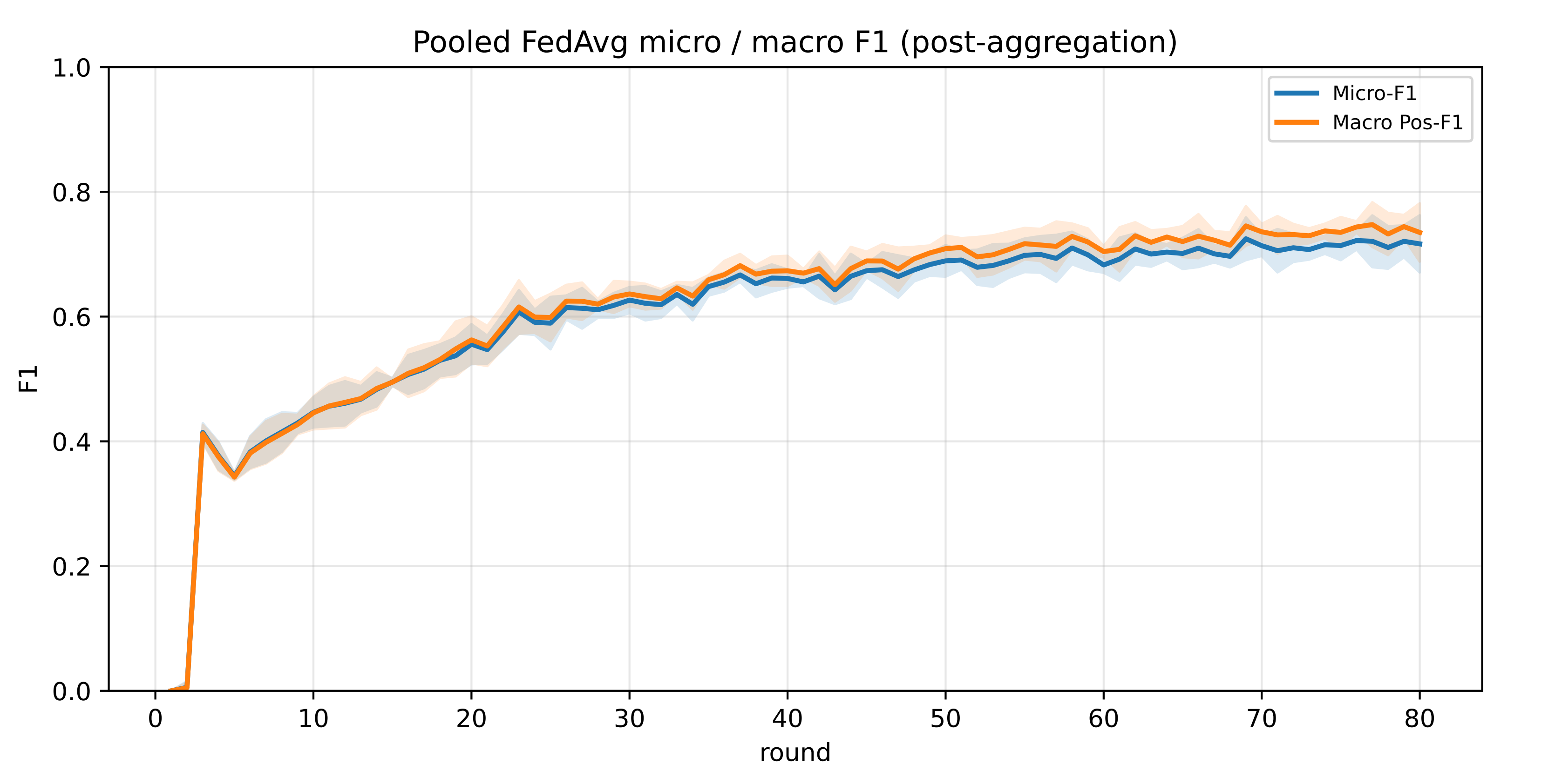
Best logged test performance (Fully Local vs Local-Centralized vs FedAvg)

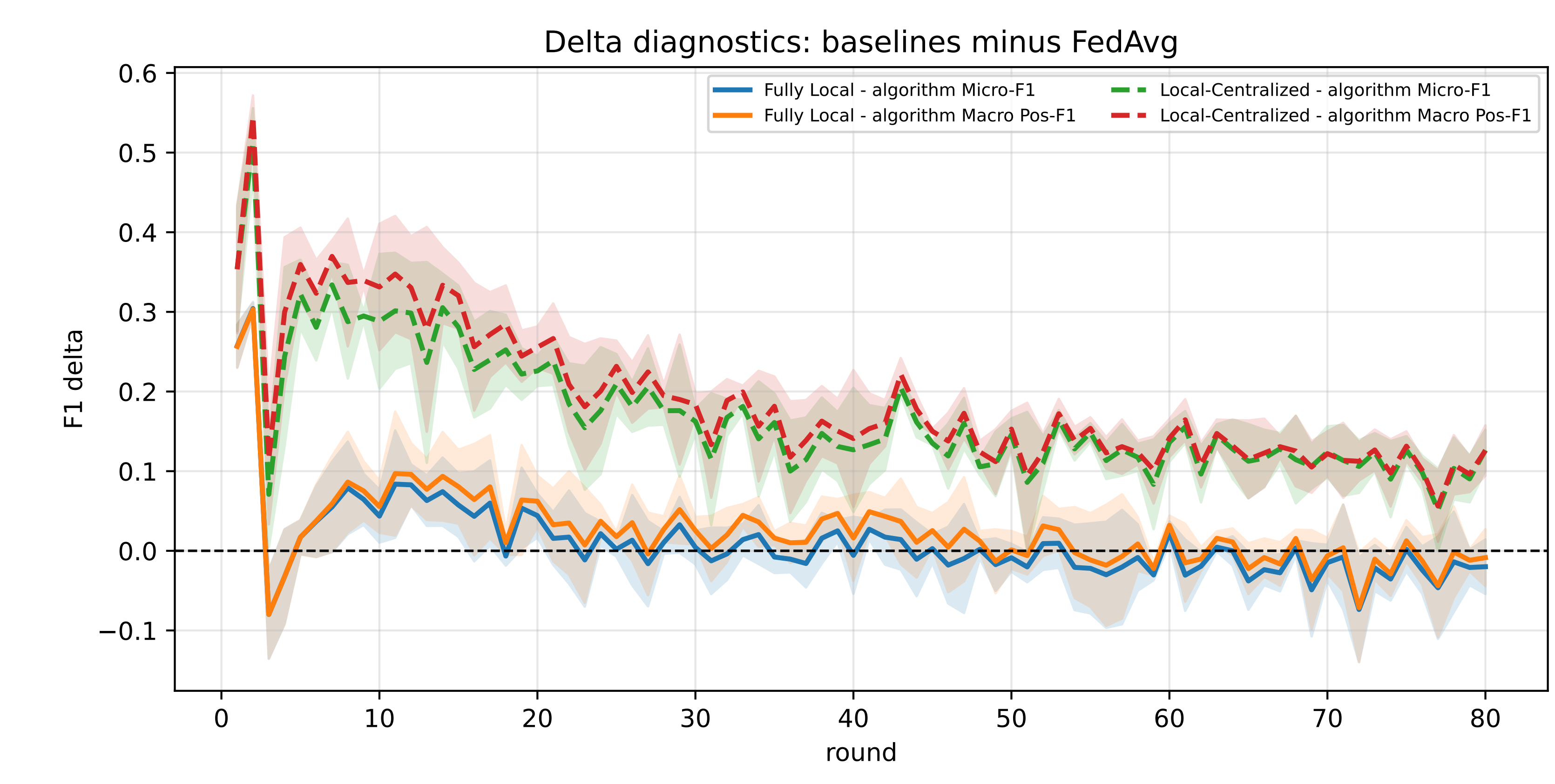
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	61.6% ± 4.0%	66.0% ± 4.3%	81.6% ± 7.0%	78.2% ± 5.1%	64.0% ± 3.6%	58.6% ± 5.2%	44.4% ± 3.0%
Local-Centralized	**84.3% ± 1.6%**	**86.3% ± 1.4%**	**96.7% ± 1.1%**	**94.7% ± 0.6%**	**87.4% ± 4.2%**	**80.9% ± 2.2%**	**71.8% ± 3.9%**
FedAvg	70.3% ± 1.4%	72.8% ± 1.1%	86.2% ± 4.2%	81.8% ± 3.9%	67.9% ± 1.8%	70.1% ± 4.1%	58.1% ± 1.9%









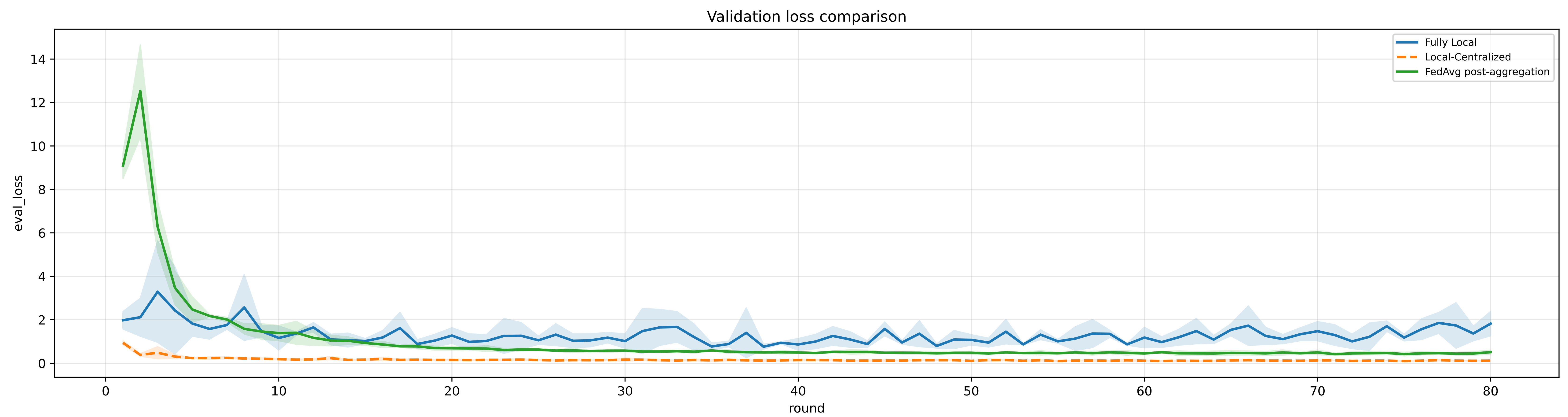


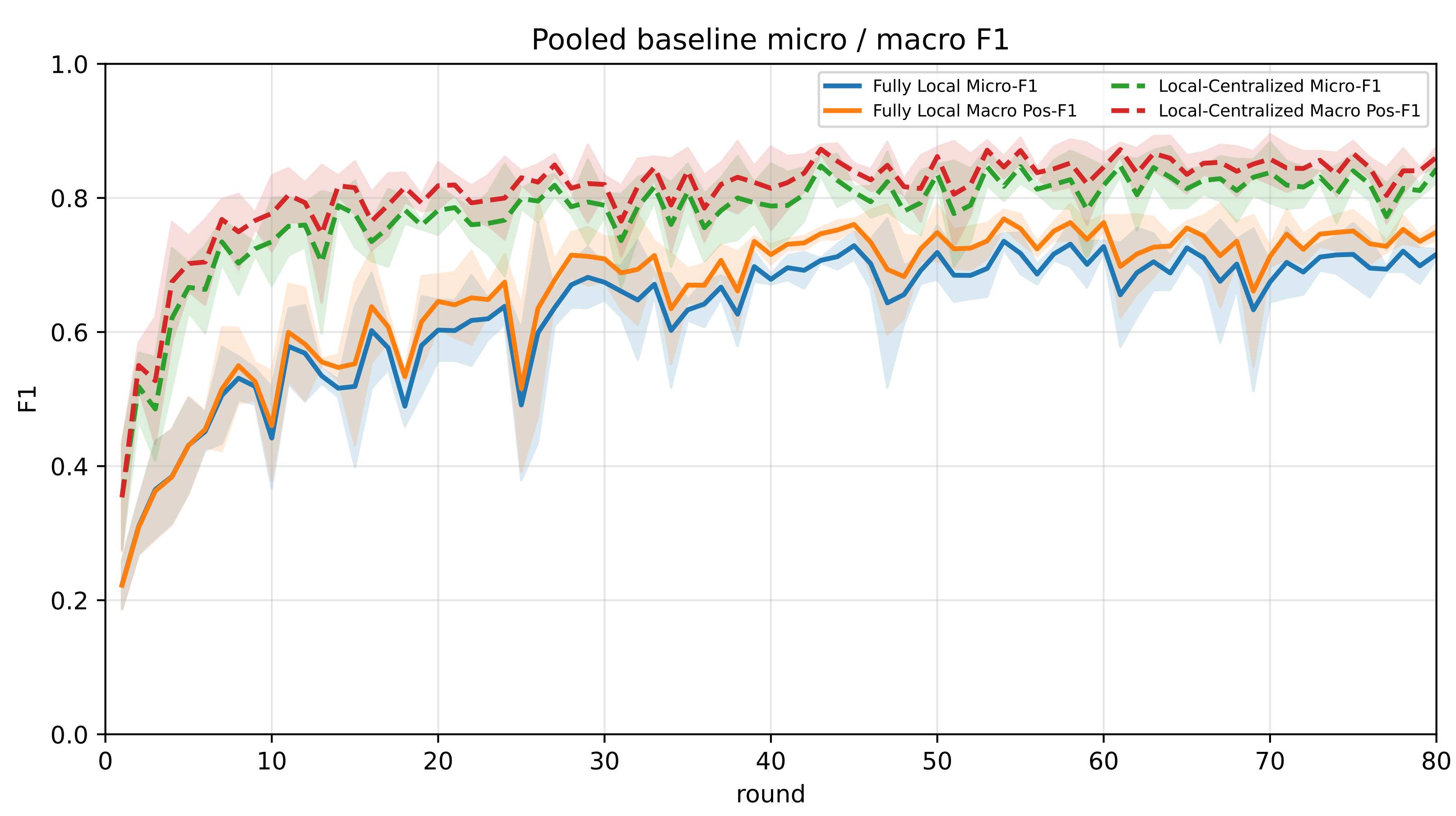
Controlled q-label heterogeneity: Mid | FedAvg diagnostics | family=q_task_cycle2 | n_clients=1 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64_headmulti

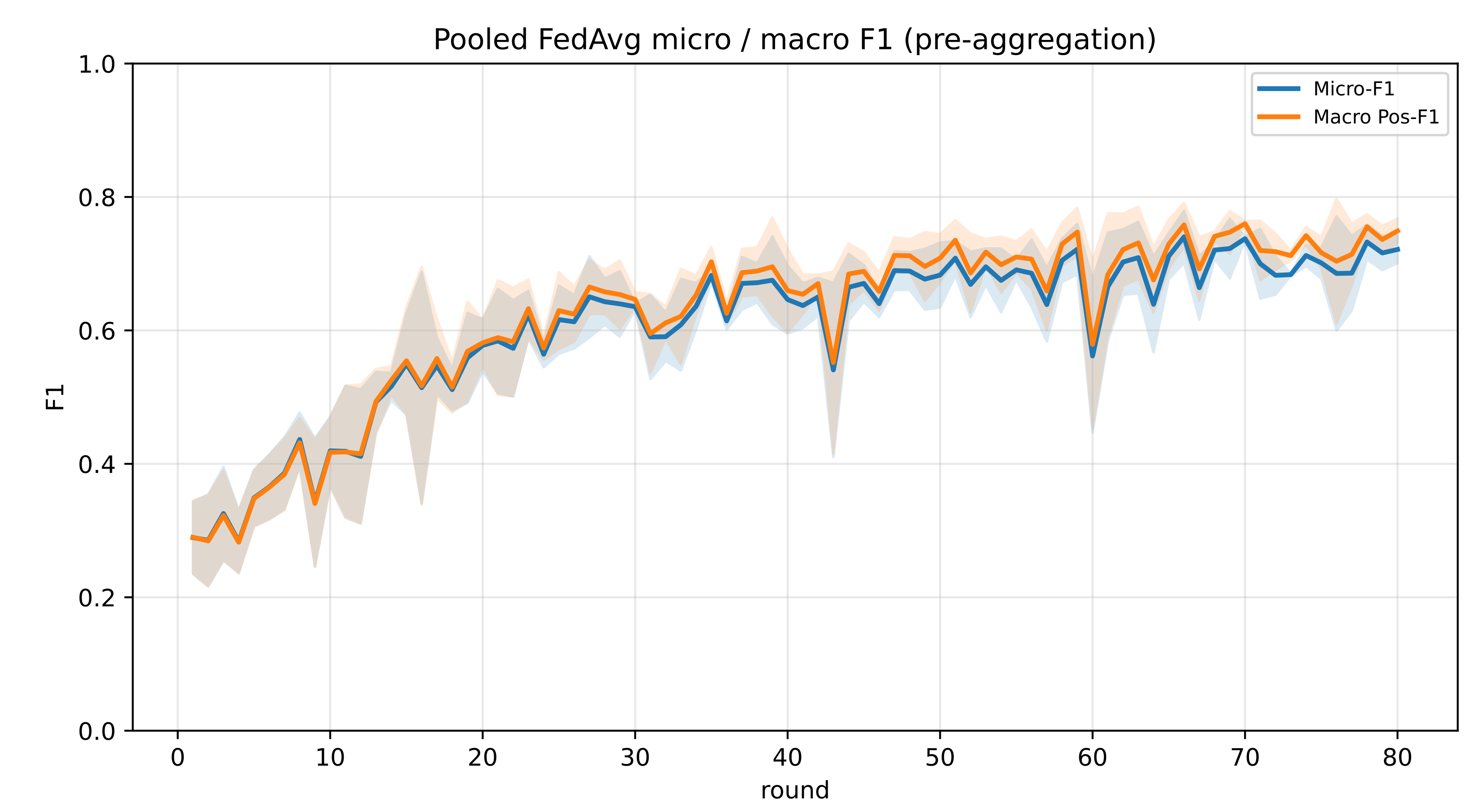
family=q_task_cycle2 | n_clients=1 | cycle2=visible 2.9%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
heterogeneity=mid | gamma=0.50 | task_jsd_mean=0.120 | task_jsd_median=0.121 | task_jsd_max=0.123 | family_centroid_jsd_mean=0.120

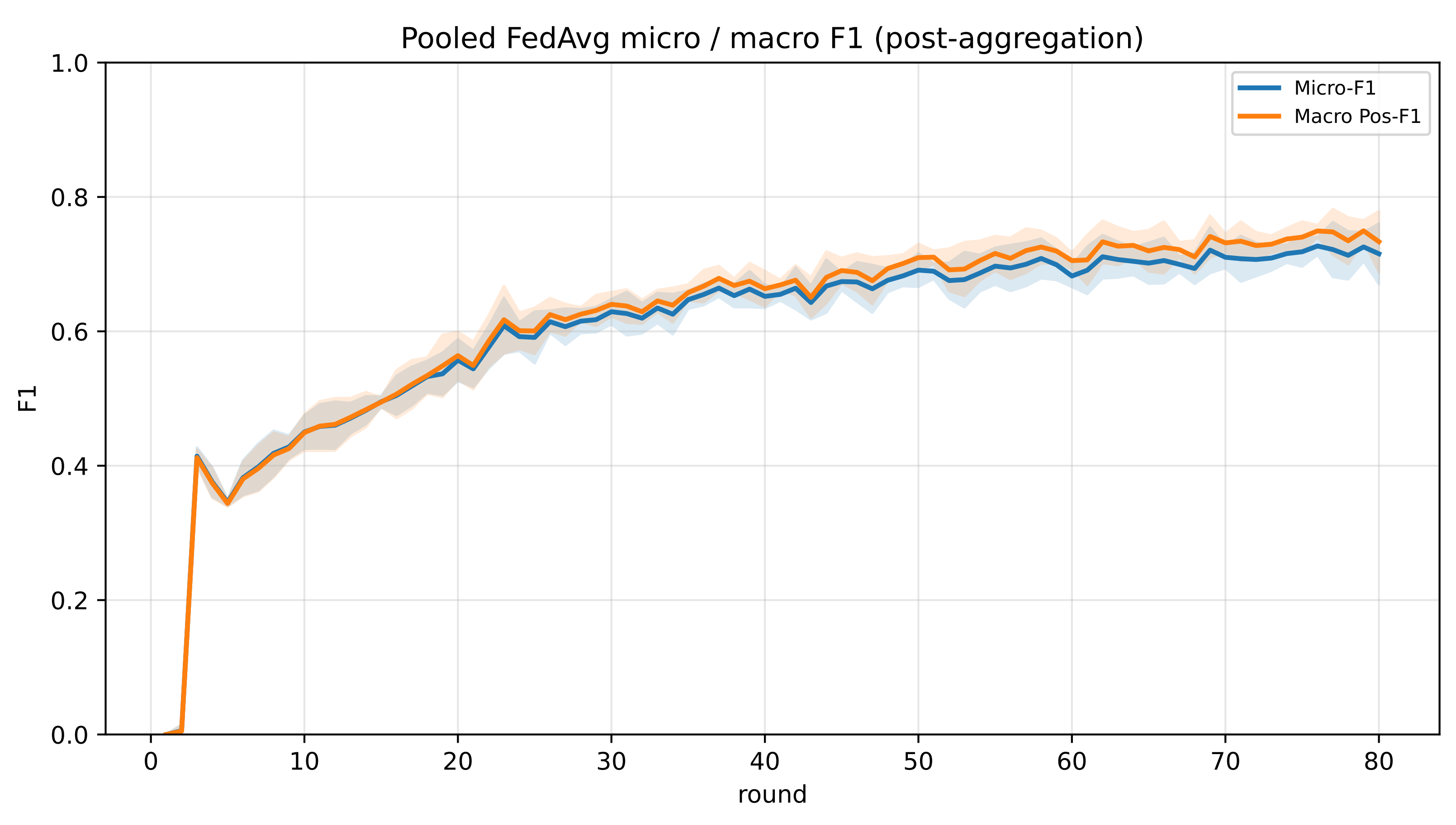
Best logged test performance (Fully Local vs Local-Centralized vs FedAvg)

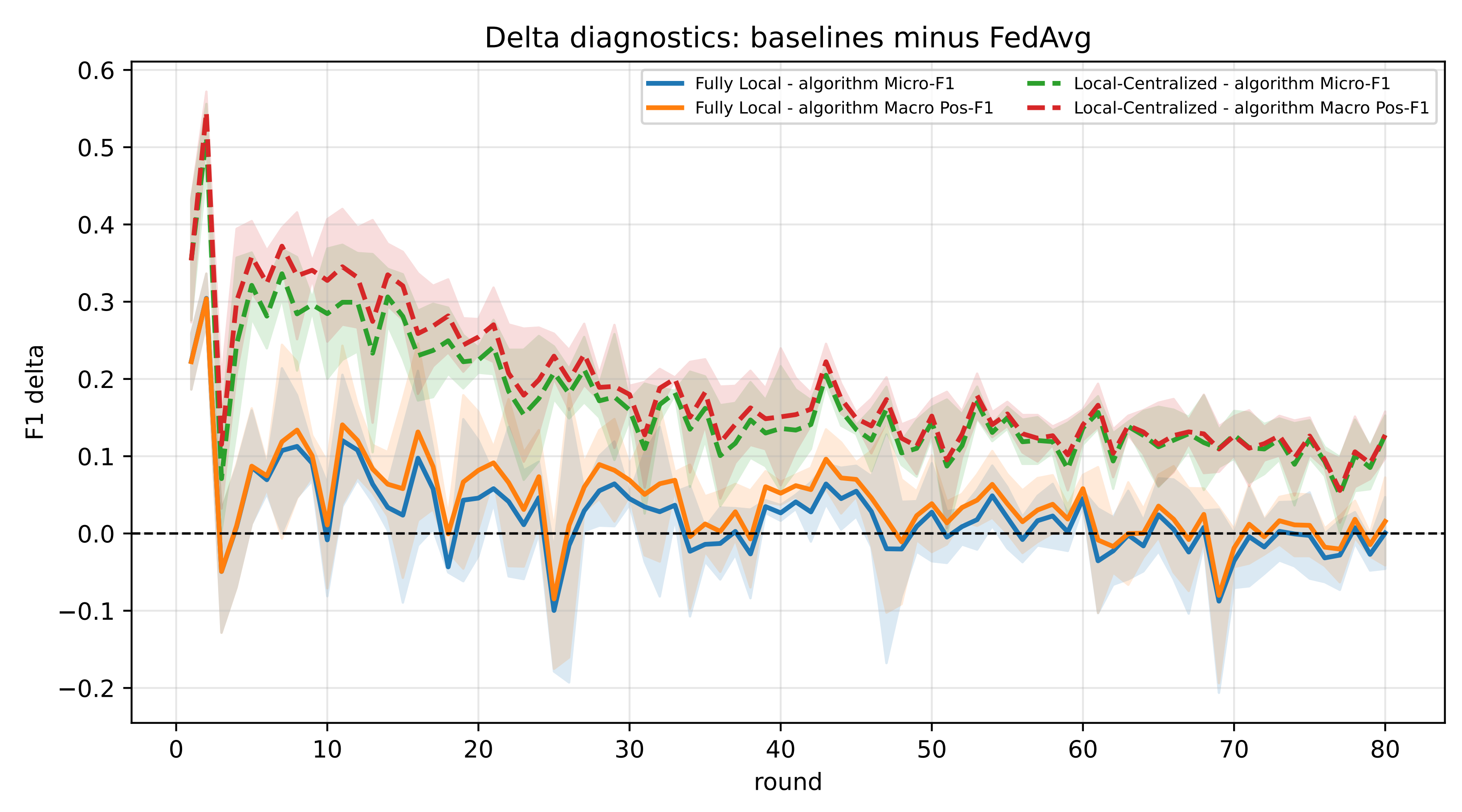
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	63.8% ± 5.4%	68.0% ± 5.8%	85.6% ± 8.0%	80.6% ± 4.3%	65.4% ± 6.2%	59.3% ± 9.8%	49.2% ± 2.7%
Local-Centralized	**84.3% ± 1.6%**	**86.3% ± 1.4%**	**96.7% ± 1.1%**	**94.7% ± 0.6%**	**87.4% ± 4.2%**	**80.9% ± 2.2%**	**71.8% ± 3.9%**
FedAvg	70.2% ± 1.5%	72.6% ± 1.3%	85.3% ± 3.3%	81.7% ± 3.7%	68.1% ± 1.3%	69.9% ± 4.0%	58.1% ± 2.0%









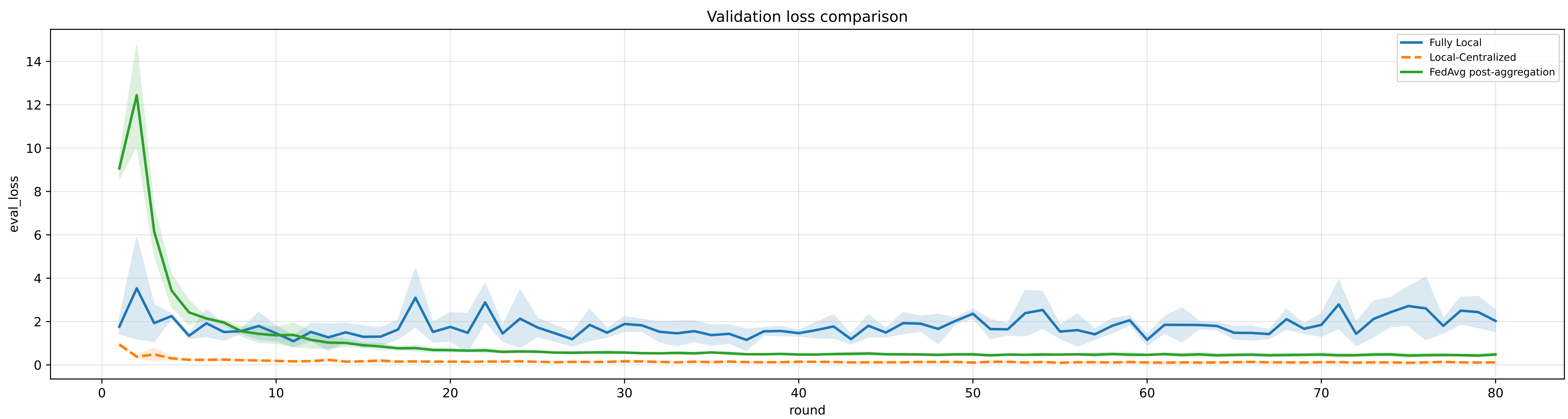


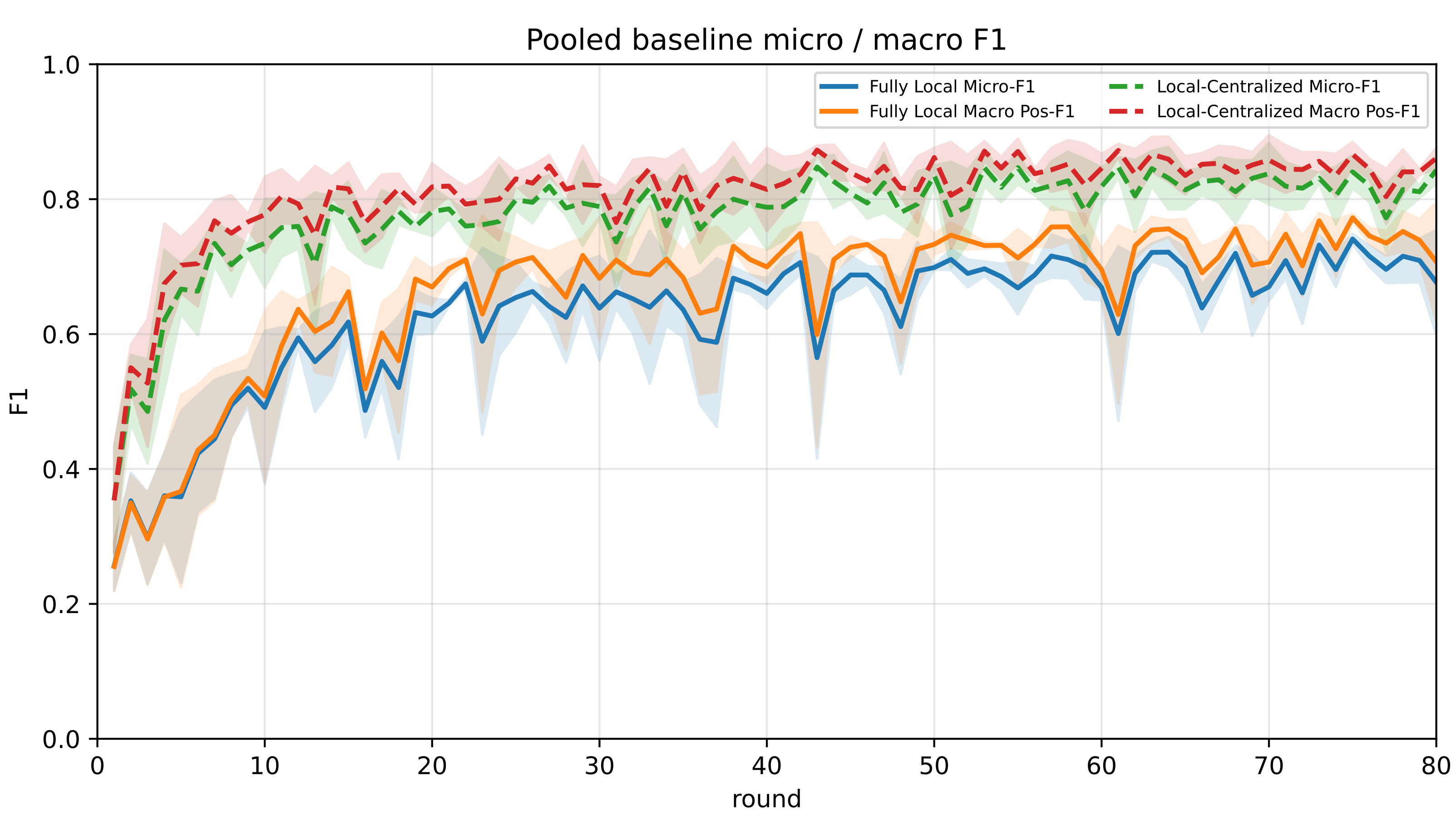
Controlled q-label heterogeneity: Mid | FedAvg diagnostics | family=q_task_cycle3 |
n_clients=1 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64_headmulti

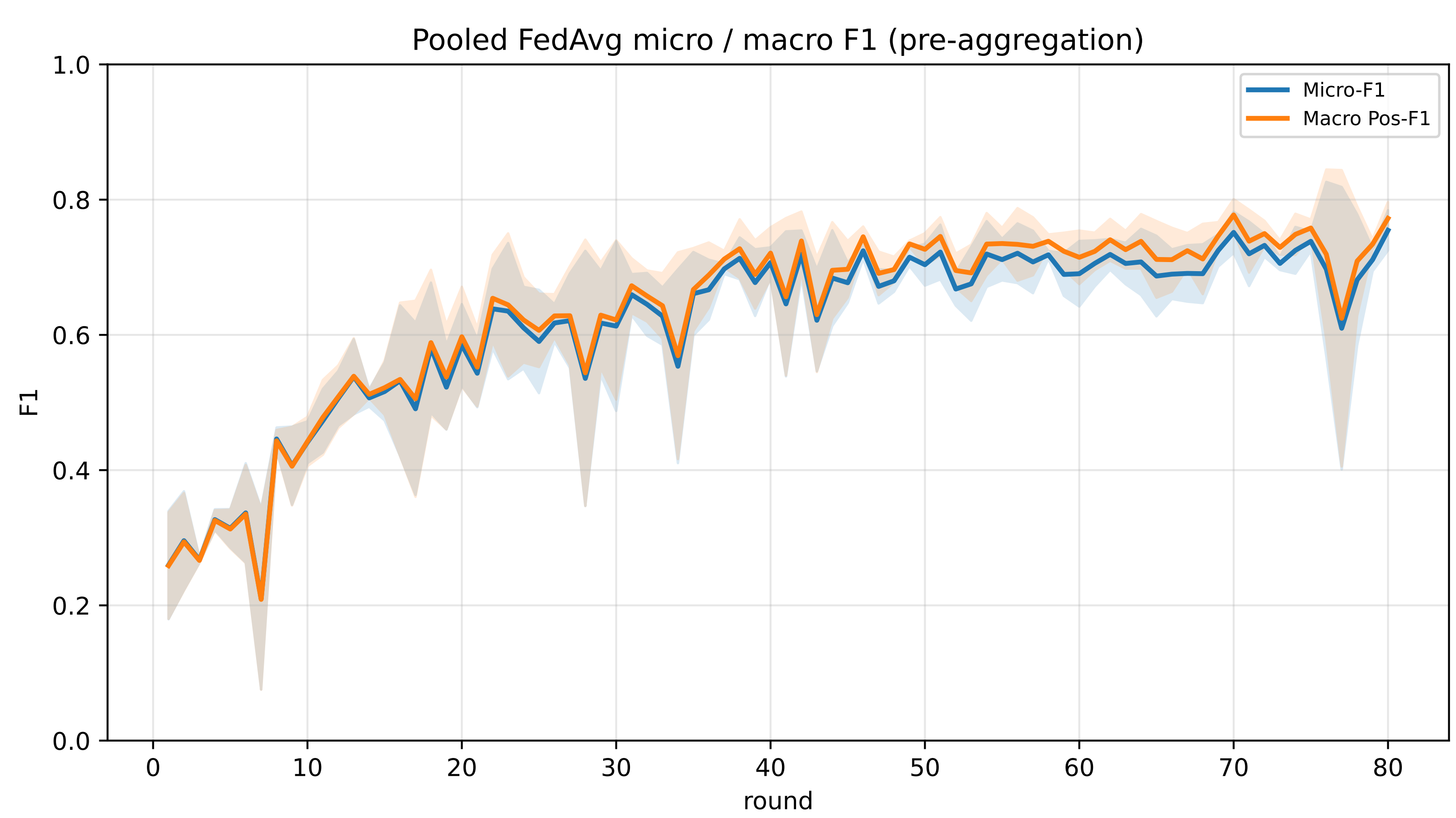
family=q_task_cycle3 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 3.1%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% |
cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
heterogeneity=mid | gamma=0.50 | task_jsd_mean=0.120 | task_jsd_median=0.121 | task_jsd_max=0.123 | family_centroid_jsd_mean=0.120

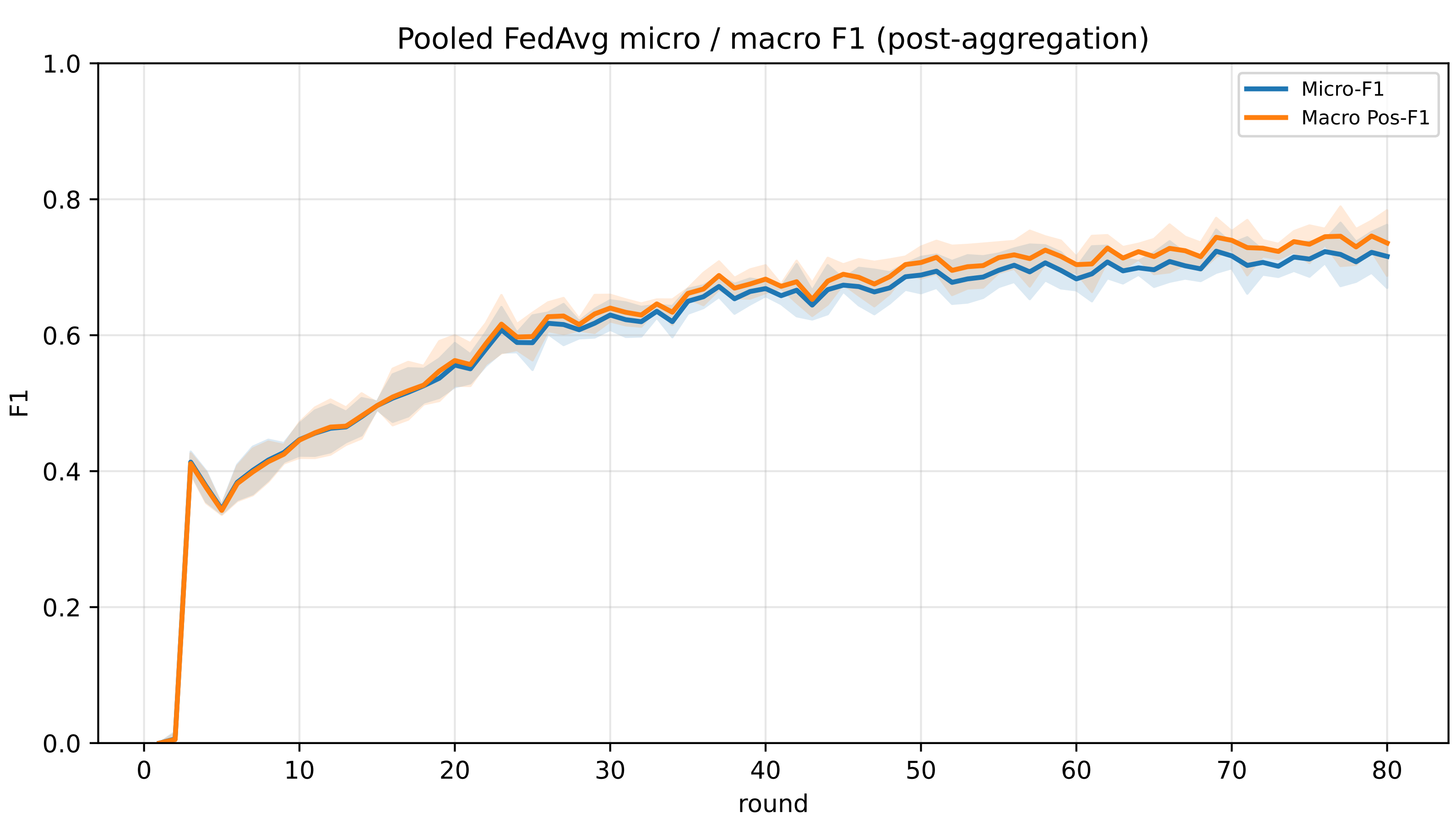
Best logged test performance (Fully Local vs Local-Centralized vs FedAvg)

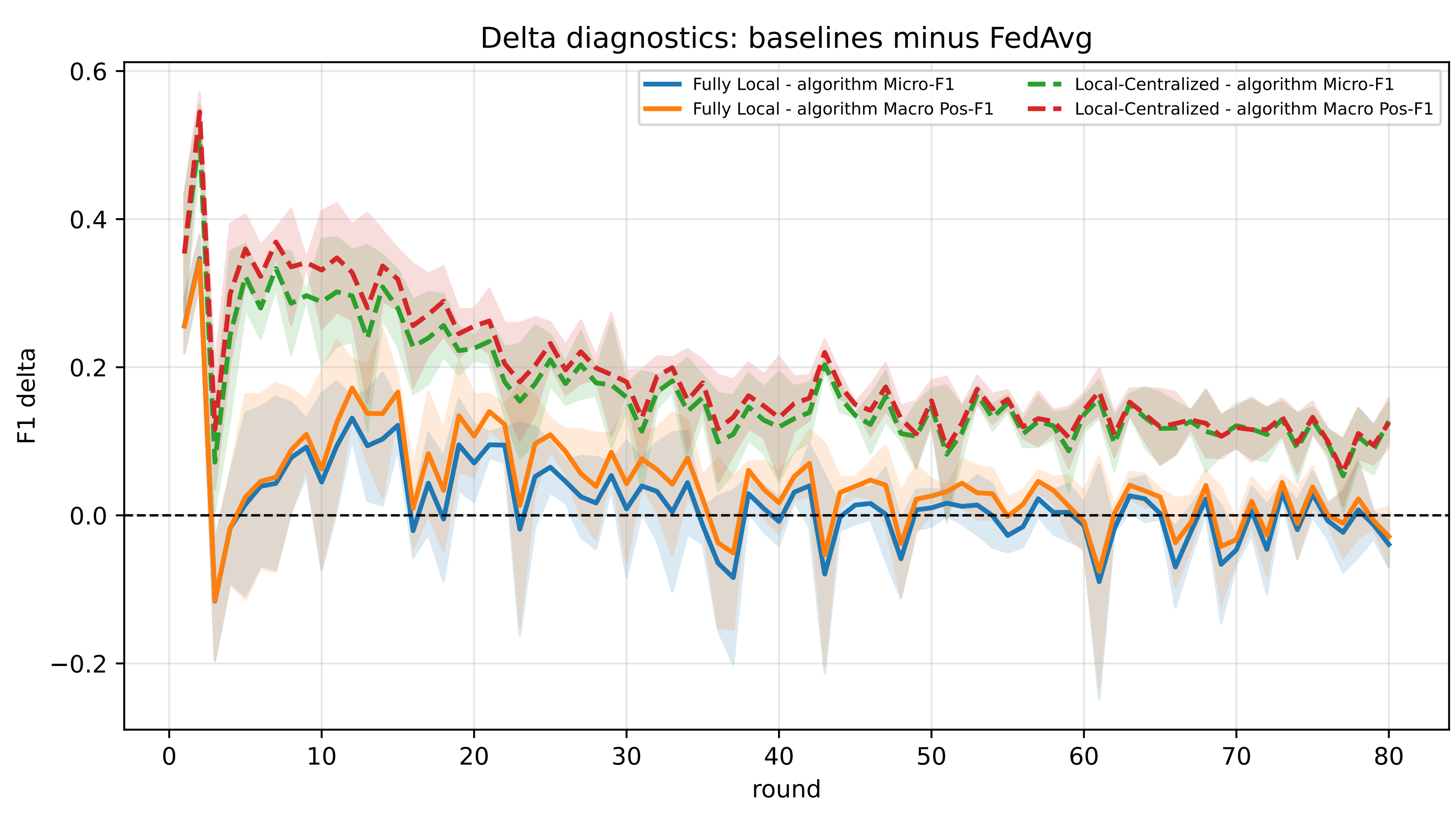
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	60.1% ± 11.6%	64.2% ± 10.9%	75.3% ± 14.5%	77.9% ± 12.0%	63.8% ± 6.1%	60.7% ± 11.5%	43.1% ± 11.1%
Local-Centralized	**84.3% ± 1.6%**	**86.3% ± 1.4%**	**96.7% ± 1.1%**	**94.7% ± 0.6%**	**87.4% ± 4.2%**	**80.9% ± 2.2%**	**71.8% ± 3.9%**
FedAvg	70.6% ± 0.8%	73.2% ± 0.5%	86.9% ± 4.4%	82.0% ± 3.1%	68.1% ± 1.9%	71.0% ± 3.3%	58.0% ± 1.5%









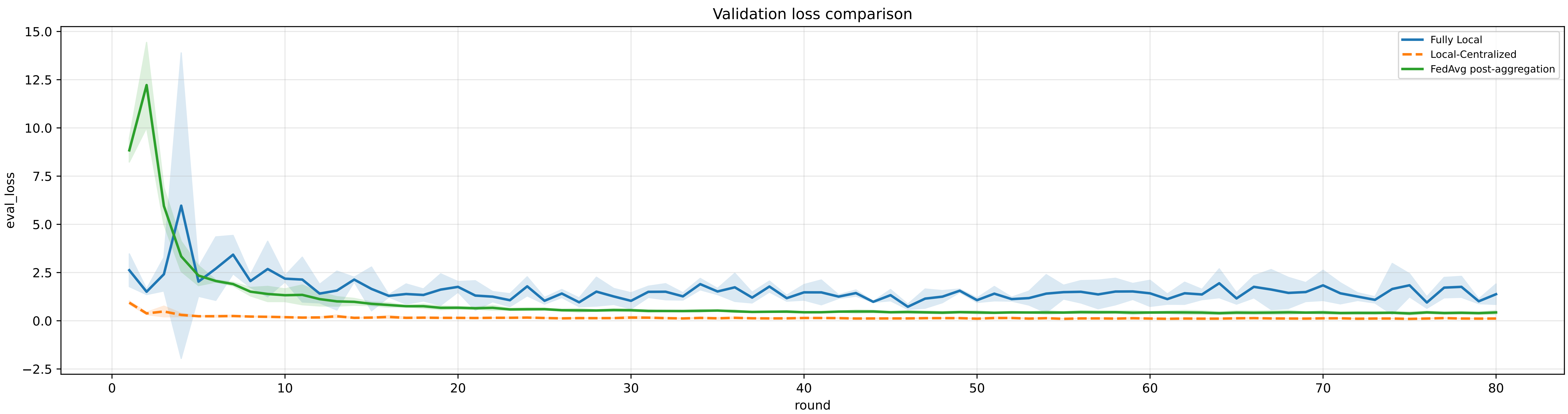


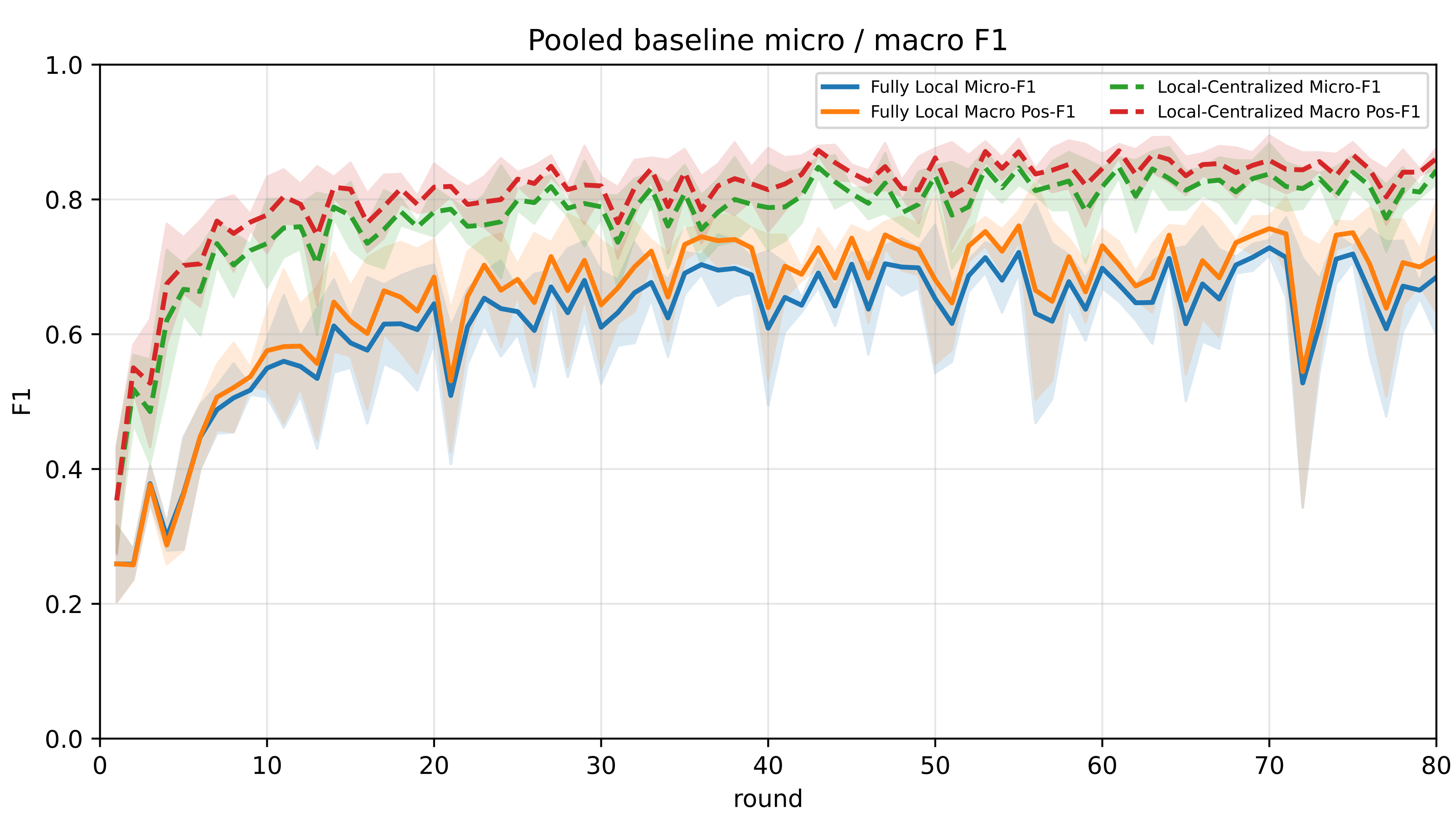
Controlled q-label heterogeneity: Mid | FedAvg diagnostics | family=q_task_cycle4 |
n_clients=1 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64_headmulti

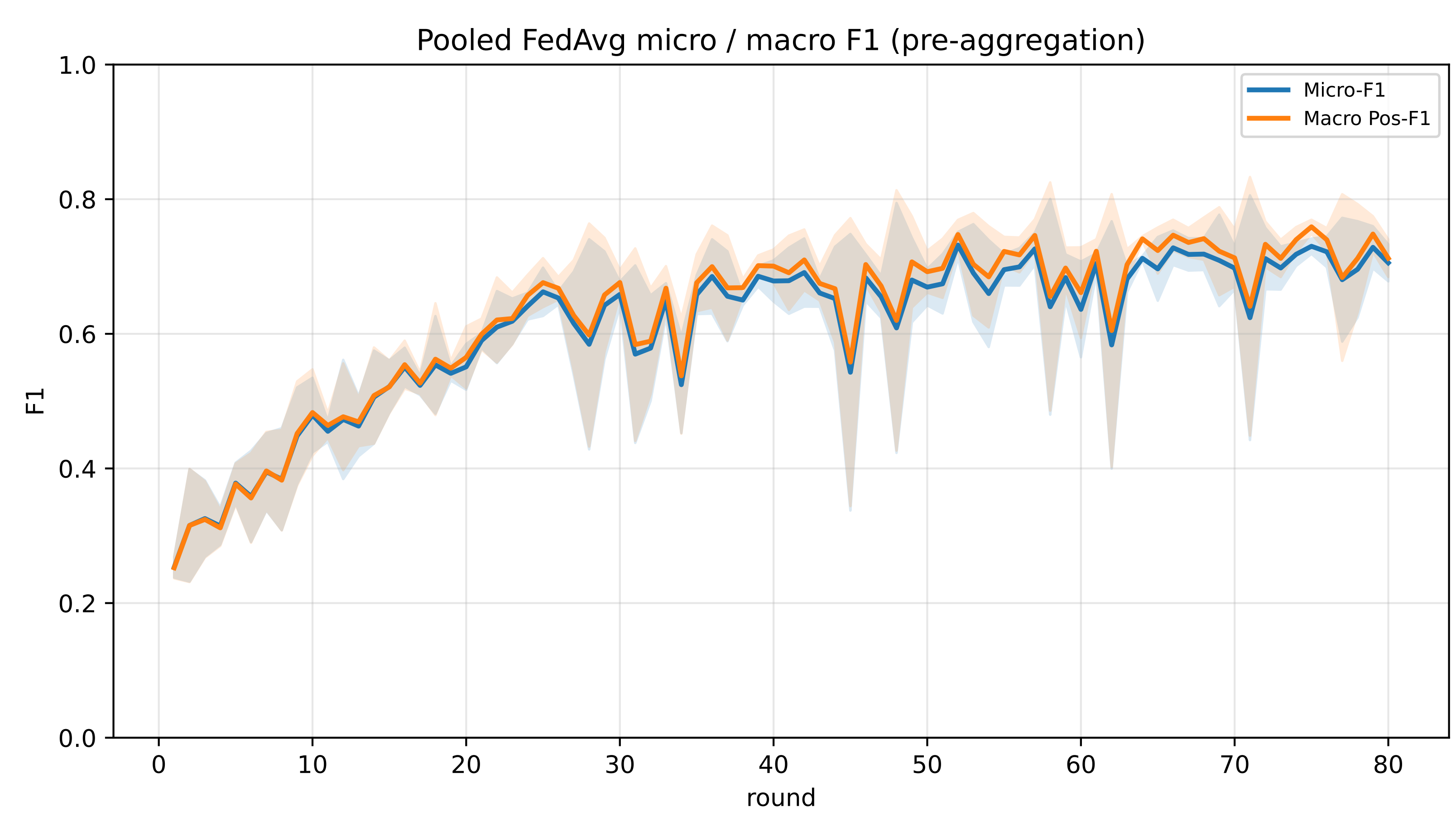
family=q_task_cycle4 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 3.3%±0.0% / true 6.6% |
cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
heterogeneity=mid | gamma=0.50 | task_jsd_mean=0.120 | task_jsd_median=0.121 | task_jsd_max=0.123 | family_centroid_jsd_mean=0.120

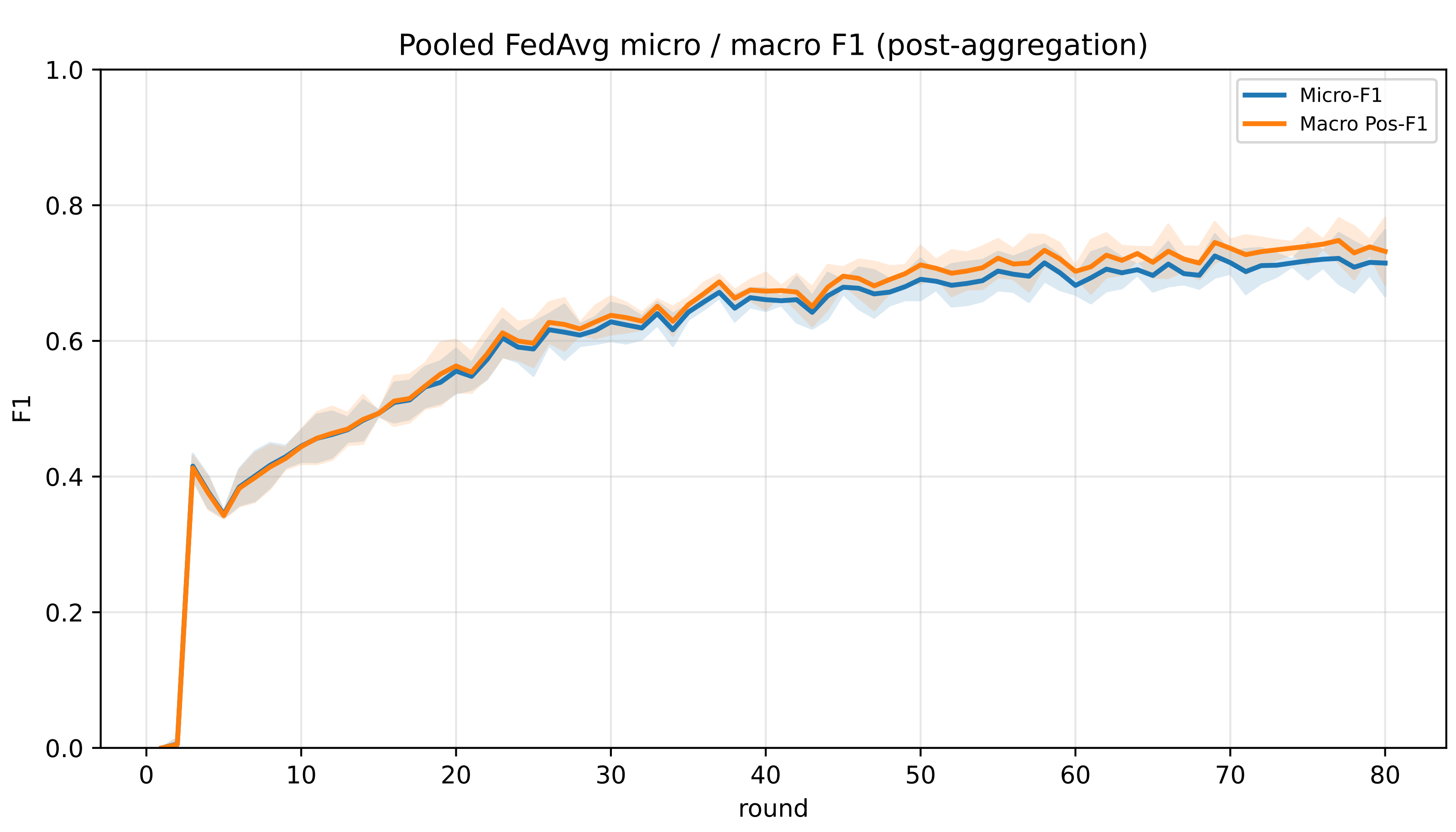
Best logged test performance (Fully Local vs Local-Centralized vs FedAvg)

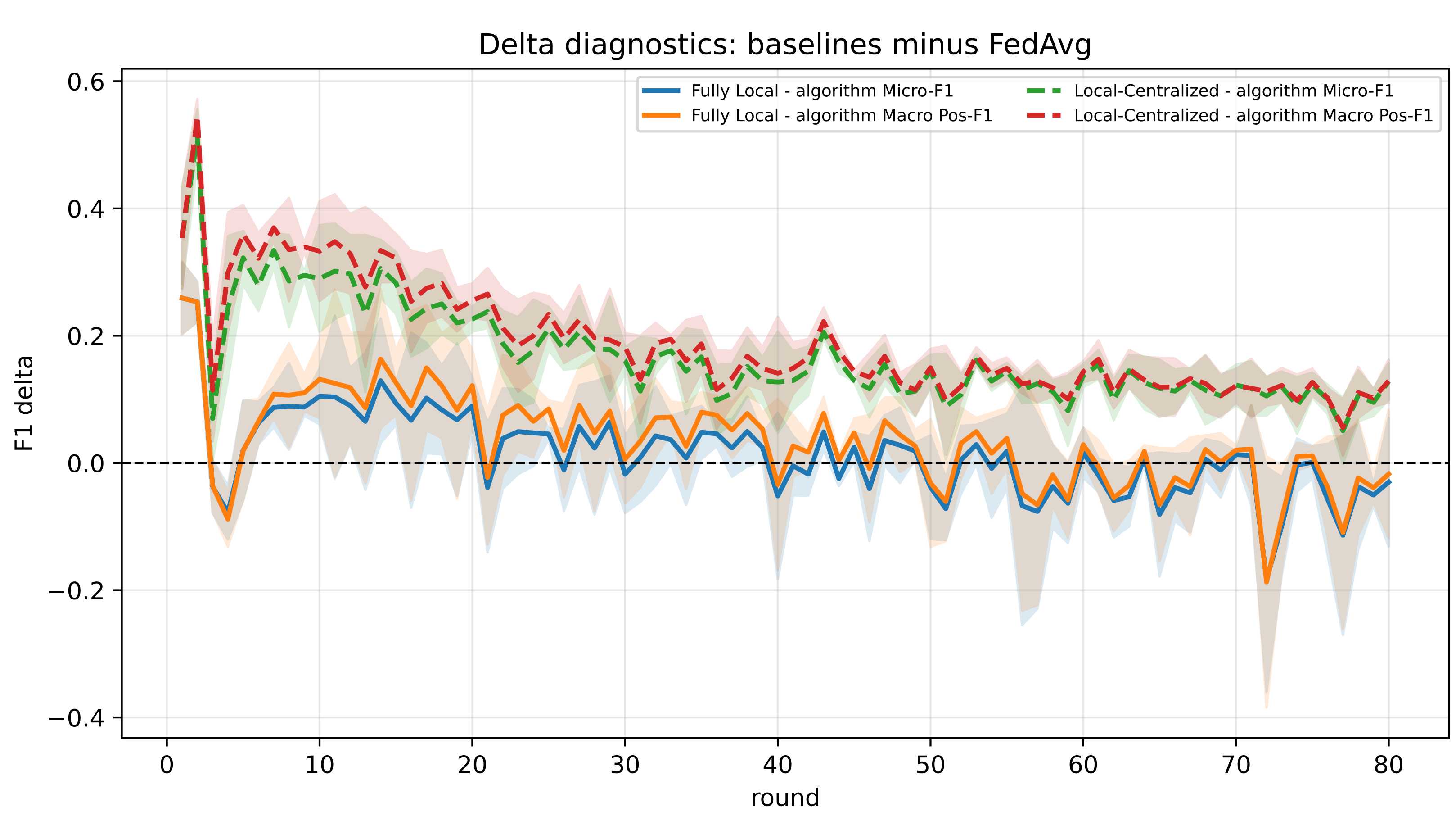
Algorithm	**Micro-F1**	**Macro Pos-F1**	**cycle2 pos-F1**	**cycle3 pos-F1**	**cycle4 pos-F1**	**cycle5 pos-F1**	**cycle6 pos-F1**
Fully Local	58.7% ± 5.5%	63.8% ± 5.1%	82.7% ± 2.7%	76.5% ± 5.8%	63.3% ± 5.5%	54.2% ± 7.8%	42.1% ± 3.8%
Local-Centralized	**84.3% ± 1.6%**	**86.3% ± 1.4%**	**96.7% ± 1.1%**	**94.7% ± 0.6%**	**87.4% ± 4.2%**	**80.9% ± 2.2%**	**71.8% ± 3.9%**
FedAvg	69.9% ± 1.5%	72.5% ± 1.1%	86.4% ± 4.4%	81.6% ± 4.2%	67.7% ± 1.9%	69.0% ± 4.4%	57.9% ± 1.9%











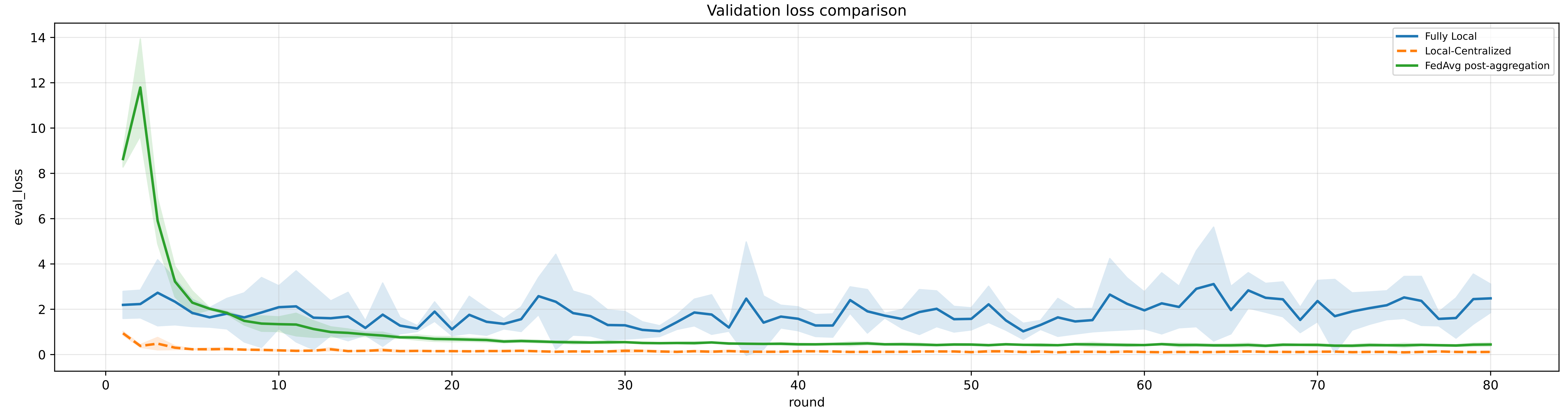
Controlled q-label heterogeneity: Mid | FedAvg diagnostics | family=q_task_cycle5 | n_clients=1 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64_headmulti

family=q_task_cycle5 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% | cycle5=visible 3.2%±0.0% / true 6.4% | cycle6=visible 0.8%±0.0% / true 6.3%
heterogeneity=mid | gamma=0.50 | task_jsd_mean=0.120 | task_jsd_median=0.121 | task_jsd_max=0.123 | family_centroid_jsd_mean=0.120

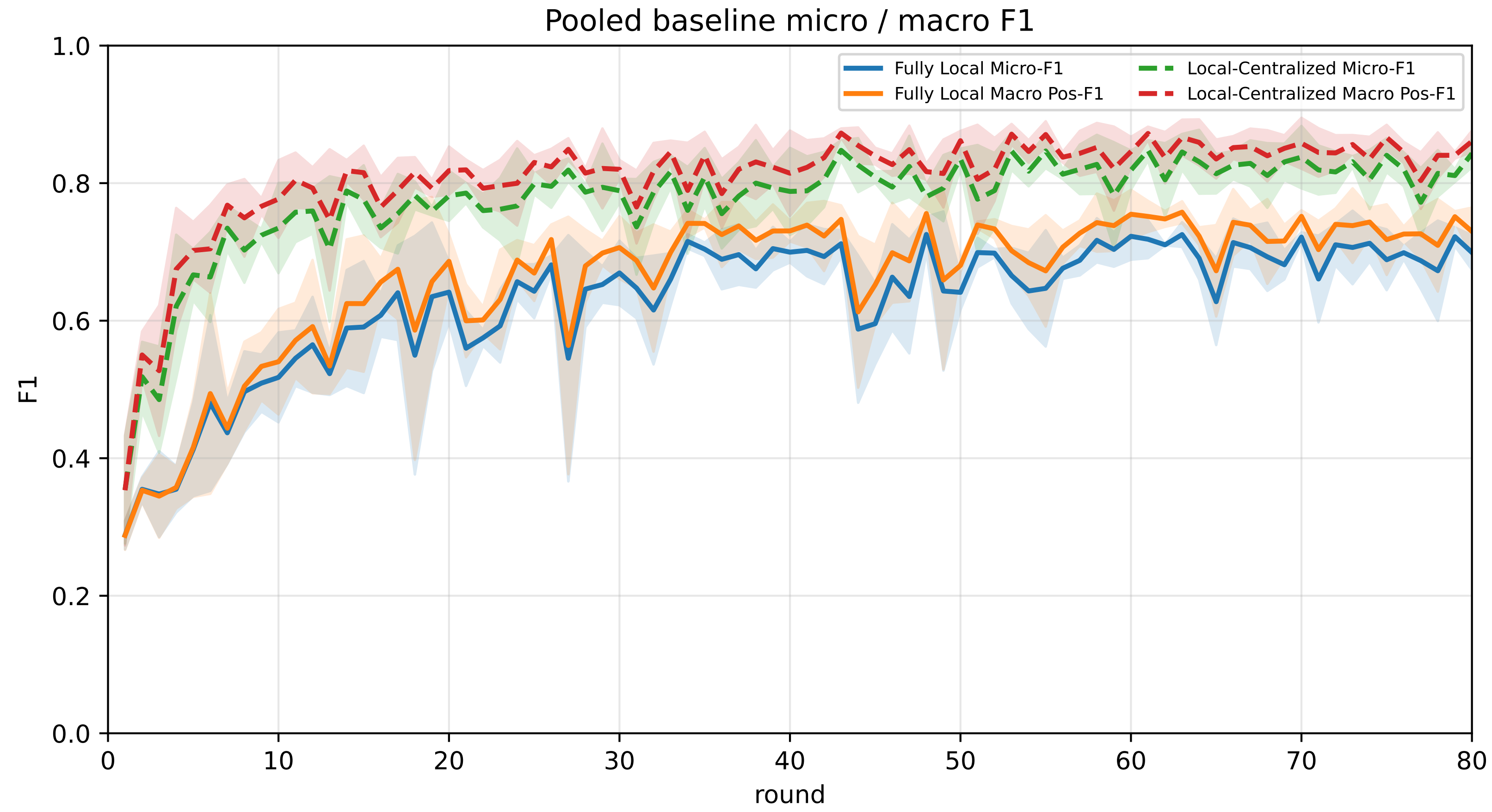
Best logged test performance (Fully Local vs Local-Centralized vs FedAvg)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	61.2% ± 0.9%	66.3% ± 1.4%	86.3% ± 4.7%	79.8% ± 5.8%	63.6% ± 1.5%	58.1% ± 1.4%	43.8% ± 3.4%
Local-Centralized	**84.3% ± 1.6%**	**86.3% ± 1.4%**	**96.7% ± 1.1%**	**94.7% ± 0.6%**	**87.4% ± 4.2%**	**80.9% ± 2.2%**	**71.8% ± 3.9%**
FedAvg	70.5% ± 1.4%	73.0% ± 1.2%	85.8% ± 4.7%	82.2% ± 4.3%	68.2% ± 2.1%	70.4% ± 4.3%	58.5% ± 2.0%

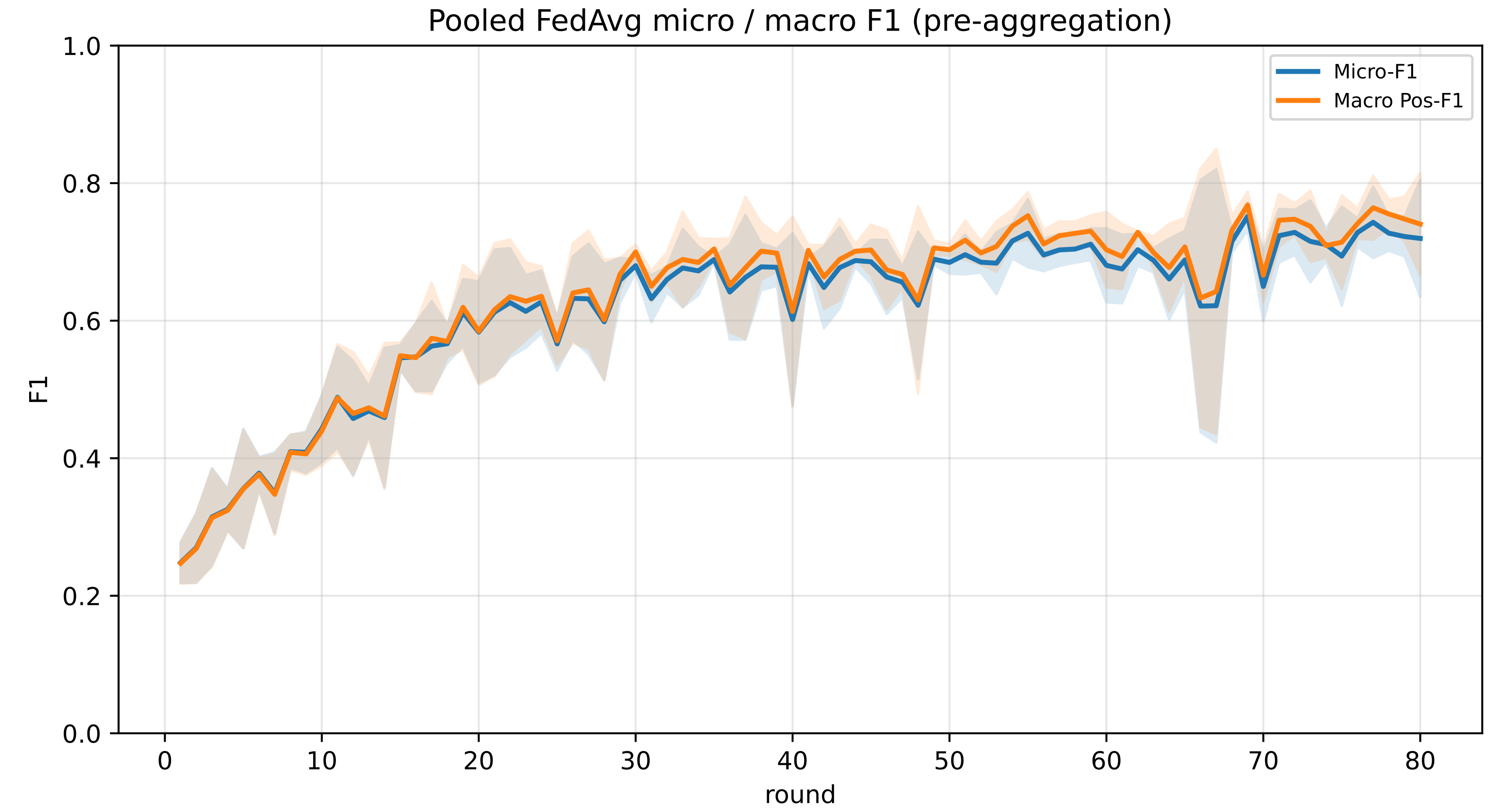
Validation loss comparison



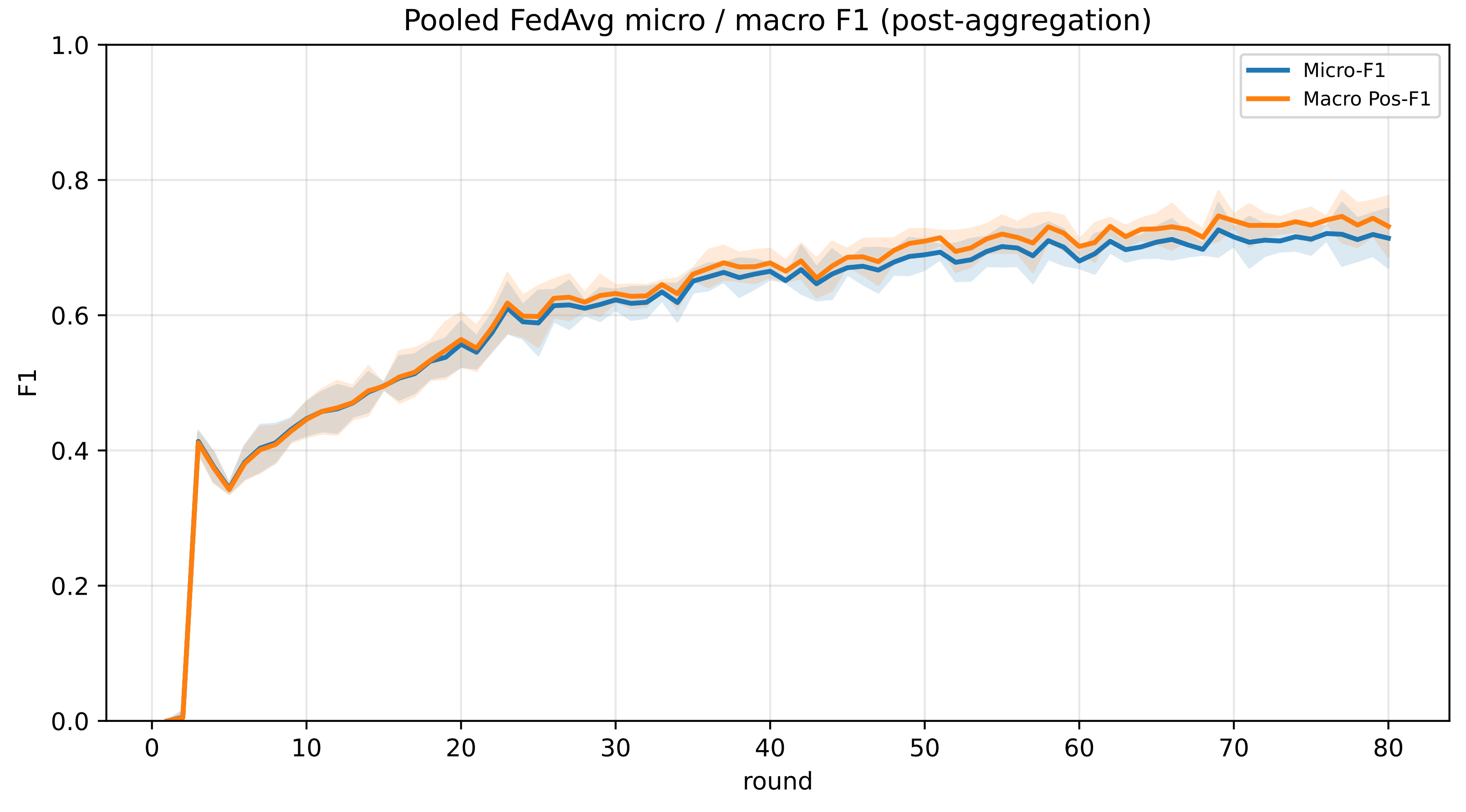
Pooled baseline micro / macro F1



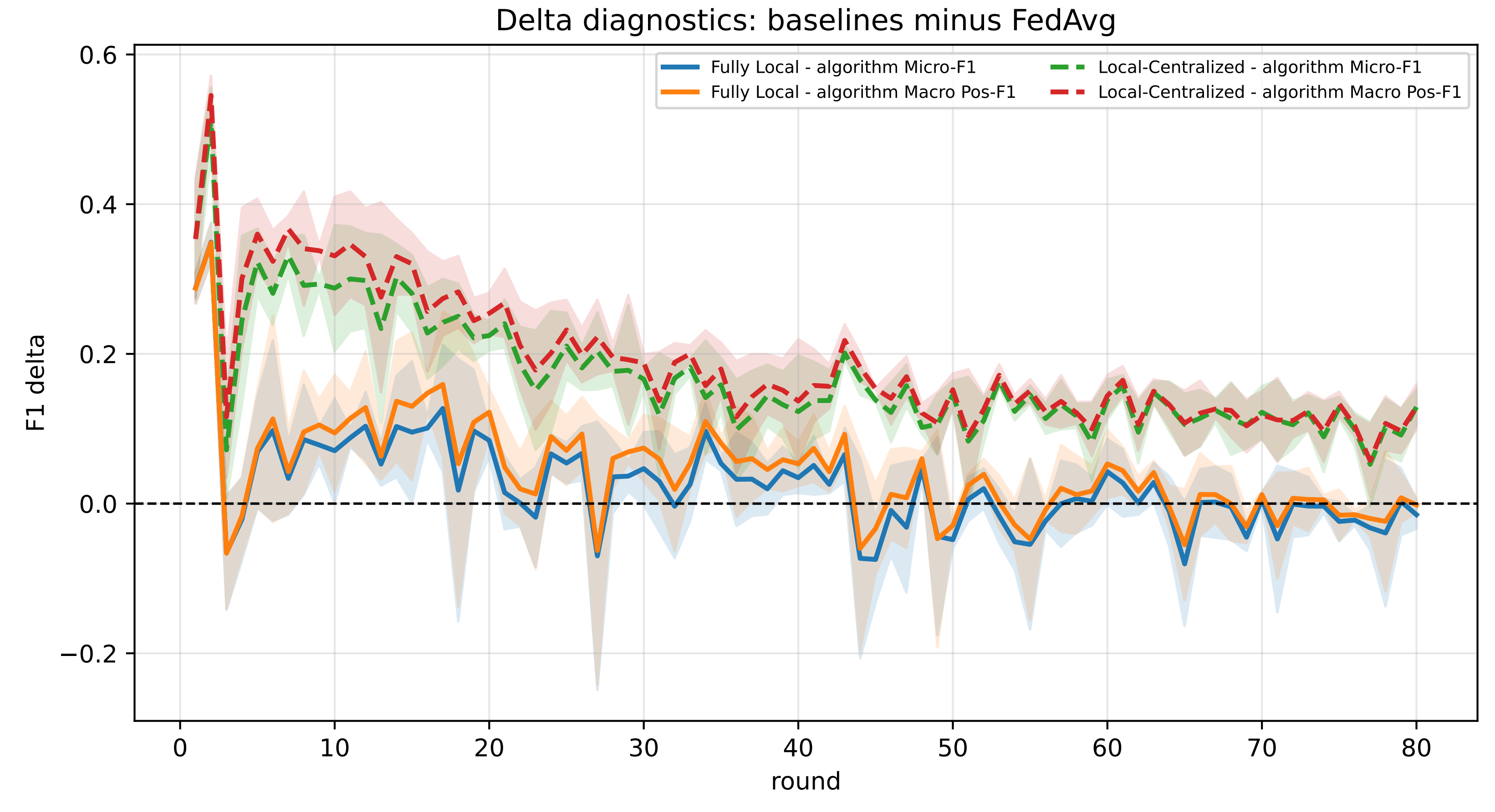
Pooled FedAvg micro / macro F1 (pre-aggregation)



Pooled FedAvg micro / macro F1 (post-aggregation)



Delta diagnostics: baselines minus FedAvg

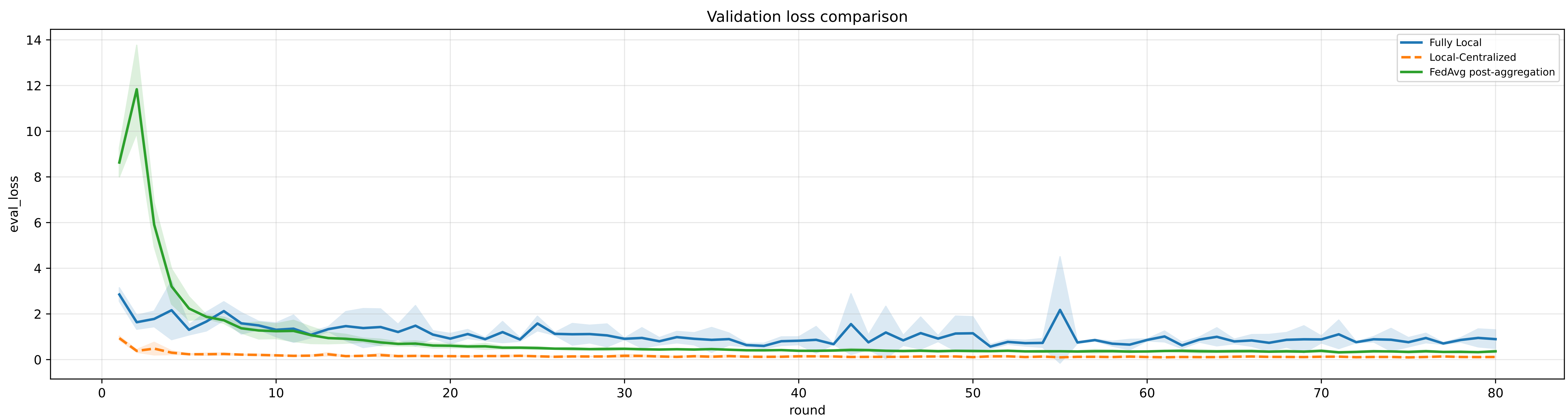


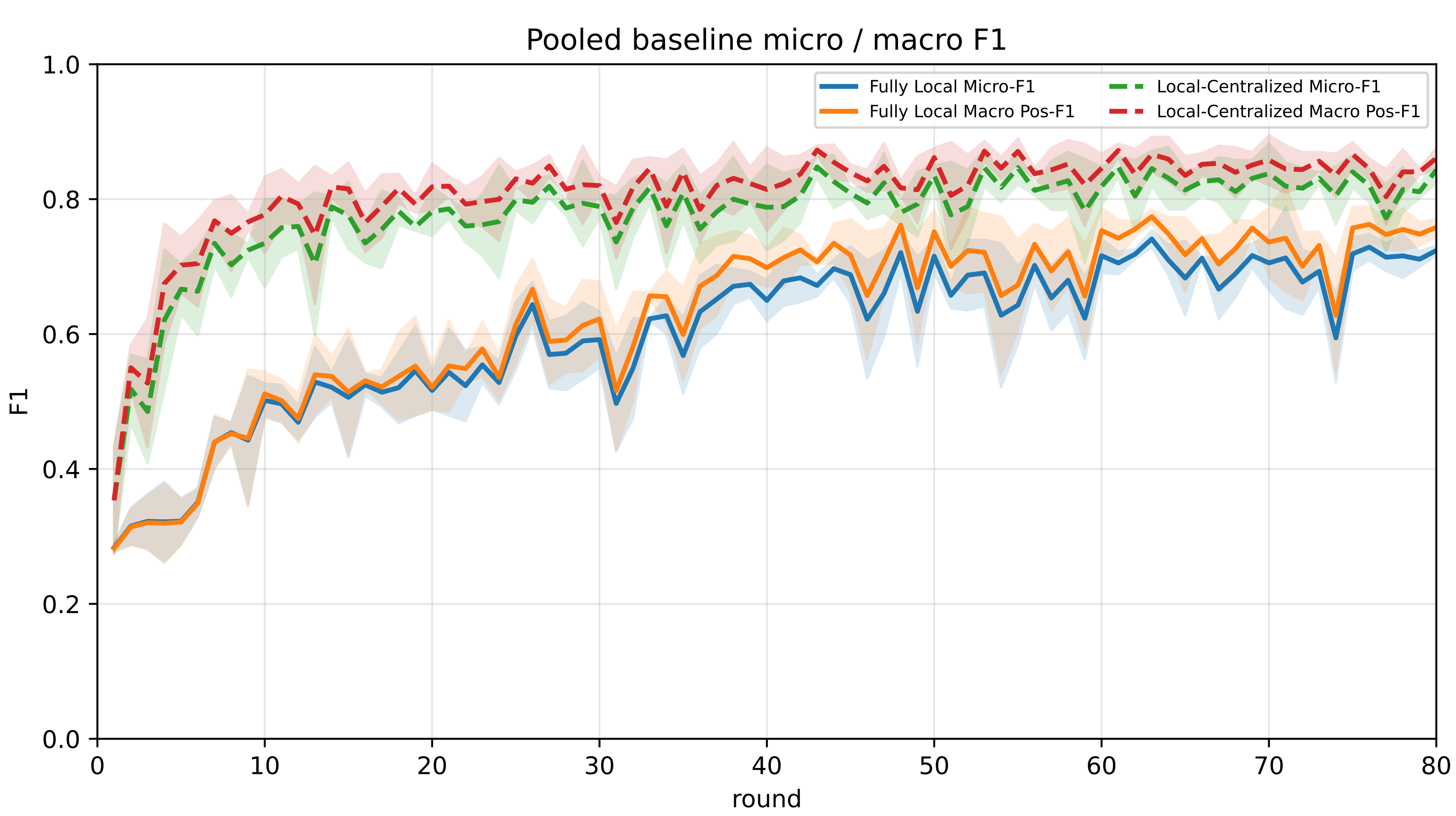
**Controlled q-label heterogeneity: Mid | FedAvg diagnostics | family=q_task_cycle6 |
n_clients=1 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64_headmulti**

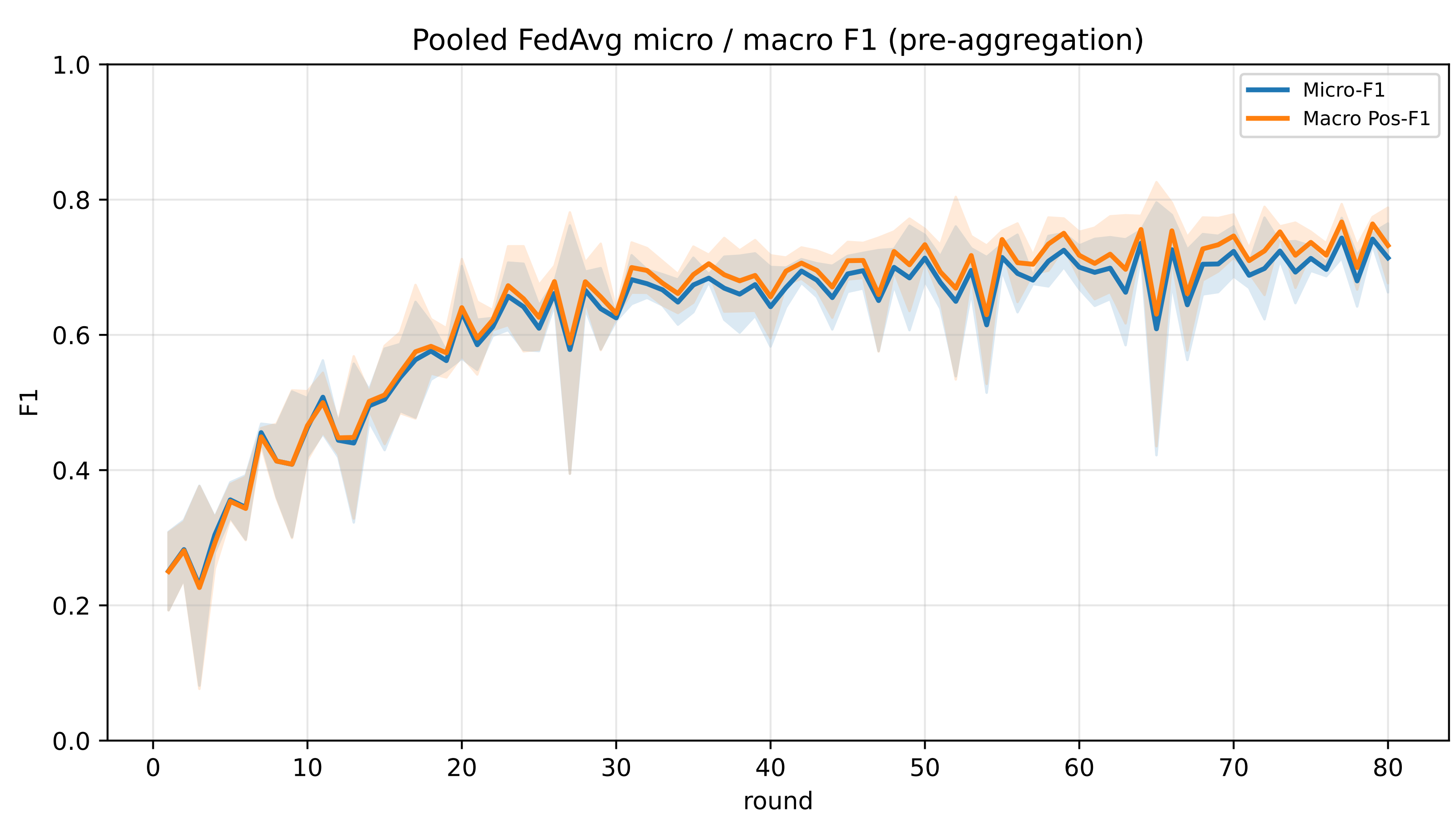
family=q_task_cycle6 | n_clients=1 | cycle2=visible 0.7%±0.0% / true 5.8% | cycle3=visible 0.8%±0.0% / true 6.2% | cycle4=visible 0.8%±0.0% / true 6.6% |
cycle5=visible 0.8%±0.0% / true 6.4% | cycle6=visible 3.1%±0.0% / true 6.3%
heterogeneity=mid | gamma=0.50 | task_jsd_mean=0.120 | task_jsd_median=0.121 | task_jsd_max=0.123 | family_centroid_jsd_mean=0.120

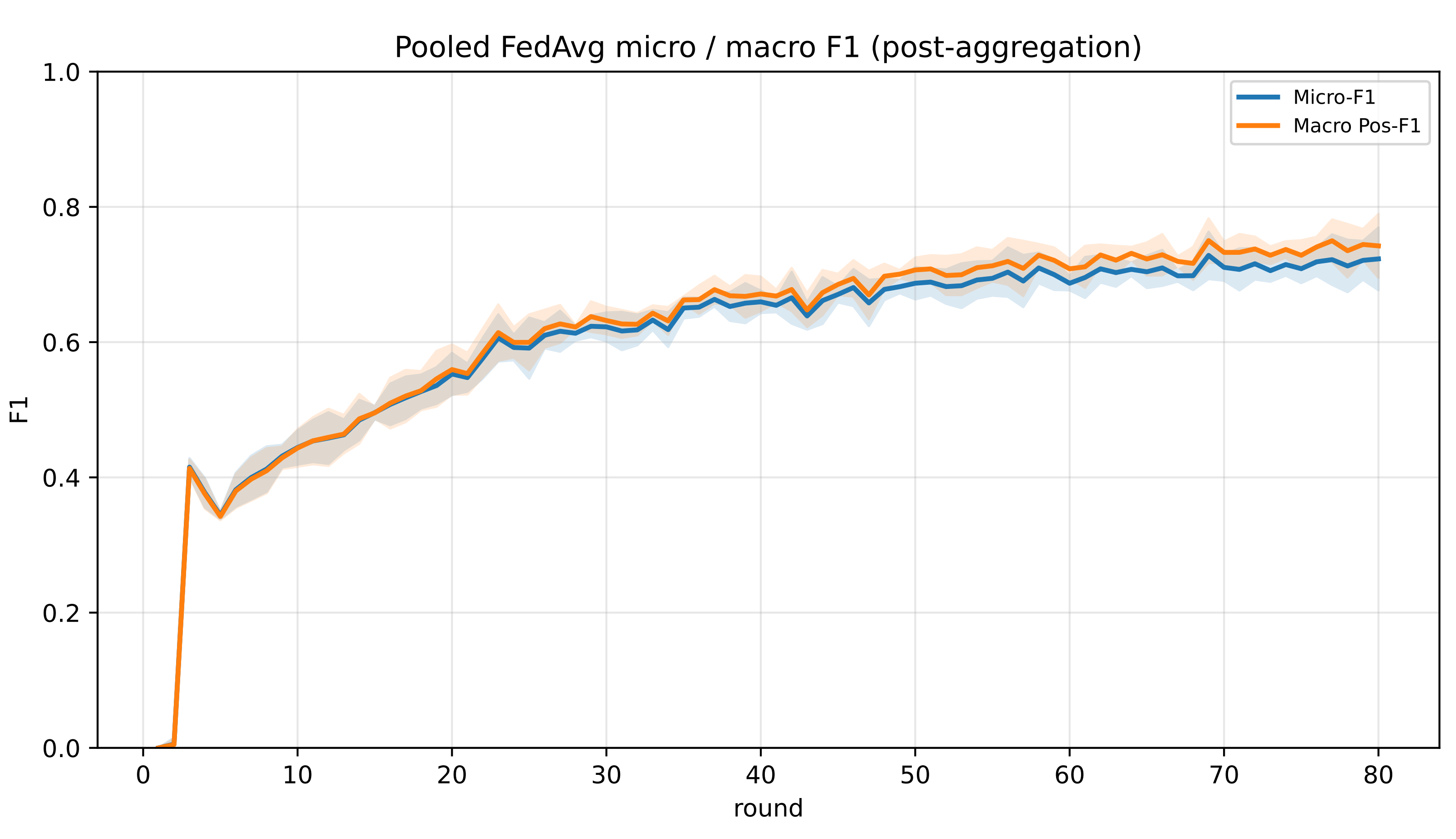
Best logged test performance (Fully Local vs Local-Centralized vs FedAvg)

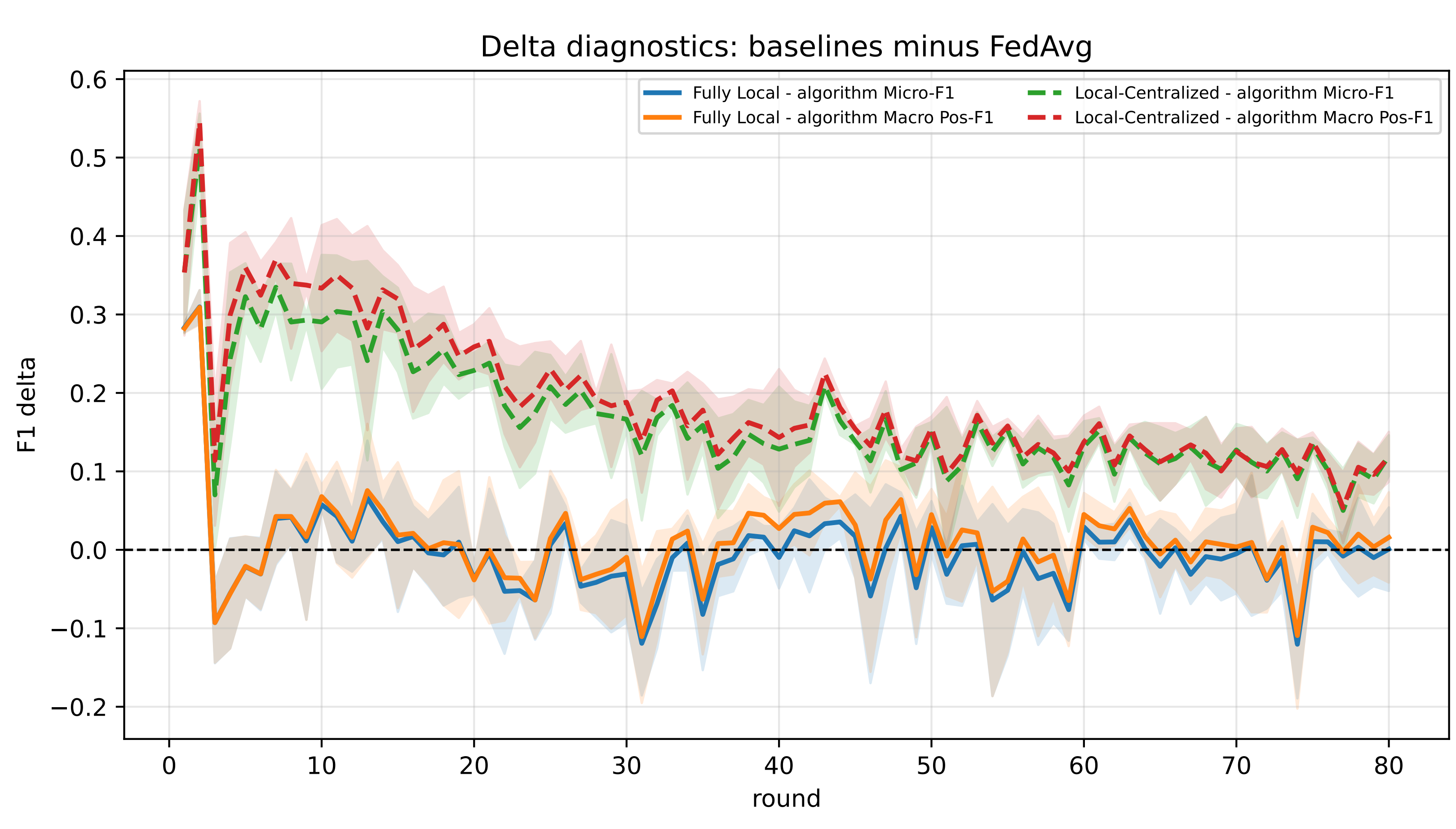
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	64.1% ± 1.8%	67.8% ± 2.4%	81.8% ± 8.2%	78.3% ± 6.8%	64.9% ± 1.4%	64.9% ± 4.4%	49.0% ± 3.2%
Local-Centralized	**84.3% ± 1.6%**	**86.3% ± 1.4%**	**96.7% ± 1.1%**	**94.7% ± 0.6%**	**87.4% ± 4.2%**	**80.9% ± 2.2%**	**71.8% ± 3.9%**
FedAvg	70.2% ± 2.0%	72.8% ± 1.6%	86.5% ± 4.6%	81.6% ± 4.4%	67.6% ± 2.3%	70.2% ± 4.4%	58.0% ± 2.4%











Controlled q-label heterogeneity: High | FedAvg diagnostics | aggregated / pooled over 5 clients | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64_headmulti

family=q_task_cycle2 | n_clients=1 | cycle2=visible 4.6%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle3 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 5.0%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle4 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 5.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle5 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 5.1%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
family=q_task_cycle6 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 5.0%±0.0% / true 6.3%
heterogeneity=high | gamma=0.80 | task_jsd_mean=0.399 | task_jsd_median=0.399 | task_jsd_max=0.403 | family_centroid_jsd_mean=0.399

Best logged test performance (Fully Local vs Local-Centralized vs FedAvg)

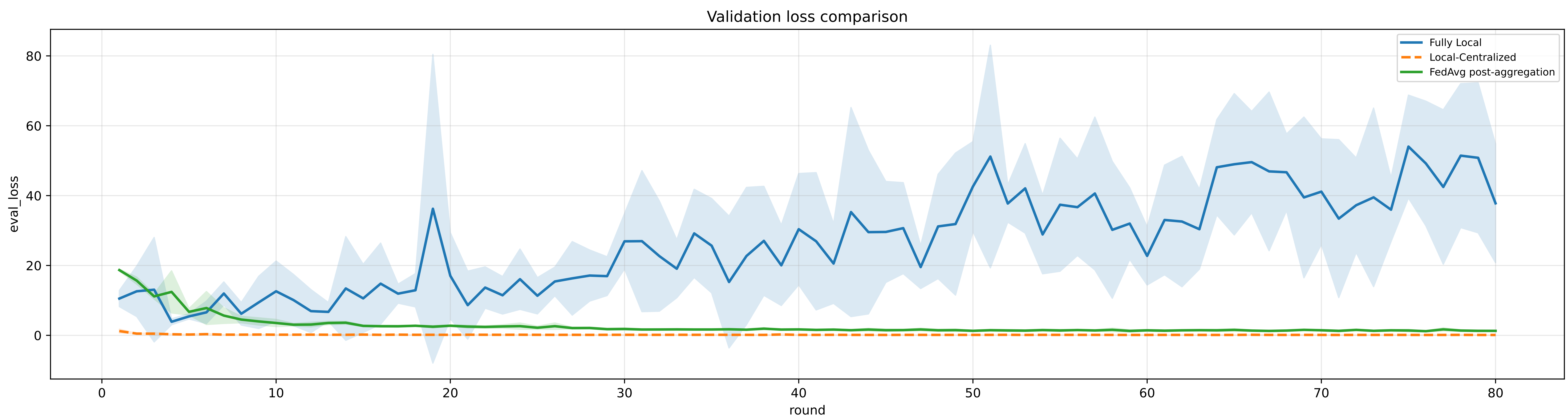
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	27.6% ± 1.8%	27.6% ± 1.8%	22.4% ± 1.5%	27.5% ± 2.1%	27.3% ± 1.2%	26.7% ± 1.2%	30.2% ± 1.5%
Local-Centralized	**84.9% ± 1.3%**	**86.7% ± 0.8%**	**96.8% ± 1.6%**	**95.1% ± 1.4%**	**86.6% ± 2.5%**	**81.4% ± 1.3%**	**73.7% ± 5.8%**
FedAvg	59.3% ± 1.9%	59.9% ± 1.9%	60.2% ± 2.0%	64.9% ± 5.2%	61.3% ± 2.4%	59.8% ± 2.6%	53.2% ± 3.1%

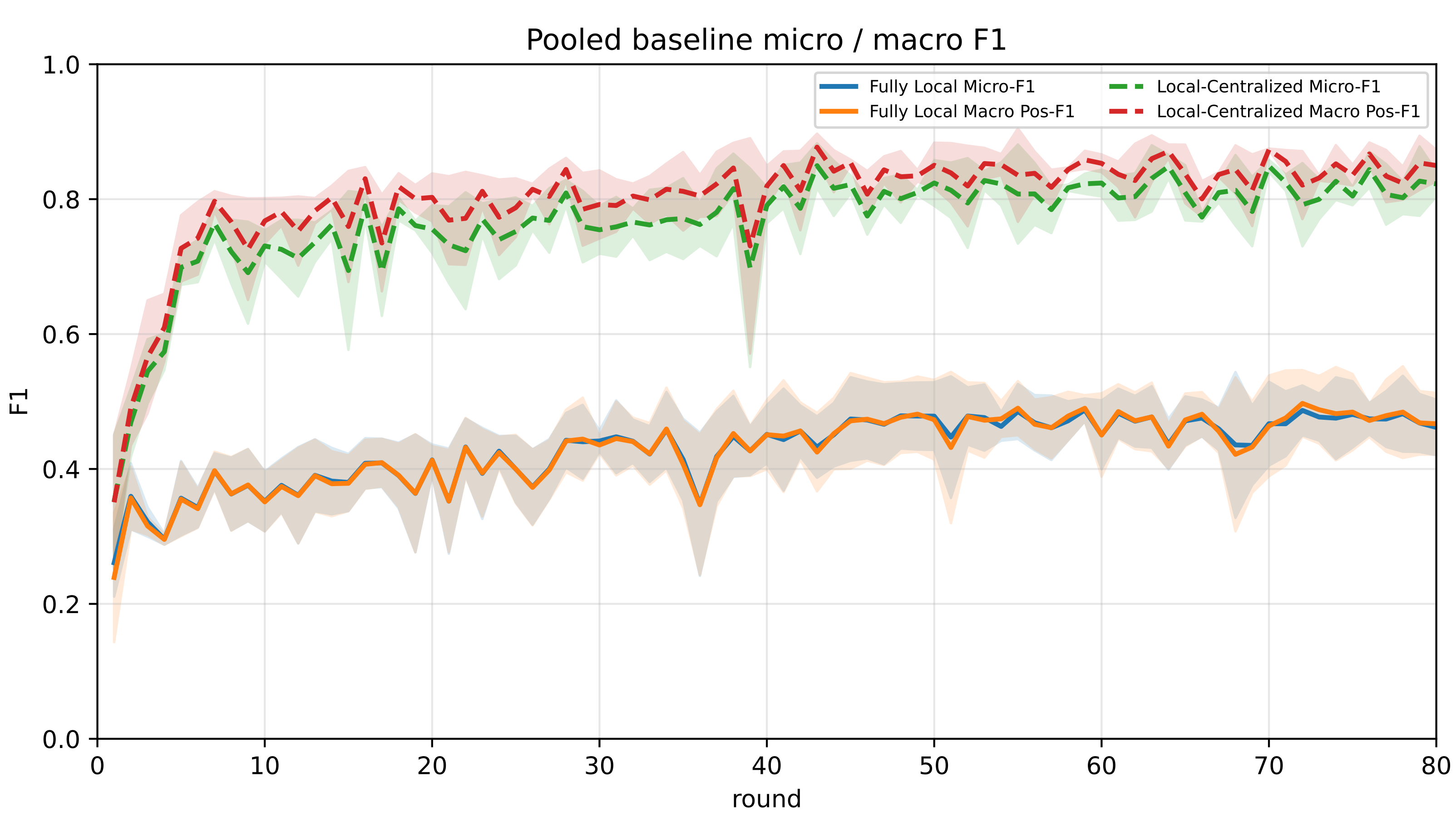
Controlled q-label heterogeneity: High | FedAvg diagnostics | family=q_task_cycle2 | n_clients=1 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64_headmulti

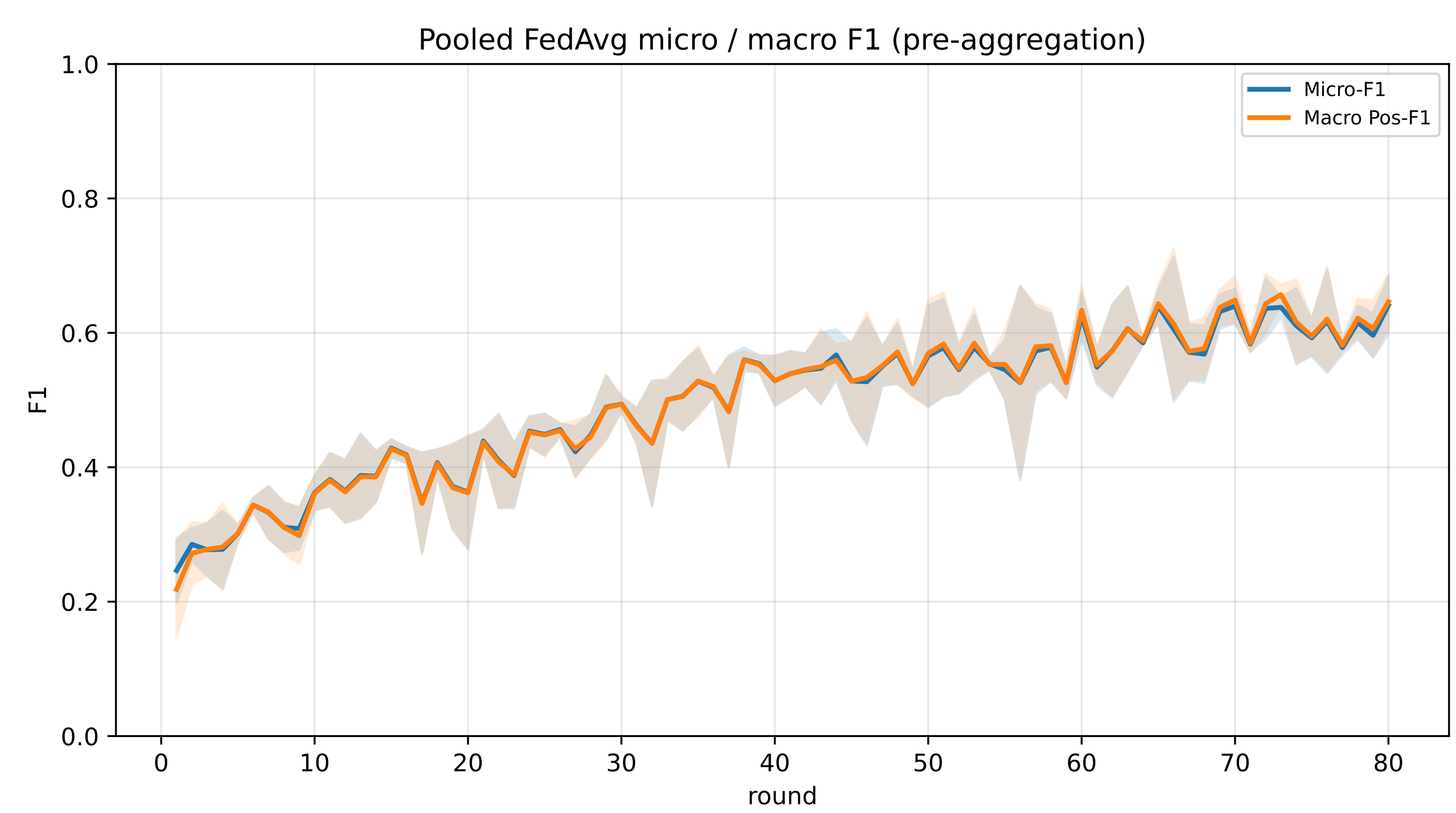
family=q_task_cycle2 | n_clients=1 | cycle2=visible 4.6%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
heterogeneity=high | gamma=0.80 | task_jsd_mean=0.399 | task_jsd_median=0.399 | task_jsd_max=0.403 | family_centroid_jsd_mean=0.399

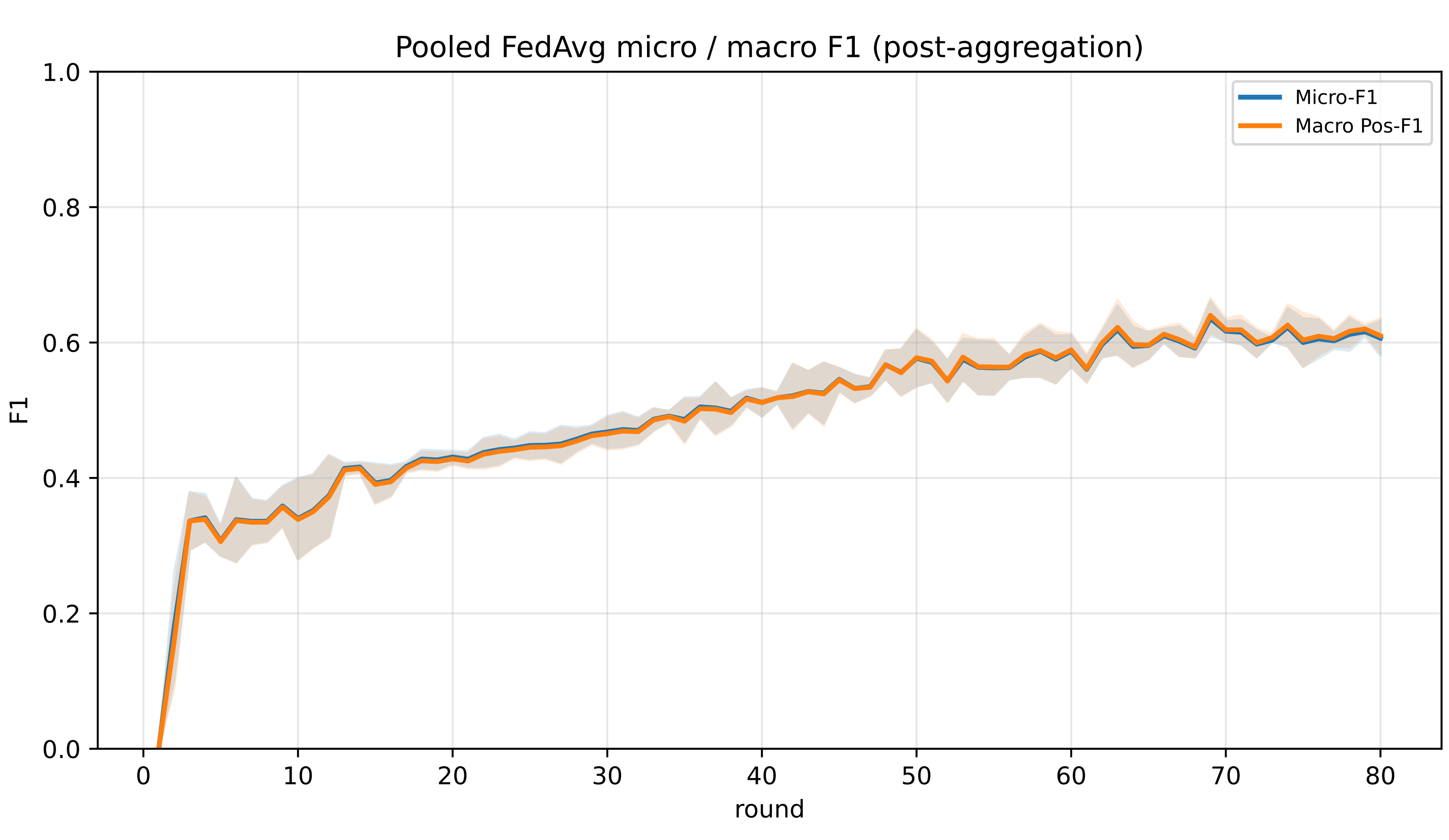
Best logged test performance (Fully Local vs Local-Centralized vs FedAvg)

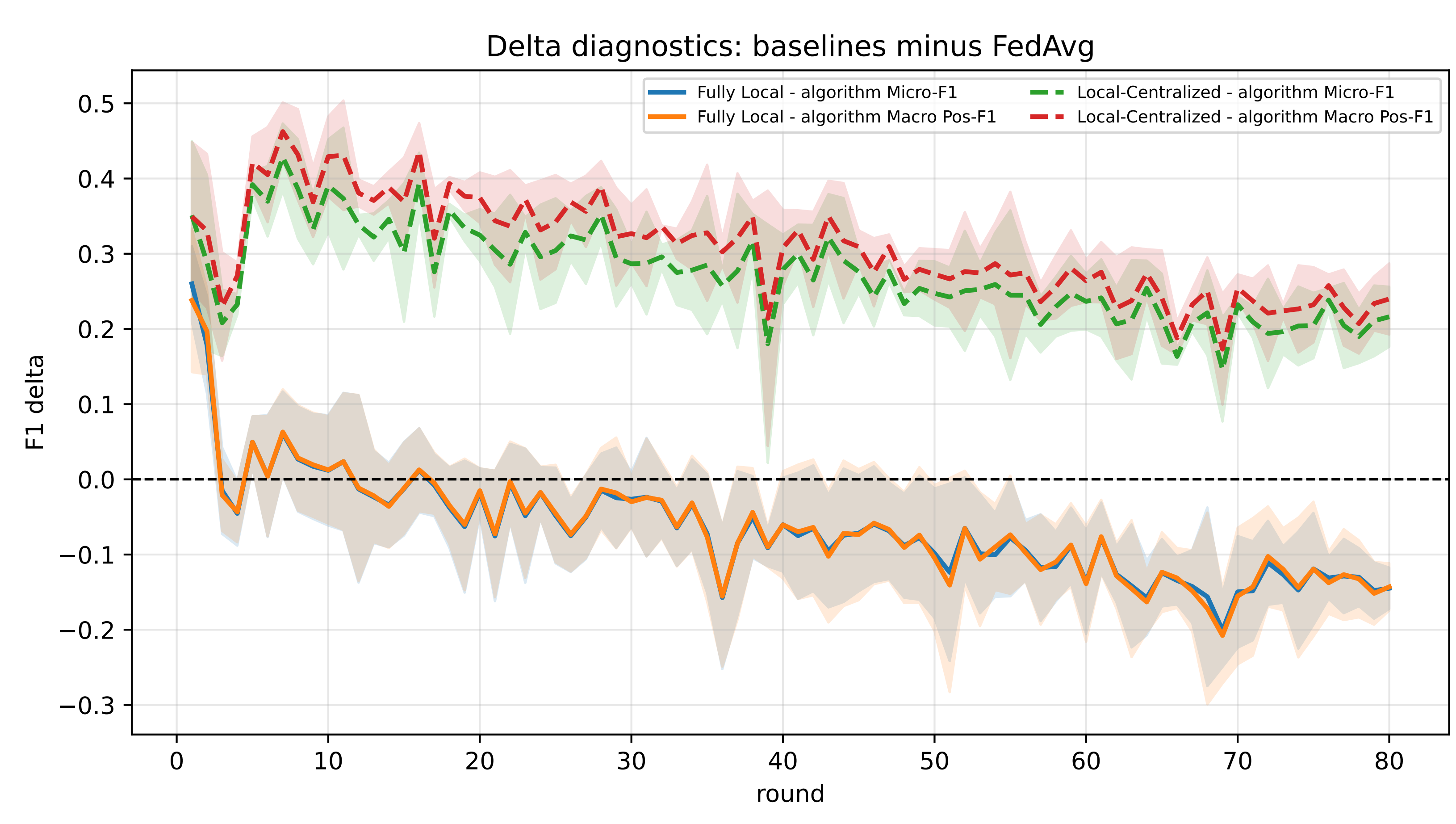
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	28.6% ± 0.7%	28.6% ± 0.7%	23.4% ± 1.7%	29.1% ± 1.2%	30.2% ± 0.8%	28.5% ± 1.4%	32.0% ± 1.3%
Local-Centralized	**84.9% ± 1.3%**	**86.7% ± 0.8%**	**96.8% ± 1.6%**	**95.1% ± 1.4%**	**86.6% ± 2.5%**	**81.4% ± 1.3%**	**73.7% ± 5.8%**
FedAvg	59.3% ± 1.6%	59.8% ± 1.5%	59.5% ± 1.4%	64.5% ± 4.3%	61.3% ± 3.0%	60.3% ± 2.8%	53.5% ± 2.6%









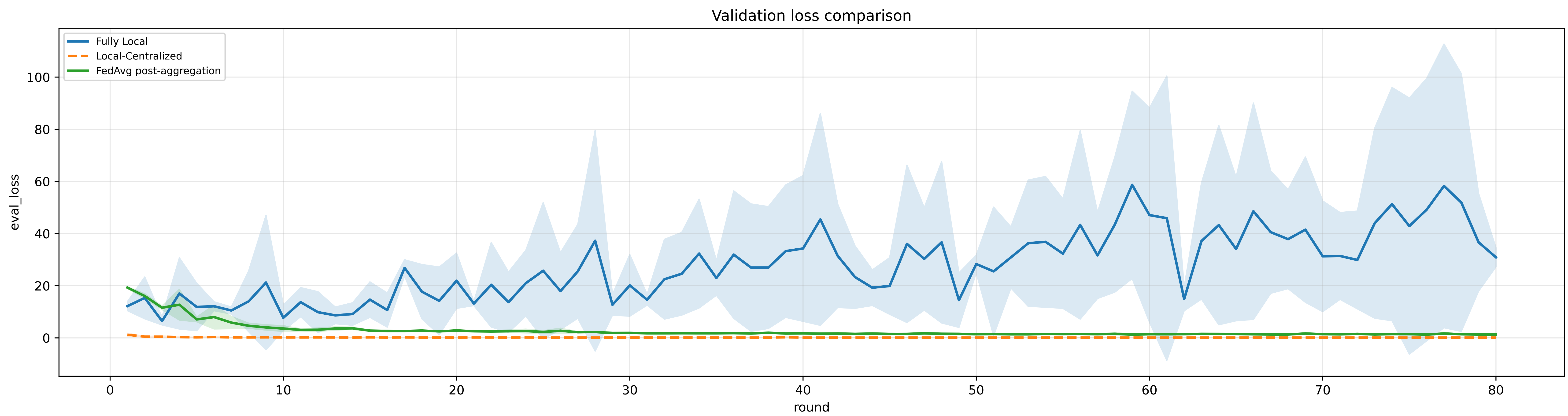


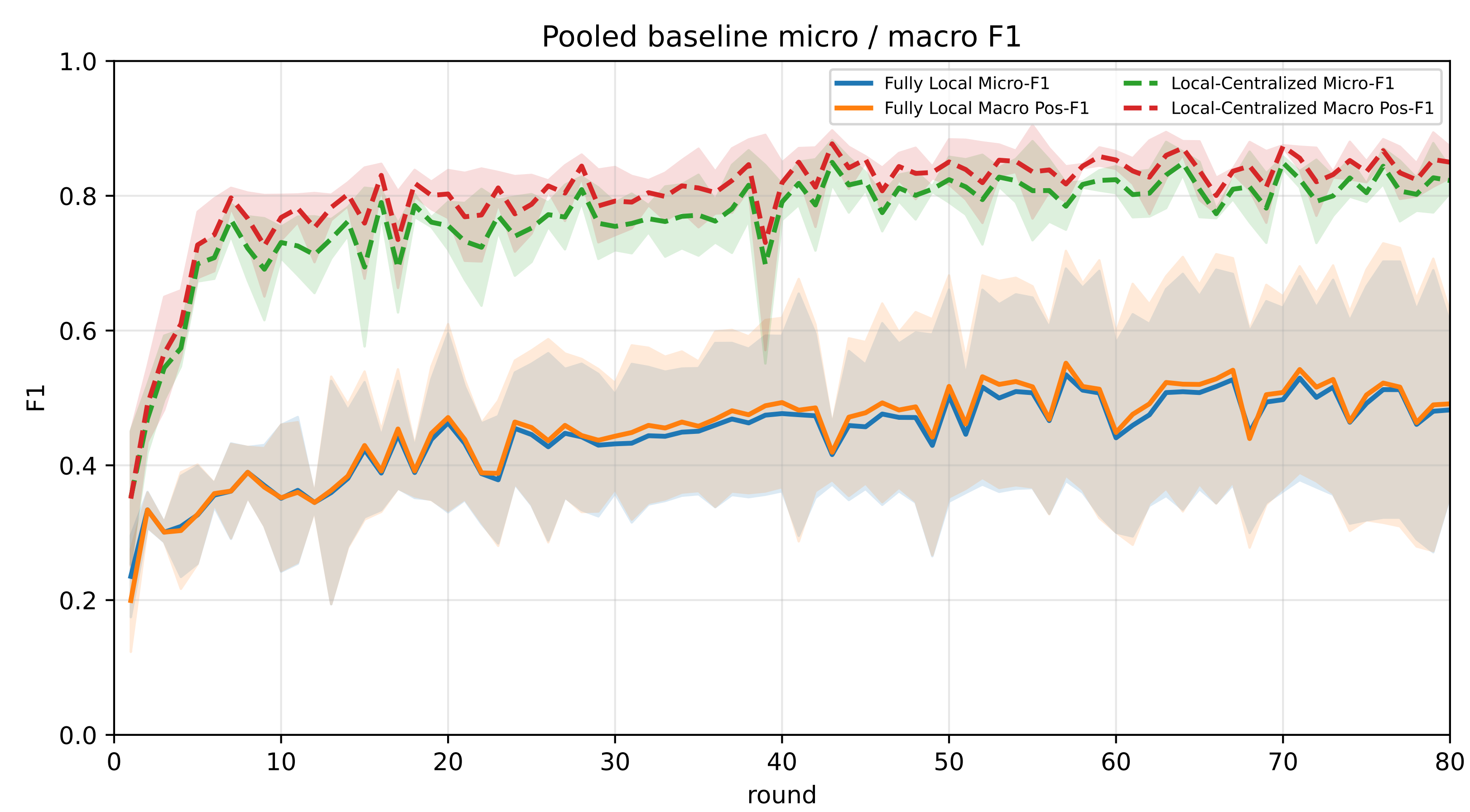
Controlled q-label heterogeneity: High | FedAvg diagnostics | family=q_task_cycle3 | n_clients=1 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64_headmulti

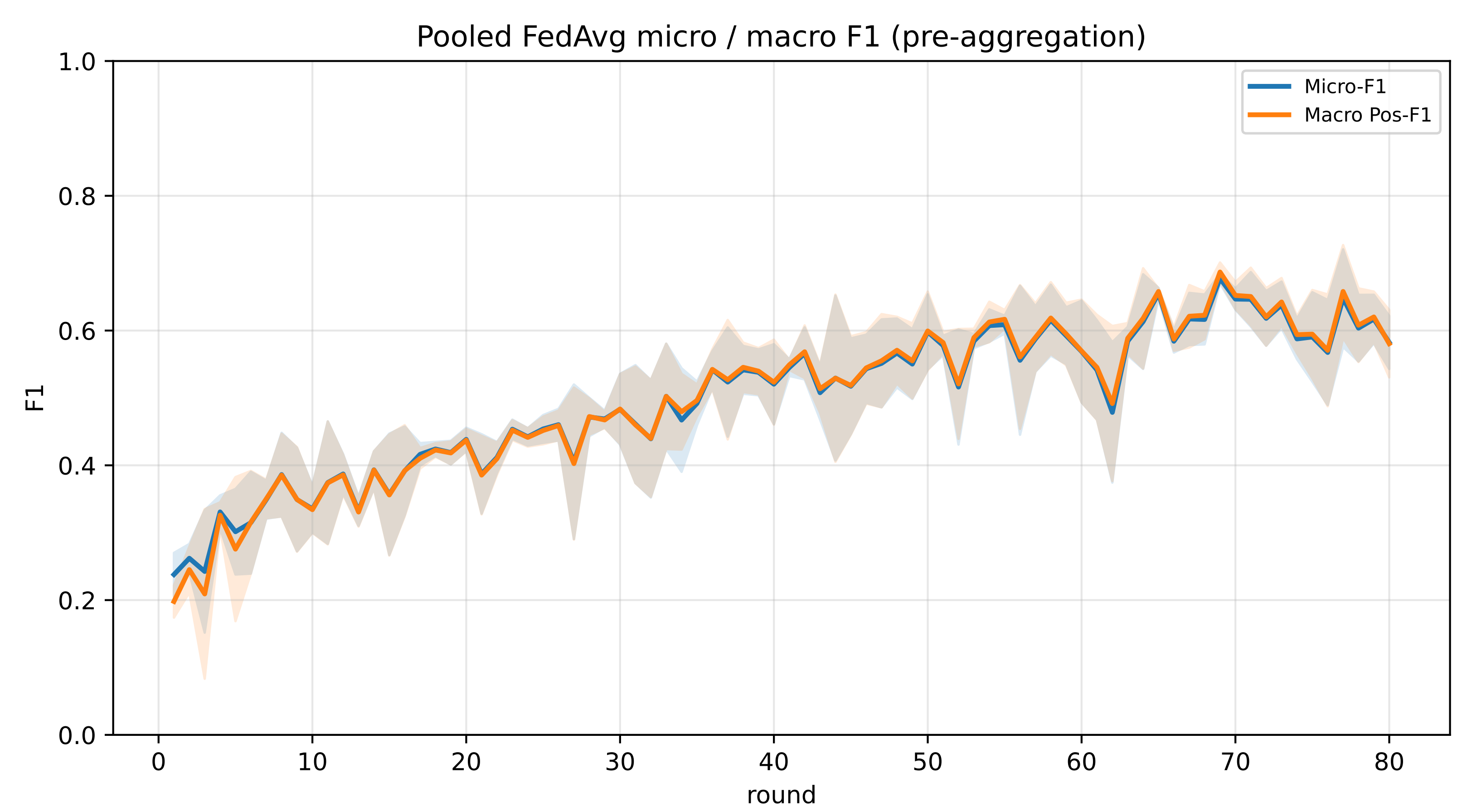
family=q_task_cycle3 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 5.0%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% |
cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
heterogeneity=high | gamma=0.80 | task_jsd_mean=0.399 | task_jsd_median=0.399 | task_jsd_max=0.403 | family_centroid_jsd_mean=0.399

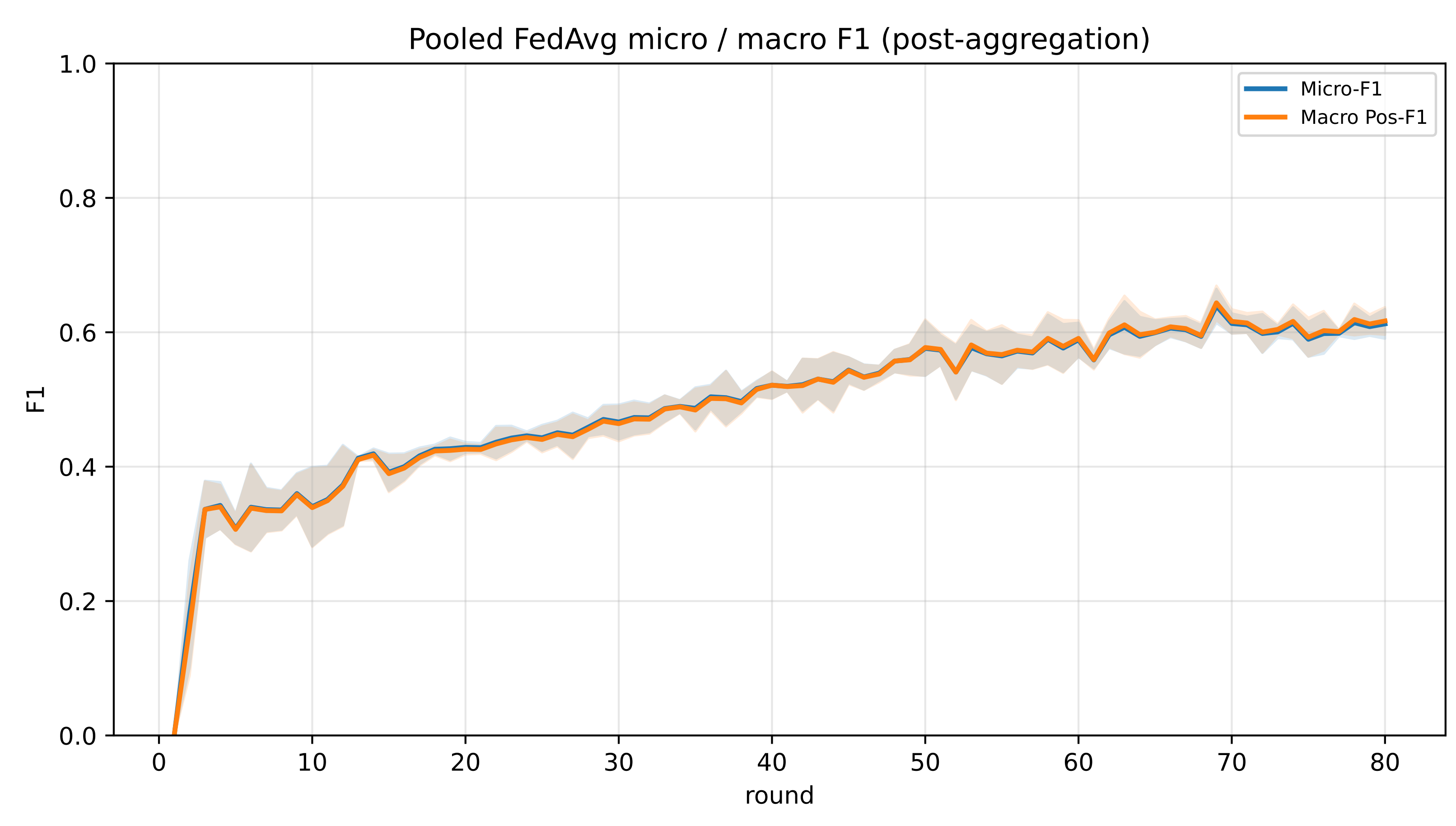
Best logged test performance (Fully Local vs Local-Centralized vs FedAvg)

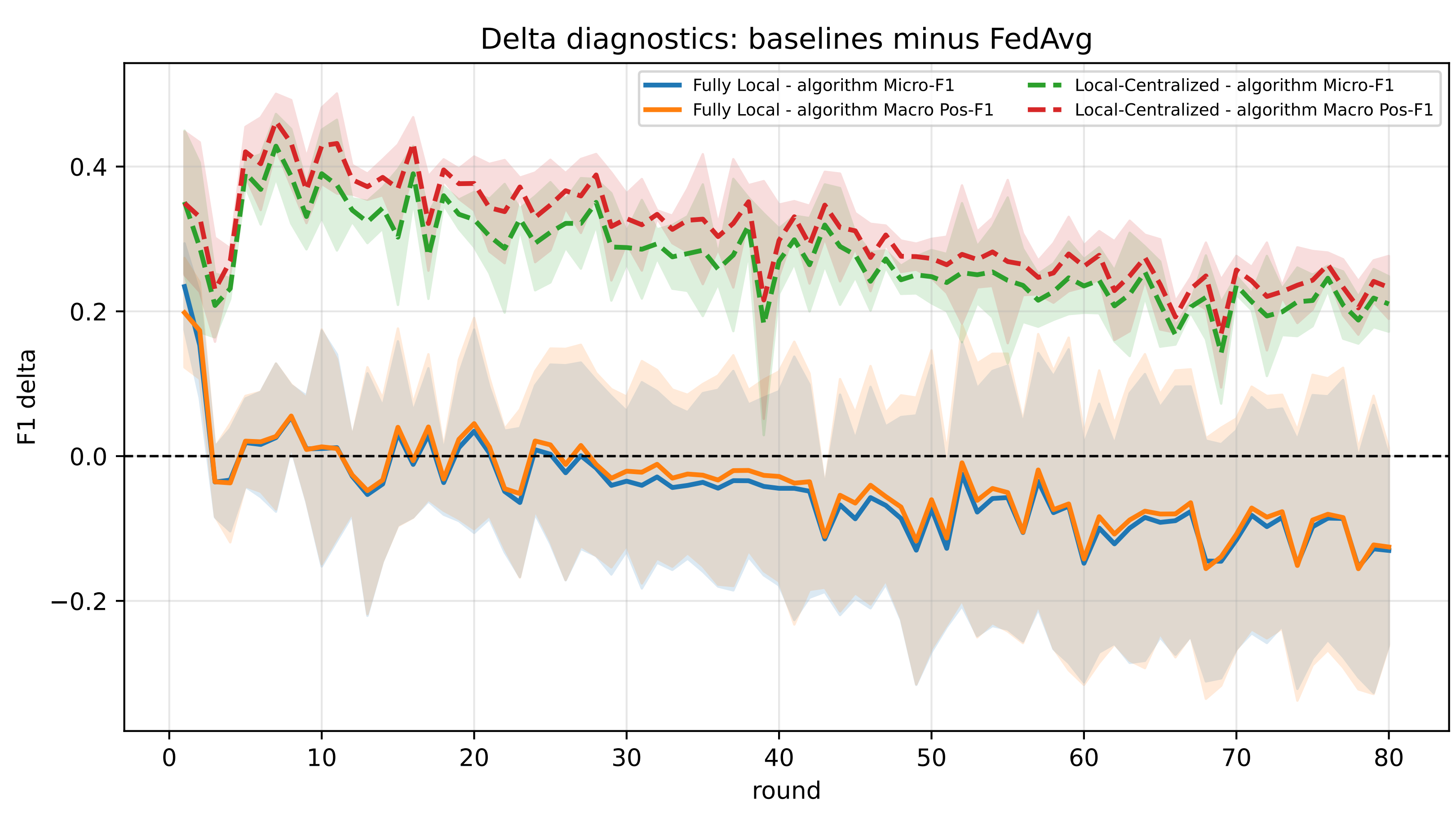
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	26.5% ± 4.7%	26.6% ± 4.7%	22.3% ± 4.2%	27.0% ± 7.0%	26.9% ± 5.3%	27.0% ± 4.1%	29.5% ± 3.2%
Local-Centralized	**84.9% ± 1.3%**	**86.7% ± 0.8%**	**96.8% ± 1.6%**	**95.1% ± 1.4%**	**86.6% ± 2.5%**	**81.4% ± 1.3%**	**73.7% ± 5.8%**
FedAvg	59.6% ± 1.7%	60.2% ± 1.6%	60.7% ± 1.9%	64.7% ± 5.0%	61.9% ± 2.6%	60.2% ± 1.9%	53.3% ± 3.1%









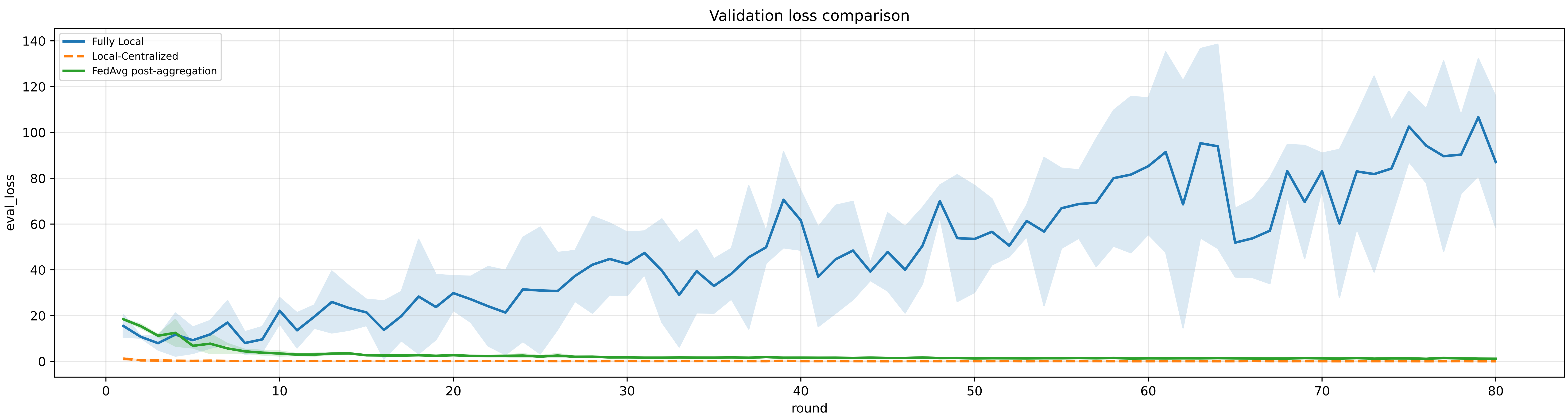


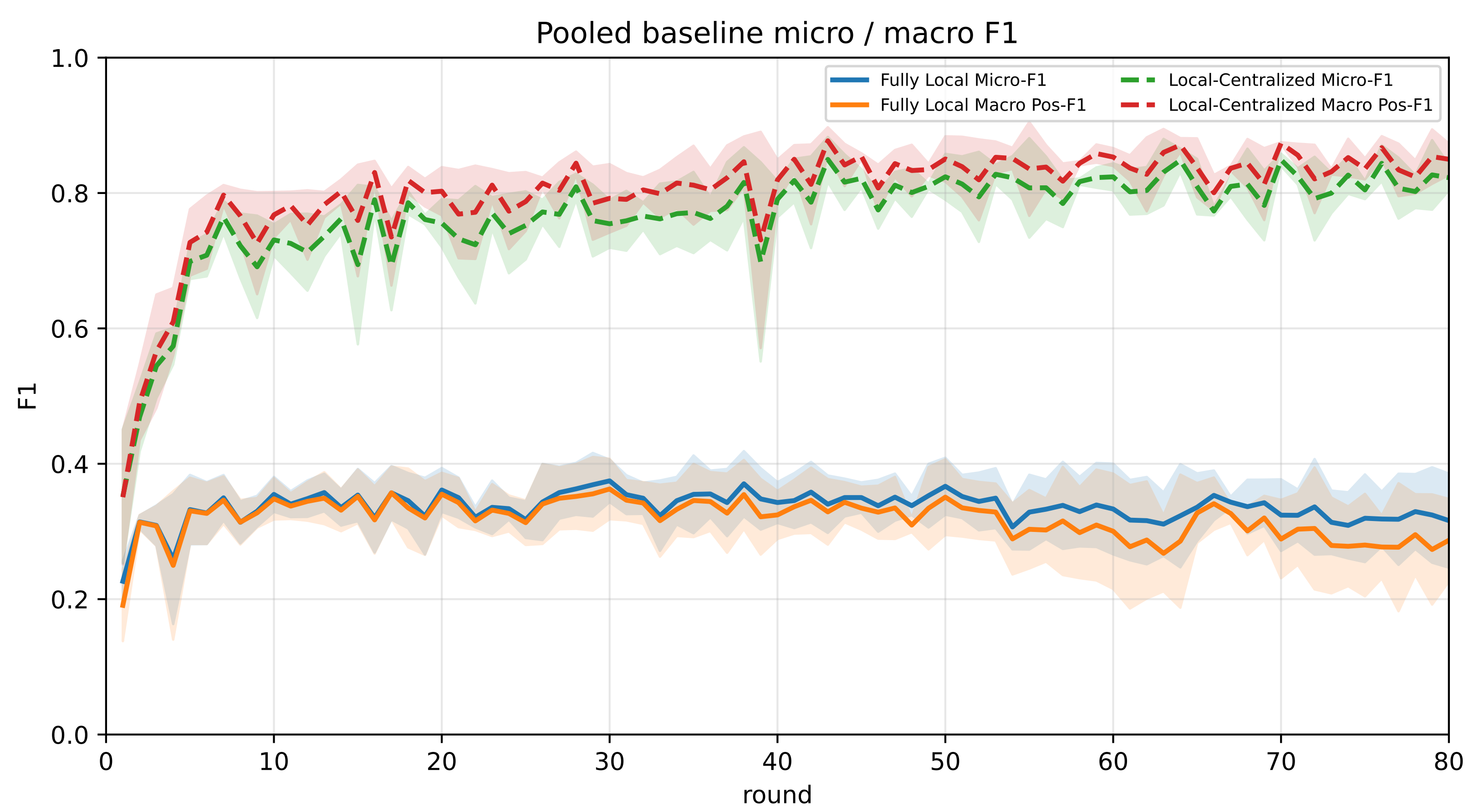
Controlled q-label heterogeneity: High | FedAvg diagnostics | family=q_task_cycle4 |
n_clients=1 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64_headmulti

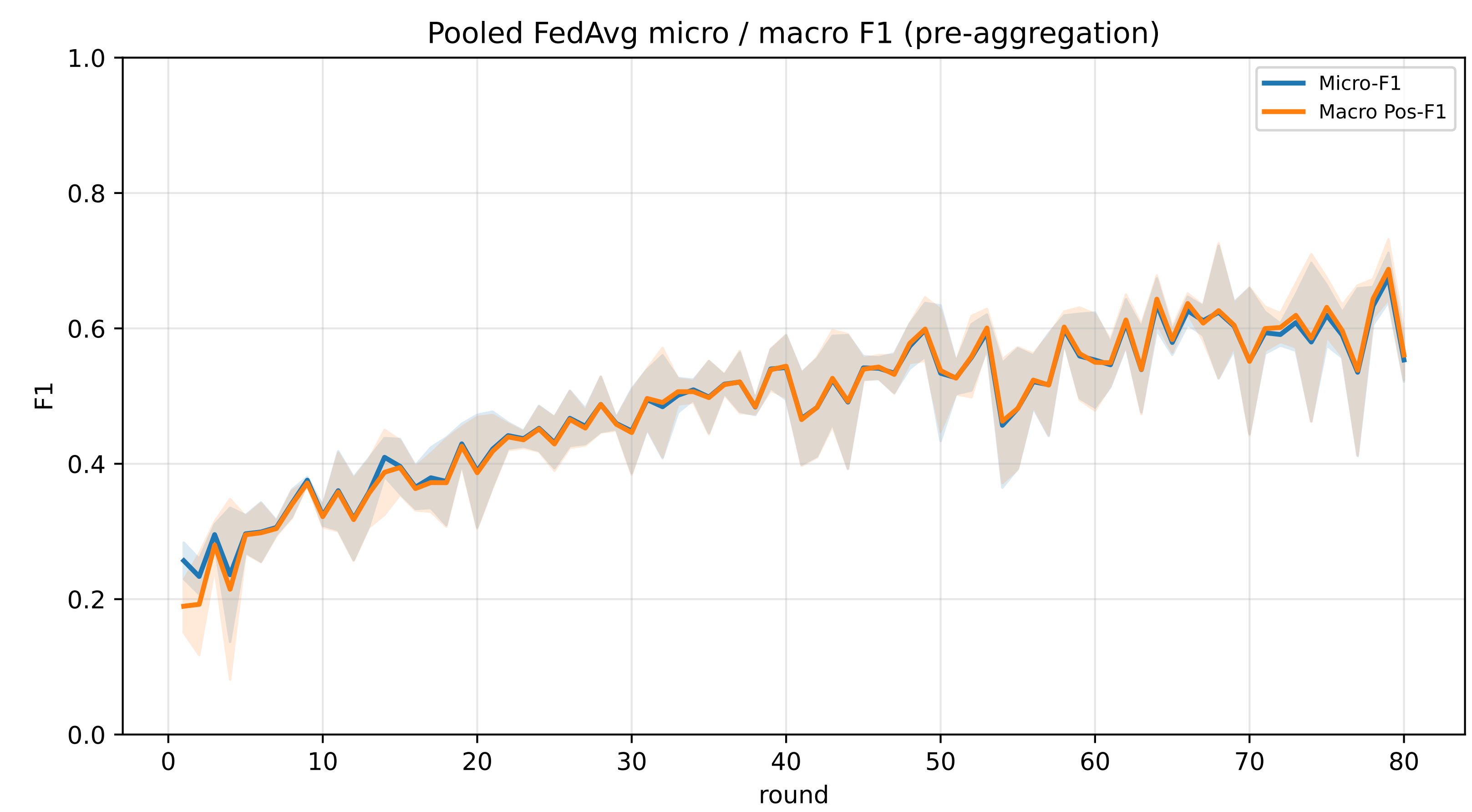
family=q_task_cycle4 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 5.3%±0.0% / true 6.6% |
cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
heterogeneity=high | gamma=0.80 | task_jsd_mean=0.399 | task_jsd_median=0.399 | task_jsd_max=0.403 | family_centroid_jsd_mean=0.399

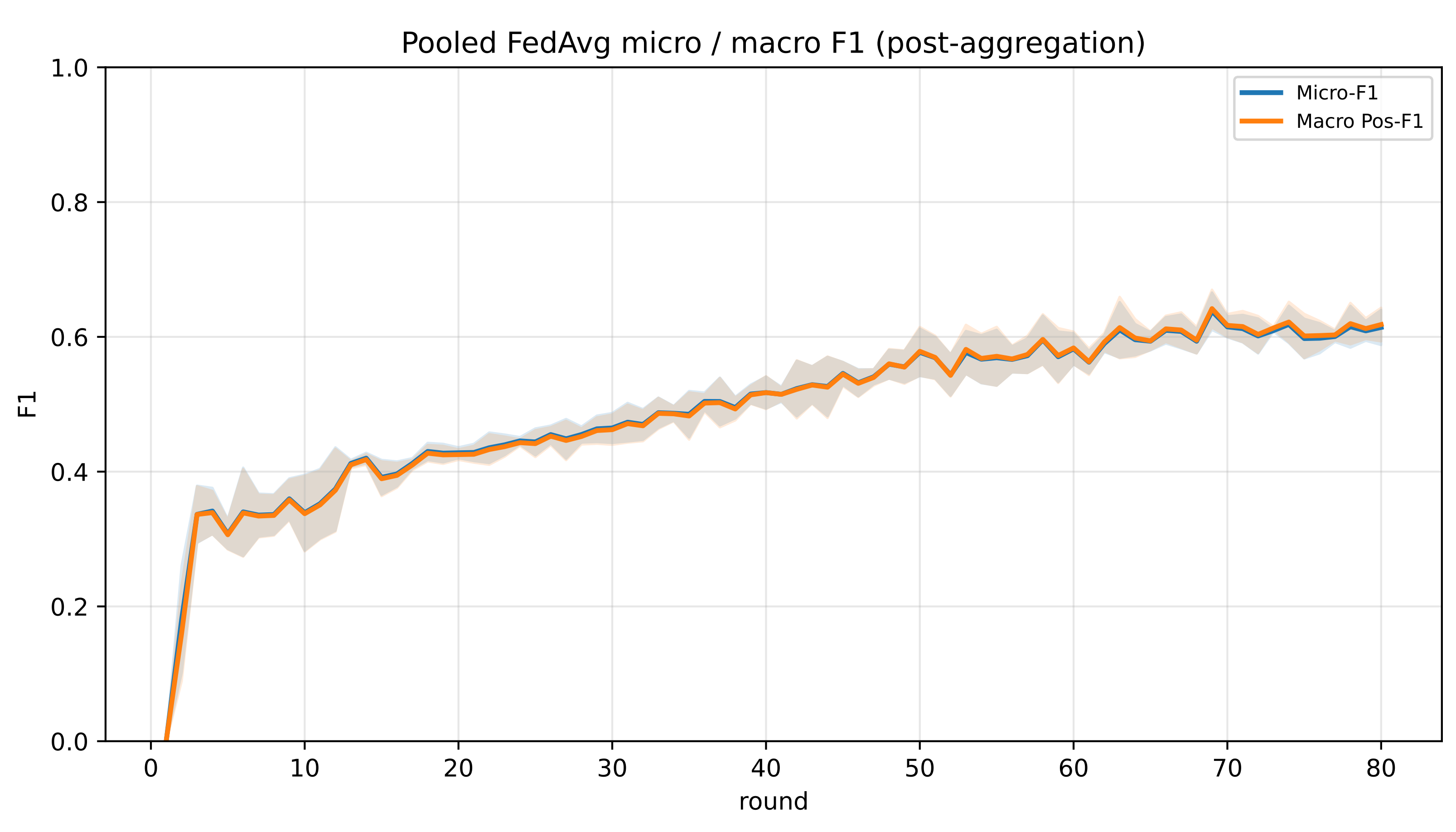
Best logged test performance (Fully Local vs Local-Centralized vs FedAvg)

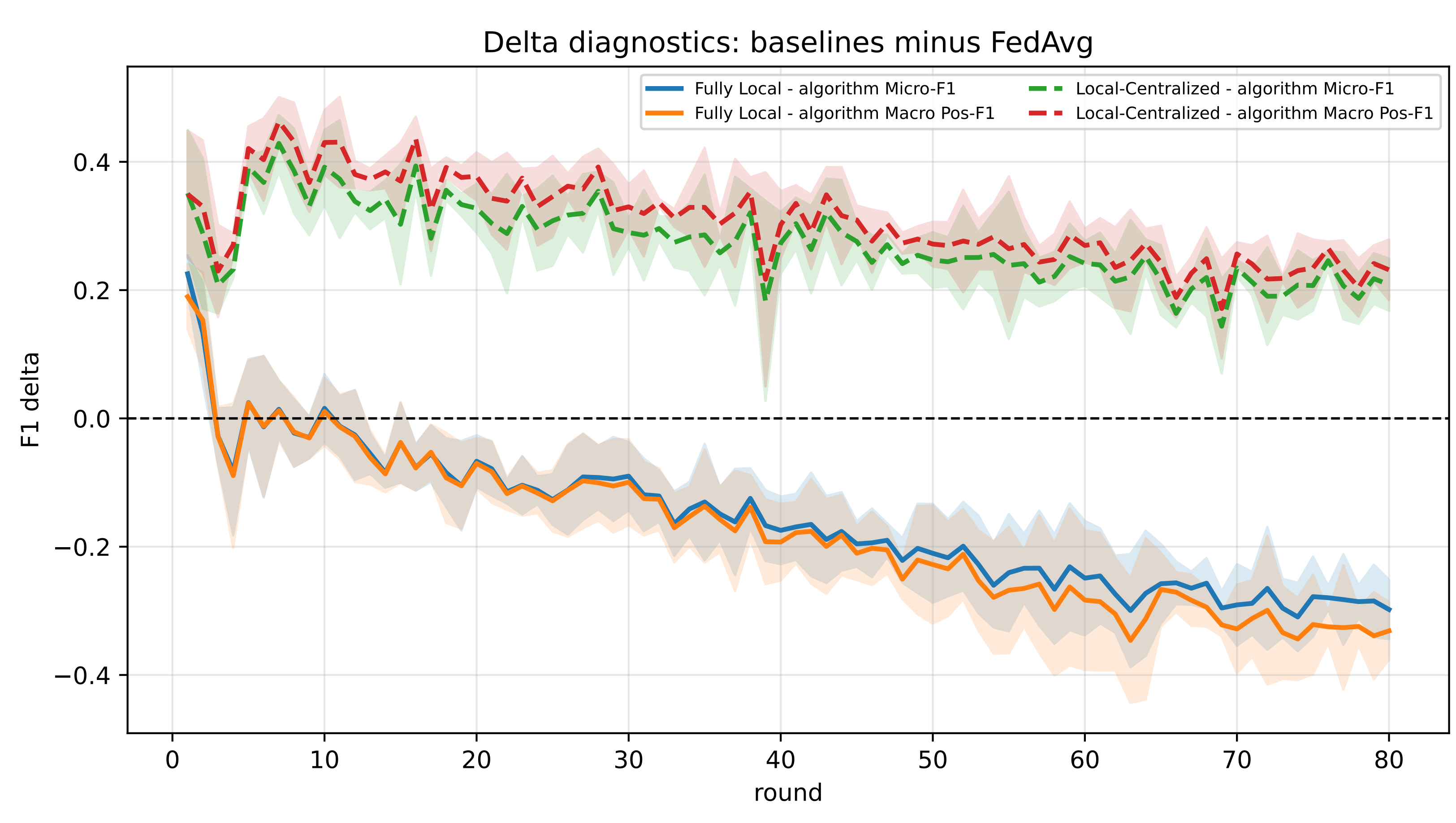
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	27.5% ± 1.5%	27.5% ± 1.4%	23.1% ± 1.8%	28.2% ± 0.7%	26.4% ± 1.8%	28.3% ± 1.7%	31.5% ± 2.1%
Local-Centralized	**84.9% ± 1.3%**	**86.7% ± 0.8%**	**96.8% ± 1.6%**	**95.1% ± 1.4%**	**86.6% ± 2.5%**	**81.4% ± 1.3%**	**73.7% ± 5.8%**
FedAvg	58.9% ± 2.1%	59.5% ± 2.1%	60.3% ± 2.1%	64.8% ± 5.2%	60.6% ± 1.9%	59.1% ± 3.0%	53.0% ± 2.9%









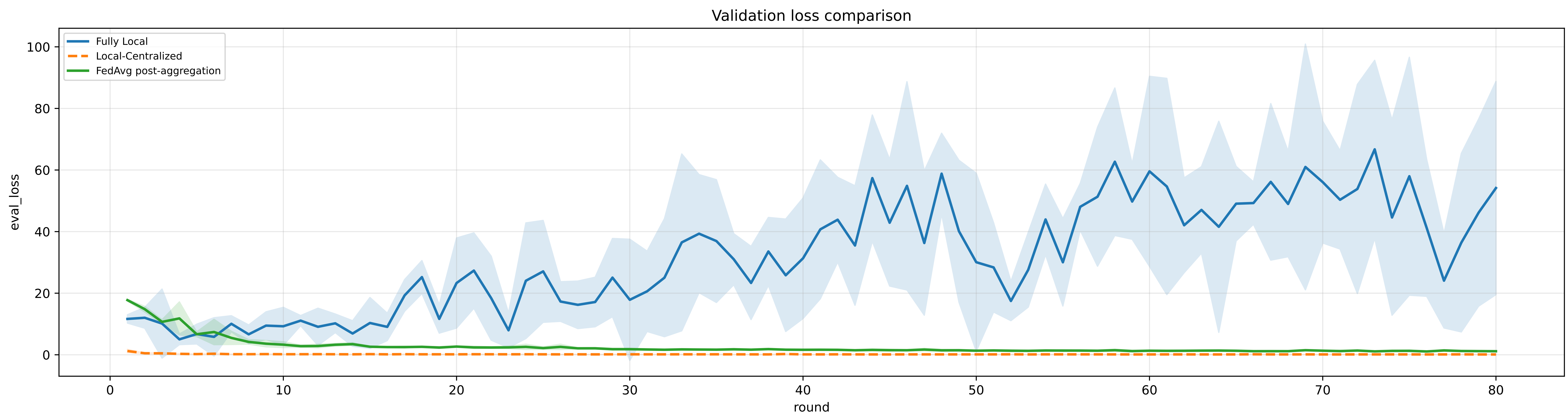


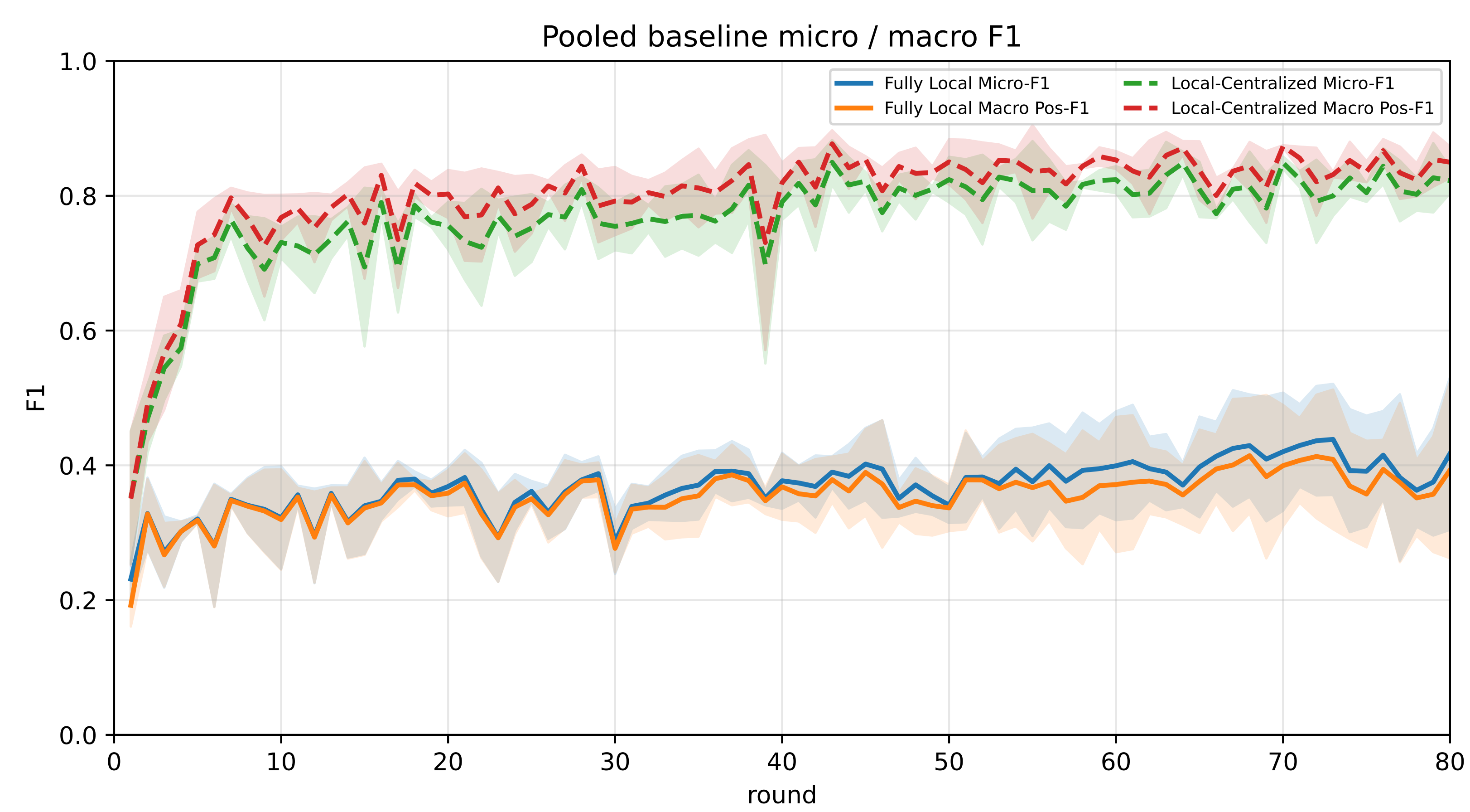
Controlled q-label heterogeneity: High | FedAvg diagnostics | family=q_task_cycle5 |
n_clients=1 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64_headmulti

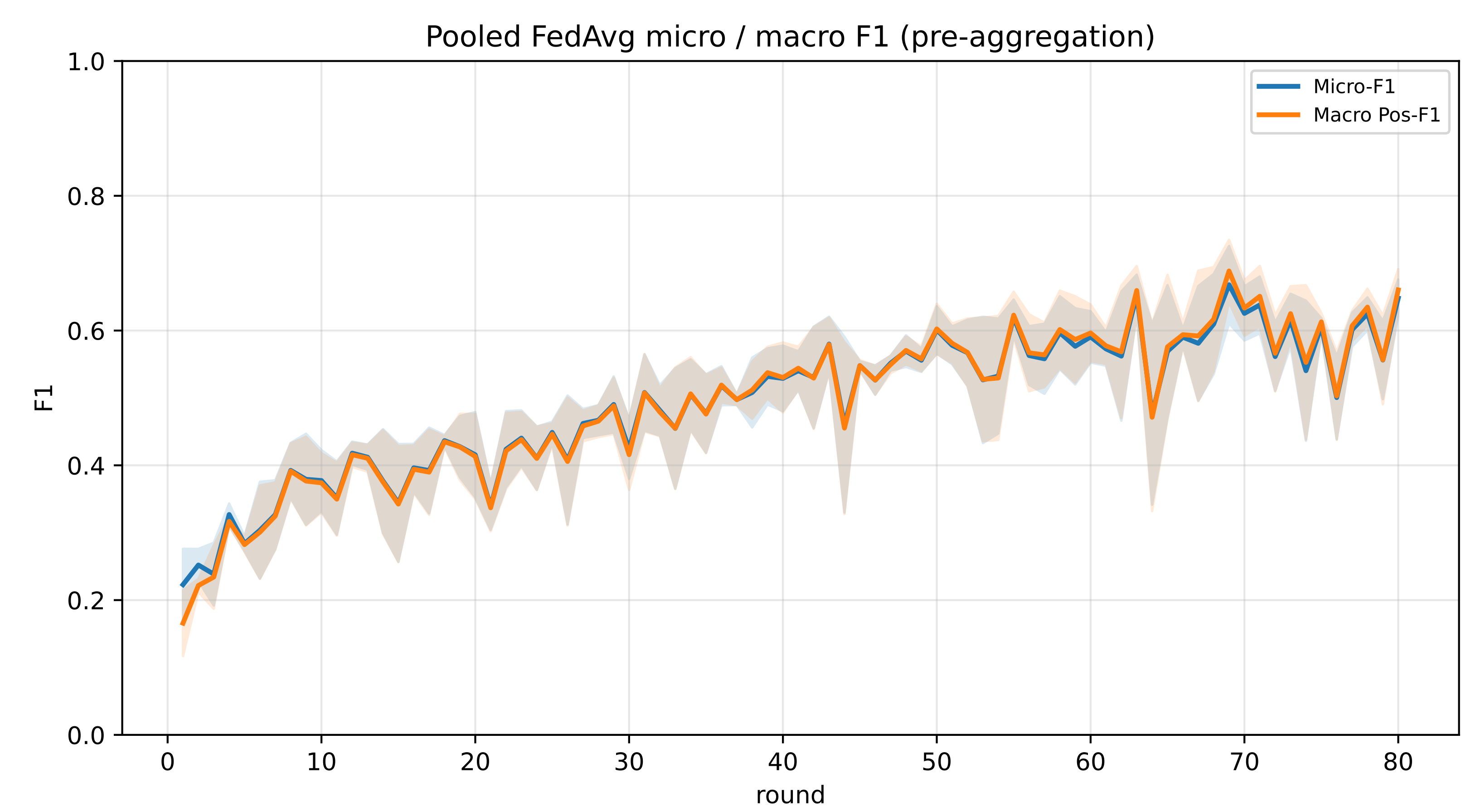
family=q_task_cycle5 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% |
cycle5=visible 5.1%±0.0% / true 6.4% | cycle6=visible 0.3%±0.0% / true 6.3%
heterogeneity=high | gamma=0.80 | task_jsd_mean=0.399 | task_jsd_median=0.399 | task_jsd_max=0.403 | family_centroid_jsd_mean=0.399

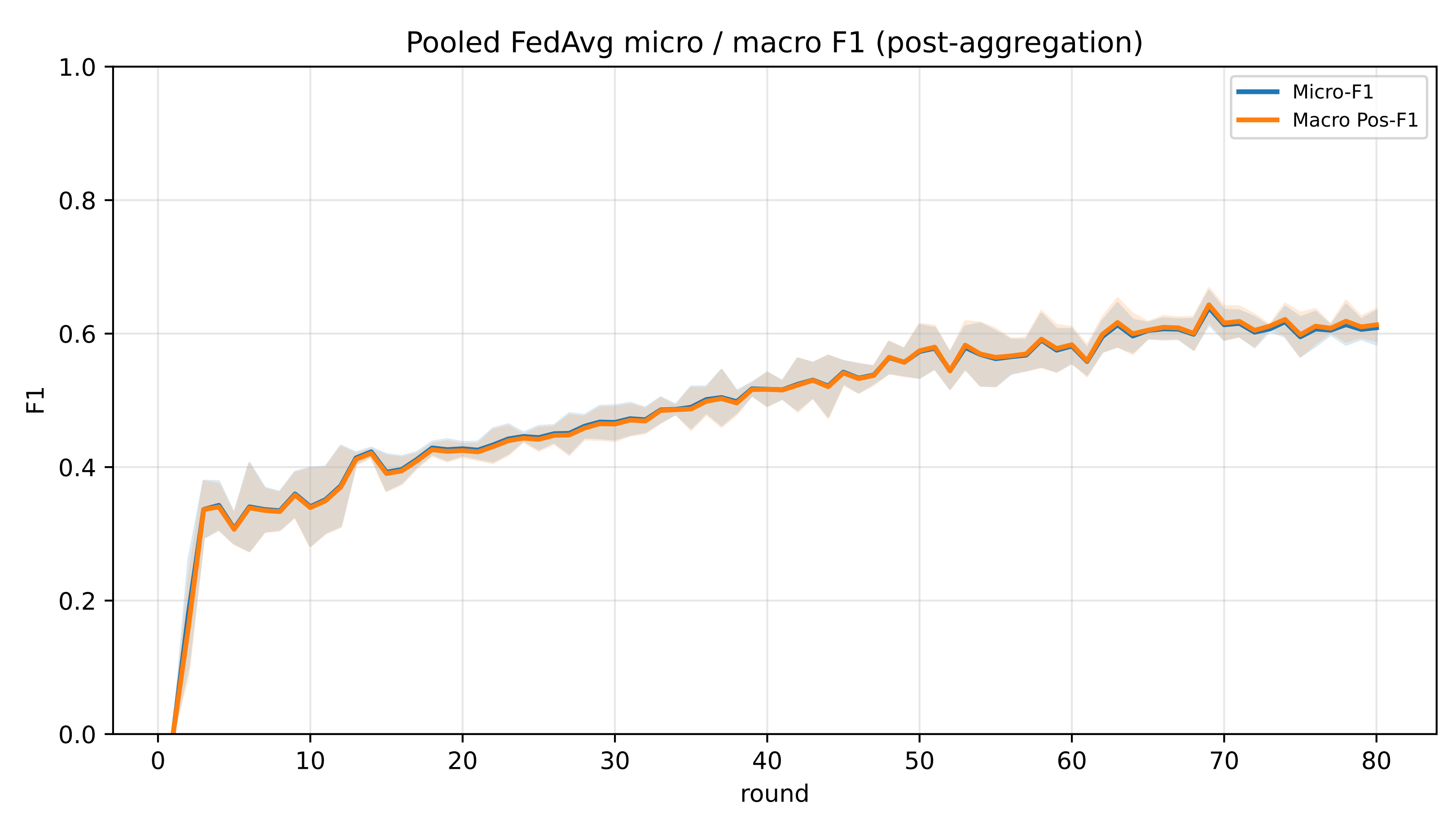
Best logged test performance (Fully Local vs Local-Centralized vs FedAvg)

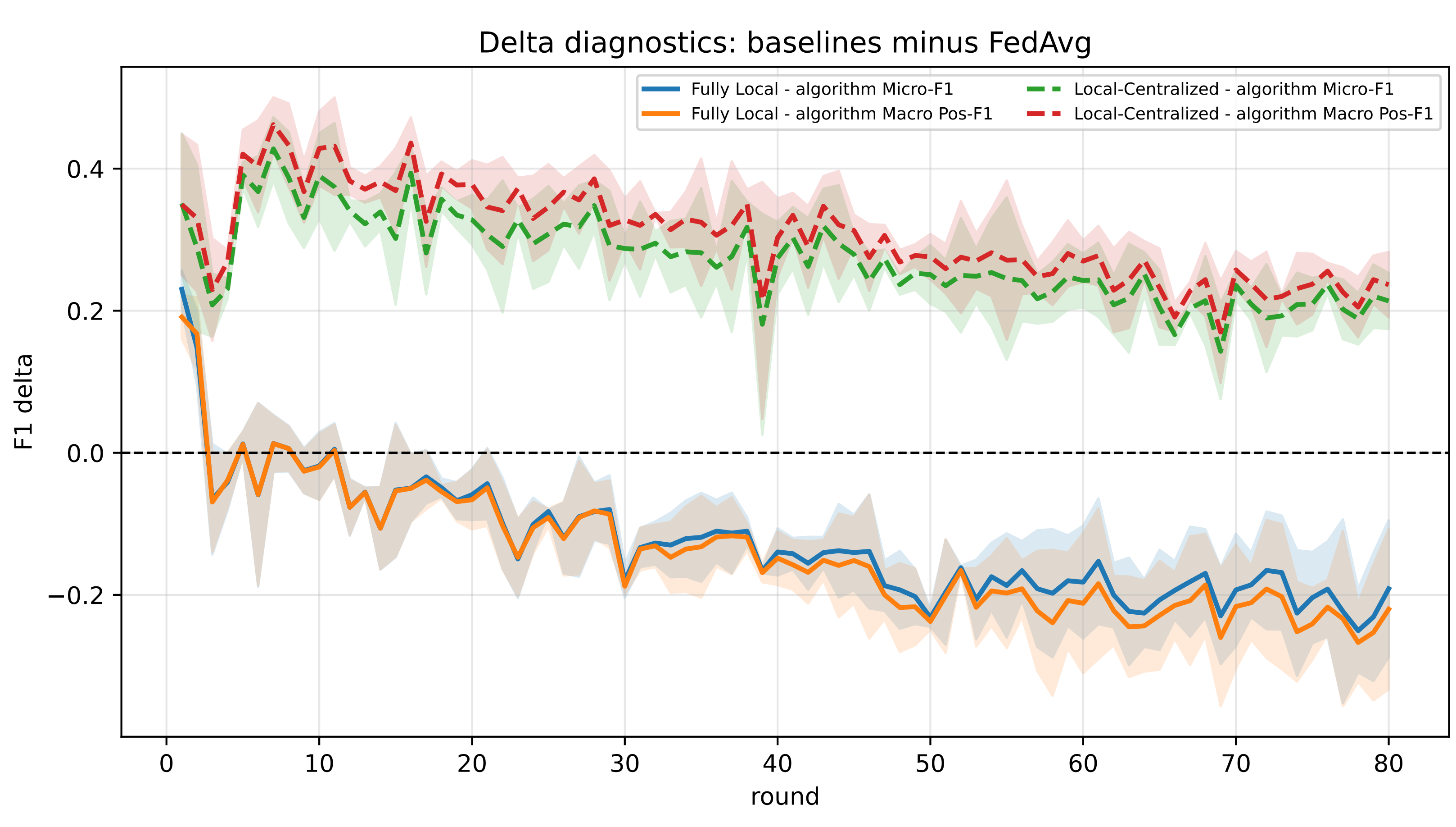
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	24.2% ± 1.1%	24.2% ± 1.1%	20.1% ± 0.7%	25.3% ± 1.1%	24.5% ± 1.6%	22.9% ± 1.3%	28.3% ± 1.1%
Local-Centralized	**84.9% ± 1.3%**	**86.7% ± 0.8%**	**96.8% ± 1.6%**	**95.1% ± 1.4%**	**86.6% ± 2.5%**	**81.4% ± 1.3%**	**73.7% ± 5.8%**
FedAvg	59.3% ± 1.8%	59.9% ± 1.8%	60.0% ± 2.5%	65.6% ± 5.4%	61.2% ± 2.5%	59.6% ± 2.5%	53.2% ± 3.2%











Controlled q-label heterogeneity: High | FedAvg diagnostics | family=q_task_cycle6 | n_clients=1 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64_headmulti

family=q_task_cycle6 | n_clients=1 | cycle2=visible 0.3%±0.0% / true 5.8% | cycle3=visible 0.3%±0.0% / true 6.2% | cycle4=visible 0.3%±0.0% / true 6.6% | cycle5=visible 0.3%±0.0% / true 6.4% | cycle6=visible 5.0%±0.0% / true 6.3%
heterogeneity=high | gamma=0.80 | task_jsd_mean=0.399 | task_jsd_median=0.399 | task_jsd_max=0.403 | family_centroid_jsd_mean=0.399

Best logged test performance (Fully Local vs Local-Centralized vs FedAvg)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Fully Local	31.2% ± 8.8%	31.1% ± 8.9%	26.7% ± 8.1%	32.1% ± 9.5%	33.6% ± 11.2%	31.3% ± 9.2%	32.1% ± 6.8%
Local-Centralized	**84.9% ± 1.3%**	**86.7% ± 0.8%**	**96.8% ± 1.6%**	**95.1% ± 1.4%**	**86.6% ± 2.5%**	**81.4% ± 1.3%**	**73.7% ± 5.8%**
FedAvg	59.4% ± 2.6%	60.1% ± 2.5%	60.8% ± 2.3%	65.2% ± 6.1%	61.7% ± 2.0%	59.8% ± 3.1%	53.0% ± 3.9%